



Wrocław University of Technology

# **Natural Price Resources Modeling Using Time Series Methods**

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# Abstract

This thesis is written to analyze and predict natural gas prices using different econometric models. The models utilized can be divided into two groups: univariate and multivariate.

**Univariate** analysis consist of **ARIMA (Autoregressive Integrated Moving Average)** models. **Multivariate** analysis approaches models like **ARIMA** with **TRANSFER FUNCTION (ARMAX)** and **Error Correction Model (ECM)**. Models are evaluated on various time horizons (different identification and estimation sets as well as predicting different numbers of values). Data set is also divided into local and global data. Analyzed data are Henry Hub Natural Gas Spot Price and West Texas Intermediate Crude Oil Spot Price. Additional exogenous variables are added as well. Time horizon covers the period from August 1997 to May 2014. At the end, all models are compared and the best one is chosen. The best predictive power reveals ARIMAX/ Transfer Function models.



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# Chapter 1

## Introduction

### 1.1 Motivation

Natural resources play a significant role in today's world. The approach of using natural resources can indicate the level of modern society development. We can easily see that more developed countries are using their resources in a smart way, while less developed countries are either not using or wasting their resources. The way we handle our resources now will also impact our future (and the future of following generations). According to the Wikipedia: "The vast majority of resources are exhaustible, which means they have a finite quantity, and can be depleted if managed improperly." That is why it is crucial to use them wisely and also price them prudently. One of the manners in which natural resources can be classified is based on renewability or non-renewability. Because non-renewable resources either form slowly or do not naturally form in the environment, their amount is limited, which is why their role is so essential.

Utilization of natural resources is a very broad topic. Energy is one of our most precious natural resources and proper usage of this energy is of prime importance.

The biggest energy consumer in the world is North America, which consumes one-quarter of the world's primary energy production [1]. 83% of the energy utilized in North America consists of hydrocarbons (non-renewable natural resources). The energy coming from hydrocarbons incorporates oil, gas and coal. According to Hughes [1] the consumption of world's oil production in North America reached 27% in 2008, while consumption of natural gas was equivalent to 25% and consumption of world's coal production was 18%.

Natural gas is the world's fastest-growing fossil fuel, with consumption increasing from 113.0 trillion cubic feet in 2010 to 185.0 trillion cubic feet in 2040. [IEO2013 Reference case][28]

Because of its "characteristics" natural gas is favored as an environmentally attractive fuel compared with other hydrocarbon fuels. Its generation is projected to surpass coal generation by 2035. Because of the growth of demand and production of natural gas, it is meaningful to be able to predict its price. Since the United States is currently the biggest natural gas producer and consumer, the data used in this paper is derived from the United States of America. An important property of the natural gas price is that it is continentally priced (not globally priced like oil) due to the expense and logistical difficulty of moving it across the ocean as liquefied natural gas (LNG). This fact also explains the choice of the data. For all the above reasons, the natural resource analyzed in this paper is natural gas. Because of the locally priced feature, Henry Hub (HH) Natural Gas Spot Price is elected for the main analysis.

### 1.2 Research objective

The intention of this thesis is to model and predict natural gas prices using different time series methods and compare prediction so the best model can be chosen. The goal is also to identify the relevant variables influencing the natural gas price. For this purpose univariate (ARIMA) and multivariate (ARMAX, ECM) models are used. Each method is utilized for all data sets. Analysis concerns not only different time series methods, but also different time horizons, different exogenous variables (multivariate analysis), and different regions (for exogenous variables). For each method, time horizon and region best parameters are chosen to obtain the best results.

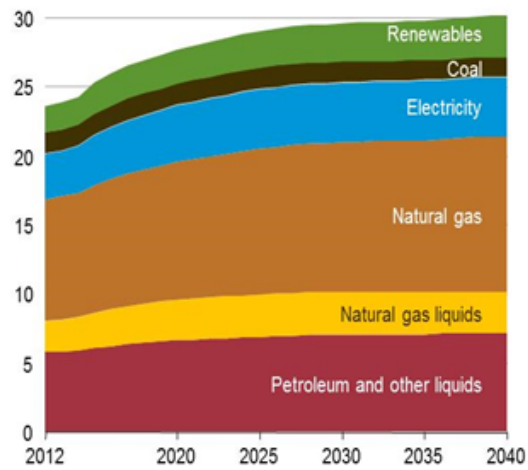


Figure 1.1: Energy sources prediction for the United States 2005-2040 (source:U.S. Energy Information Administration (EIA))

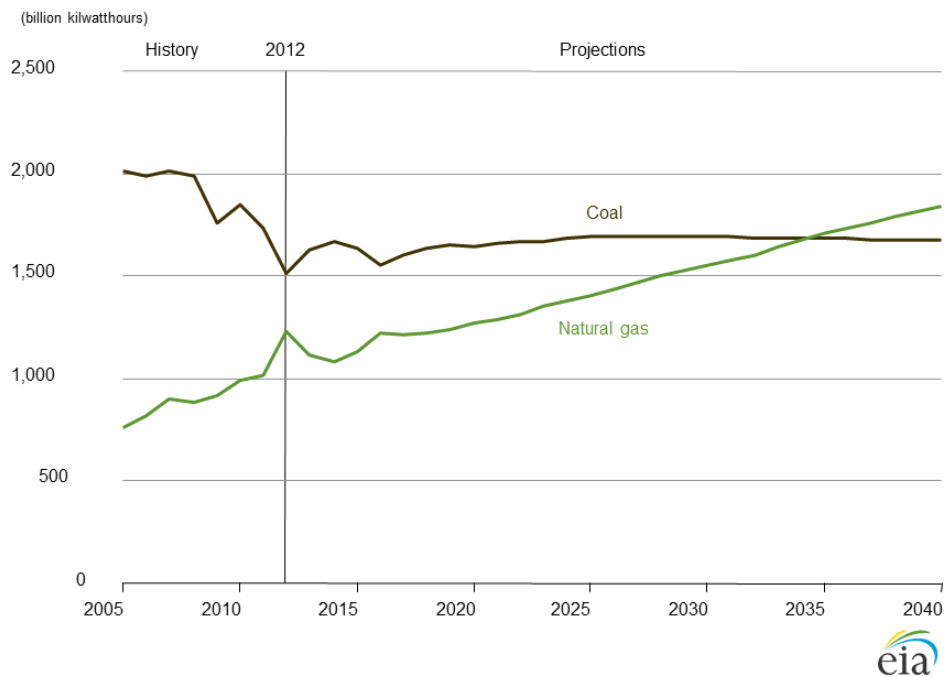


Figure 1.2: Electricity generation from natural gas and coal 2005-2040 (source:U.S. Energy Information Administration (EIA))

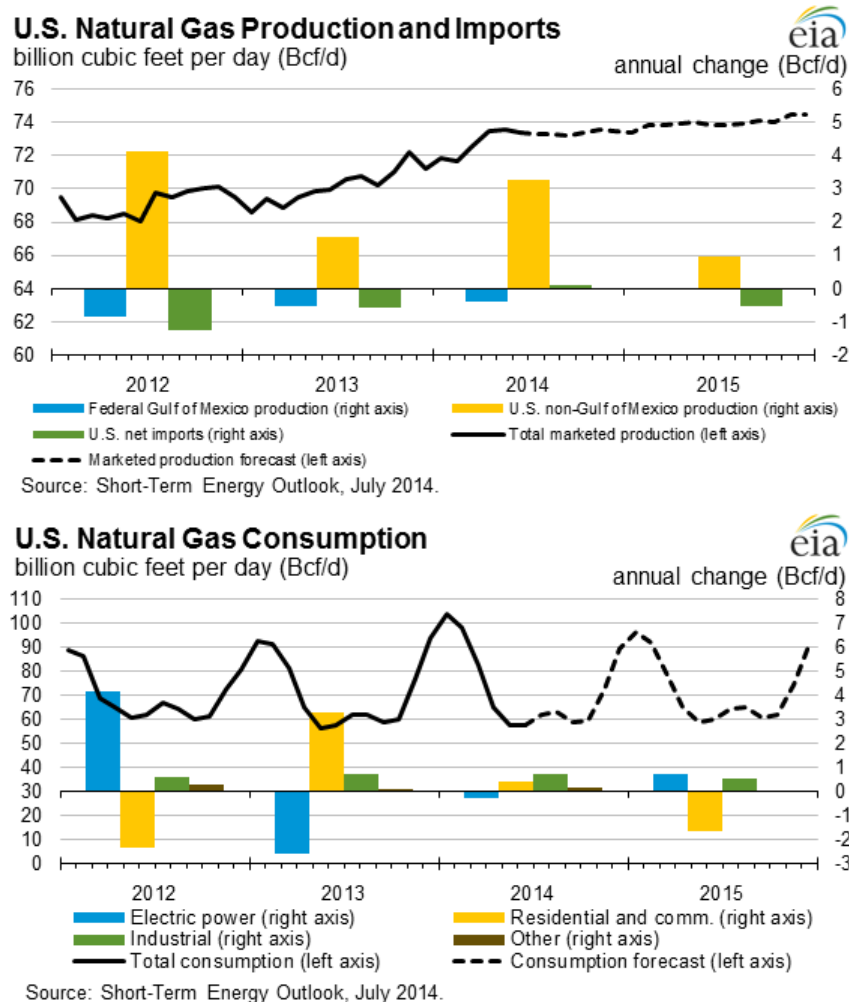


Figure 1.3: U.S. natural gas production and consumption (source:U.S. Energy Information Administration (EIA))

### 1.3 Organization of the thesis

This thesis is divided into 6 chapters.

*Chapter 1* explains the motivation and main objective of this paper and highlights main points of the analysis. It also discloses the importance of natural gas price predictions.

In *Chapter 2* different approaches concerning natural gas price predictions are described. Each of the further mentioned papers has some impact on this thesis and this influence is also explored.

*Chapter 3* describes used data, its sources, time range and its statistical properties. Different variables are reported.

In *Chapter 4* theoretical basics as well as the whole research process is detailed presented.

Finally, *Chapter 5* reveals the results of the conducted analysis. Different models and their variations are compared, and the best are chosen.

*Chapter 6* concludes and closes the whole thesis, giving fresh ideas of possible expansion of the subject.



## Chapter 2

# Literature review

Since natural gas price plays a significant role in today's economics there are many different approaches to investigated problem. Almost every paper on natural gas price prediction links natural gas price with crude oil price using different models, time horizons and also variety of other exogenous factors.

There are numerous papers predicting natural gas prices using different time series method, but I will mention only those that influenced this paper the most.

Villar and Joutz (2006) [2] use West Texas Intermediate (WTI) crude oil price to predict Henry Hub (HH) natural gas price. They employ monthly data series from January 1989 through December 2005, focusing on decomposing the short run and long run effects of changes in crude oil and natural gas prices. Monthly data selection is justified by having sufficient texture to capture short - run movements over time, without adding unnecessary complexity to the analysis. In order to remove the scale effect in the variables and reduce the possible effect of heteroscedasticity, data are transformed into natural logarithmic scales. Vector autoregressive model (VAR) is implemented. Log-likelihood, Bayesian Schwarz Criterion (BSC), Hannan-Quinn Criterion (HQ) and Akaike Information Criterion (AIC) was utilized for Model Reduction for VAR. Authors use Johansen test for cointegration analysis. Since cointegrating relation influences WTI crude oil price, VECM can be expressed as error correction mechanism (ECM) and the final model can be reduced to Conditional Error Correction Model.

Brown and Yücel (2008) [3] also employed Henry Hub natural gas price and West Texas Intermediate crude oil price. They applied weekly data from June 13, 1997 to July 14, 2006 to establish the relationship between natural gas and crude oil prices. In their paper they presented different models (driving natural gas prices), like simple rules of thumb (10-to-1, 6-to-1, Burner-tip -parity), but also focused on more complicated models like Simple Linear Regression and Error Correction Model. Additional exogenous variables (like weather, seasonality, natural gas storage conditions and disruption in natural gas production) were taken into consideration. The time period regarding those factors had to be reduced to June 13,1997 till July 14,2006. Because of different time periods and additional exogenous variables, 3 different models were implemented (long period without the exogenous variables, short period without the exogenous variables and short period with the exogenous variables). Schwarz and Akaike Information Criterion were used. They concluded that in the long term natural gas price depends on crude oil price, and also in the short time horizon natural gas price is influenced by weather and inventory factors.

Ramberg and Parson (2012) [4] described similar Vector Error Correction (VECR) and Conditional Error Correction Model (conditional ECM). They used weekly day-ahead price of commodities - Henry Hub natural gas price (volume weighted averages) and West Texas Intermediate crude oil price (arithmetic averages). Cointegrating relationship was fitted to the data from June 13,1997 to December 31, 2010. As a result, they reported the character of relationship between natural gas and crude oil prices.

Nick and Thoenes (2013) [5] proposed a structural Vector Autoregressive (VAR) approach using data from Germany to clarify the effects of different fundamental influences on natural gas price during supply interruptions. The data set utilizes weekly data from January 2008 till June 2012. An interesting fact is their choice of varies forecast horizons (1,2,4,8,12,26 and 52 weeks). Variables included in the model are: NetConnect Germany (NCG) - natural gas price - spot price, Brent crude oil price, the North-Western European coal price, the deviation from the historical

average heating degree days in Germany, German natural gas storage data, shortfalls of the natural gas supplies to the European market and European LNG import data. Forecasting errors were measured in Mean Square Error (MSE) and forecast error variance decomposition. Similar to the previous approaches, the natural gas, crude oil and coal price time series are transformed into their natural logarithms. They concluded (again, similar to the previous research) that natural gas is affected in the short term by temperature, storage, supply shortfalls and imports of liquefied natural gas (LNG), while long term development is closely dependent on crude oil prices and coal prices.

Mishra (2012) [6] provided natural gas price forecasting using time series (ARIMA) and a nonparametric approach - nonparametric regression Alternating Conditional Estimation (ACE). Monthly data from January 1976 to September 2011 were studied. The dependent variable was US Natural Gas well head price. Independent variables were represented by US Crude Oil First Purchase Price and Gold Average Price. Error metrics used for model evaluation and comparison consist of Mean Squared Error (MSE) and Akaike Information Criterion (AIC). In the final comparison, all time series models (except GARCH) provided suitable forecast.

Krichene (2005) [7] introduced the Simultaneous Equations Model for World Crude Oil and Natural Gas Markets. Time horizon in this analysis was very broad, from 1918 to 2004, with a midpoint in 1974 (which divided data set into two regimes 1918-1973 and 1974-2004). A vector autoregressive model (VAR) is used to analyze cointegration between crude oil prices, changes in the nominal effective exchange rate (NEER) of the U.S. dollar and the U.S. interest rate for monthly, quarterly and annual data.

Ababio (2012) [8] employs the univariate and multivariate ARIMA model (ARIMA and ARIMAX) to evaluate in-sample forecasting accuracy. The monthly adjusted close price of four stocks in the oil and gas industry (London Stock Exchange) in time horizon 2005-2010 were analyzed. Model forecasting was evaluated by few errors measurements (Mean Square Error (MSE), Mean Absolute Error (MAE) and Root Mean Square Error (RMSE)). Linear correlation between independent and dependent variables as well as Akaike Information Criterion (AIC) were also taken into consideration. The best models occurred to be ARIMAX models.

There are many other sources that contributed to this work. Not all of them are predicting natural gas prices. Bosler (2010) [11] in his Master's thesis investigates different models for oil price prediction and forecasting. Also multitude of research on predicting natural gas consumption were useful.

As can be seen, two most popular models to predict or estimate natural gas price prediction are the Vector Autoregressive model (VAR) and Error Correction Model (ECM). This paper introduces also the Error Correction Model but more as a comparison model for ARIMA models, without getting into details about ECM model and cointegration analysis. The main focus of this work is the ARIMA model (univariate and multivariate). Among lots of different independent variables for natural gas price modeling, the most significant seems to be crude oil price. This is why West Texas Intermediate Crude Oil Spot Price was selected as the main independent variable. Furthermore, exogenous variables like weather conditions, storage, and disruption in production were employed. Since most of the previously mentioned papers were from 2012 or earlier, they could not account for newer data. Models presented in this analysis use data up till May 2013.



# Chapter 3

## Data

Commonly used data in Natural Gas Price Forecasting are weekly, monthly and quarterly based time series. In this paper weekly data are used.

Weekly data allow one to characterize the volatility in much greater detail than quarterly or annual data, without adding too much volatility present in daily data that is usually caused by weather conditions.

### 3.1 Data sources

Analyzed data were collected from several sources. Different sources as well as data full names and units are presented in (Table 3.1).

### 3.2 Time range

The time period used for the analysis ranged from August 1997 until May 2014. For analysis of weekly data the time series begins on August 1, 1997 and ends on May 30, 2014, and contains 878 observations.

Weekly data (if not provided) were obtained as arithmetical averages of daily data.

In order to identify and estimate the model, time series data has be to divided into two sets - *identification and estimation* set and *model forecast* set. Time series consists of weekly spot prices from the period of August 1997 - May 2014.

The data set contains 878 observations (877 for differentiated data). To obtain the best results, different time periods for both data sets (prediction and evaluation) will be used, and different ARIMA (p,d,q) models will be estimated.

Since ARIMA time series is well known for its short time predictions, first only 4 values will be predicted. This is equivalent to a month-ahead prediction (using weekly data). Second predicting horizon includes 12 values, which is equivalent to a quarter-ahead prediction (using weekly data). In order to find the best model, 5 different identification and estimation sets were applied.

In the first variant 874 observations will be used for the identification and estimation of the model, while the remaining 4 observations will be used as post-sample comparison of the model performance. In the next step, 583 observations are used to predict 4 values, then 439, 305 and finally only 244 observations.

While using 12 values for post-sample comparison of the model performance, 866 observations are employed to built ARIMA model. Similar to the previous case (predicting 4 values), the identification set is getting smaller (values of 575, 431, 297 and 236).

In Figure 3.1 green dotted lines represent the beginning of the identification and estimation periods (observations number 1, 292, 436, 570, 631, for exact dates see Tables 3.2), while green and black lines show the beginning of prediction sets (observation number 866 for predicting 12 values and observatin number 874 for predicting 4 values).

Table 3.1: Variable sources

| Variable                      | Full name                                        | Short name | Unit                                                        | Source                                 | Website                                                                                                                                                                                                                                       |
|-------------------------------|--------------------------------------------------|------------|-------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural gas price             | Henry Hub Natural Gas Spot Price                 | HH         | Dollars per million British Thermal Unit (\$\text{mmBtu}\$) | U.S. Energy Information Administration | <a href="http://www.eia.gov/dnav/ng/hist/rngwhhdm.htm">http://www.eia.gov/dnav/ng/hist/rngwhhdm.htm</a>                                                                                                                                       |
| Crude oil price               | West Texas Intermediate Crude Oil Spot Price     | WTI        | Dollars per Barrel (\$\text{bbl}\$)                         | U.S. Energy Information Administration | <a href="http://www.eia.gov">http://www.eia.gov</a>                                                                                                                                                                                           |
| Heating degree days           |                                                  | HDD        | day                                                         | Climate Prediction Center (NOAA)       | <a href="ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/">ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/</a> |
| Cooling degree days           |                                                  | CDD        | day                                                         | Climate Prediction Center (NOAA)       | <a href="ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/">ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/</a> |
| Heating degree days deviation |                                                  | HDDdev     | day                                                         | Climate Prediction Center (NOAA)       | <a href="ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/">ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/</a> |
| Cooling degree days deviation |                                                  | CDDdev     | day                                                         | Climate Prediction Center (NOAA)       | <a href="ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/">ftp://ftp.cpc.ncep.noaa.gov/htdocs/products/analysis_monitoring/cdus/degree_days/archives/degree_days/archives/</a> |
| Storage                       | Working gas in underground storage               | STOR       | Billion Cubic Feet ( <b>BCF</b> )                           | U.S. Energy Information Administration | <a href="http://www.eia.gov/dnav/ng/ng_stor_wkly_s1_w.htm">http://www.eia.gov/dnav/ng/ng_stor_wkly_s1_w.htm</a>                                                                                                                               |
| Storage differential          | difference in working gas in underground storage | STOR_ch    | Billion Cubic Feet ( <b>BCF</b> )                           | U.S. Energy Information Administration | <a href="http://www.eia.gov/dnav/ng/ng_stor_wkly_s1_w.htm">http://www.eia.gov/dnav/ng/ng_stor_wkly_s1_w.htm</a>                                                                                                                               |
| Shut in production            | Shut-in gas production                           | SHUTin     | million cubic feet of gas per day ( <b>MMCF/D</b> )         | The Minerals Management Service        | <a href="http://www.bsee.gov/Hurricanes/History/">http://www.bsee.gov/Hurricanes/History/</a>                                                                                                                                                 |

Table 3.2: Identification and estimation sets for Henry Hub Natural Gas Spot Price - weekly data

| Number of predicted values $n = 4$ |                  |                |                      |                      |
|------------------------------------|------------------|----------------|----------------------|----------------------|
| Observation number                 |                  |                | Date                 |                      |
| Identification set                 | Ident Set Length | Prediction Set | Identification set   | Prediction Set       |
| 1 - 874                            | 874              | 875 - 878      | 1-Aug-97 - 2-May-14  | 9-May-14 - 30-May-14 |
| 292 - 874                          | 583              | 875 - 878      | 28-Feb-03 - 2-May-14 | 9-May-14 - 30-May-14 |
| 436 - 874                          | 439              | 875 - 878      | 9-Dec-05 - 2-May-14  | 9-May-14 - 30-May-14 |
| 570 - 874                          | 305              | 875 - 878      | 4-Jul-08 - 2-May-14  | 9-May-14 - 30-May-14 |
| 631 - 874                          | 244              | 875 - 878      | 4-Sep-09 - 2-May-14  | 9-May-14 - 30-May-14 |

| Number of predicted values $n = 12$ |                  |                |                      |                       |
|-------------------------------------|------------------|----------------|----------------------|-----------------------|
| Observation number                  |                  |                | Date                 |                       |
| Identification set                  | Ident Set Length | Prediction Set | Identification set   | Prediction Set        |
| 1 - 866                             | 866              | 867 - 878      | 1-Aug-97 - 7-Mar-14  | 14-Mar-14 - 30-May-14 |
| 292 - 866                           | 575              | 867 - 878      | 28-Feb-03 - 7-Mar-14 | 14-Mar-14 - 30-May-14 |
| 436 - 866                           | 431              | 867 - 878      | 9-Dec-05 - 7-Mar-14  | 14-Mar-14 - 30-May-14 |
| 570 - 866                           | 297              | 867 - 878      | 4-Jul-08 - 7-Mar-14  | 14-Mar-14 - 30-May-14 |
| 631 - 866                           | 236              | 867 - 878      | 4-Sep-09 - 7-Mar-14  | 14-Mar-14 - 30-May-14 |

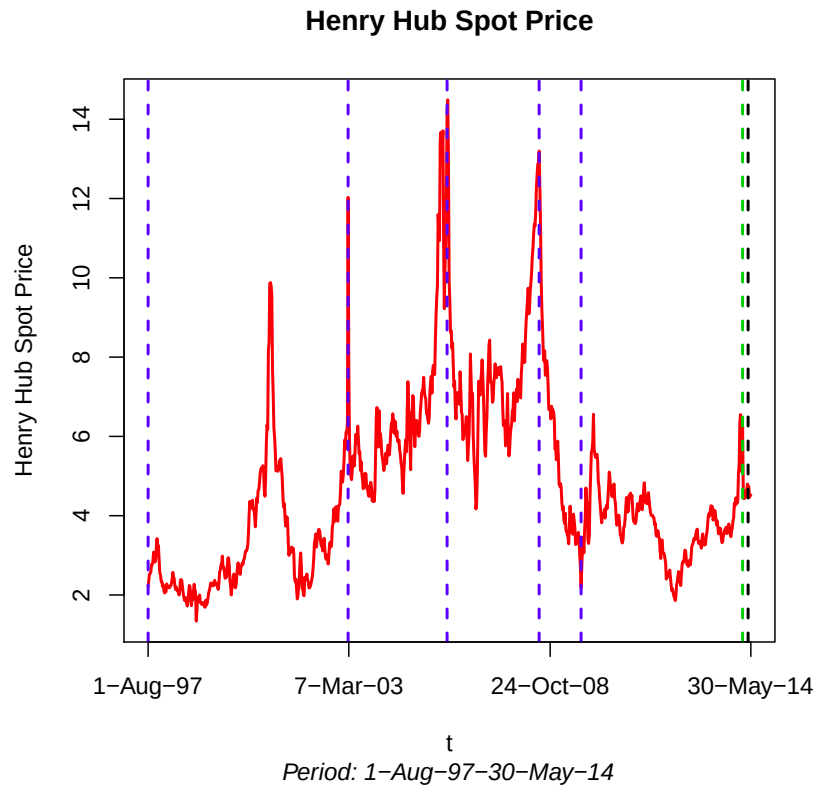


Figure 3.1: Henry Hub weekly spot prices - time horizon division

### 3.3 Variables

In case of multivariate analysis, different exogenous variables are used.

One of the most important (and common) exogenous variable driving natural gas price is crude oil price.

Additional factors are weather and storage correlated variables, such as the number of heating degree days (HDD), the number of cooling degree days (CDD), the deviation from the normal number of heating degree days (HDDdev), the deviation from the normal number of cooling degree days (CDDdev), the deviation of the amount of natural gas in storage from its five-year average (STOR\_ch), and the amount of natural gas production shut-in, e.g. due to storms and hurricanes (SHUTin)).

#### Henry Hub Natural Gas Spot Price

The time series data for natural gas price is **Henry Hub (HH) Natural Gas Spot Price** (delivery in Erath, Louisiana). It is measured in Dollars per million British Thermal Unit ( \$ \backslash \text{mmBtu} ).

Henry Hub is a distribution hub on the natural gas pipeline system in Erath, Louisiana. For further information about natural gas see (Appendix A.1.1).

#### West Texas Intermediate Crude Oil Spot Price

In multivariate analysis, **West Texas Intermediate (WTI) Spot Price** (delivery in Cushing, Oklahoma) is applied as oil prices. The unit is Dollars per Barrel ( \$ \backslash \text{bbl} ).

Since both natural gas and crude oil are energy carriers, it is said that **one barrel (1 bbl) of crude oil** has approximately the same energy content as **six million Btu (6mm Btu) of natural gas** [4].

#### Heating Degree Day Data

Heating Degree Day is a way of measuring energy requirements for heating.

According to the Climate Prediction Center of National Weather Services [30] Heating Degree Day is a measure of the coldness of the weather experienced, based on the extent to which the daily mean temperature falls below a reference temperature (65 °F, which is equivalent to 18.33°C ). Typically, heating degree days are calculated as how much colder the mean temperature at a location is than 65°F on a given day. For example, if a location experiences a mean temperature of 55 °F on a certain day, there were 10 HDD (Heating Degree Days) that day because  $65 - 55 = 10$ . A daily mean temperature represents the sum of the high and the low readings divided by two.

Since natural gas is a form of energy which is used for heating, Heating Degree Day seem to be important variable, linked with natural gas demand (consumption) which can also drive natural gas price.

#### Heating Degree Day Deviation

Heating Degree Day Deviation is simply deviation from normal Heating Degree Day at the same week (since weekly data was used).

#### Cooling Degree Day Data

Cooling Degree Days are the opposite of the Heating Degree Days in a sense that it measures the need of cooling (air conditioning).

This variable is based on the extent to which the daily mean temperature rises above a reference temperature (65 °F). For example, on a day when the mean outdoor temperature is 85 °F, there would be 20 degree-days experienced. A daily mean temperature represents the sum of the high and the low readings divided by two.

Cooling Degree Days are predicted to influence southern regions much stronger than northern because of hot summers. This fact is another reason dividing data set into local and global variables.

Considering the price of natural gas traded locally (not like price of crude oil traded globally), the price is impacted much more strongly by fluctuations in supply and demand specific to the North American Marketplace. Since Henry

Hub Natural Gas Spot Price (delivery in Erath, Louisiana) and West Texas Intermediate Crude Oil Spot Price (delivery Cushing, Oklahoma) are used, Heating and Cooling Degree Days for West South Central Census Region (Texas, Oklahoma, Louisiana and Arkansas) are used. Also, data from the whole United States is considered.

### **Cooling Degree Day Deviation**

Cooling Degree Day Deviation is simply deviation from normal Cooling Degree Day at the same week (since weekly data was used).

### **Working Gas in Storage**

**The Underground Storage Report** measures the utilization of subsurface facilities for storing gas that has been transferred from its original location and stored for the purposes of load balancing and meeting critical winter heating season gas supply requirements. The facilities are usually natural geological reservoirs such as depleted oil or gas fields or water bearing sands capped by impermeable rock. The facilities may be man-made or naturally occurring (according to American Gas Association [31]).

Storage data are divided into three regions:

1. Consuming Region East
2. Consuming Region West
3. Producing Region

Consuming Region East includes all states east of the Mississippi River less Alabama and Mississippi, plus Iowa, Nebraska and Missouri.

Consuming Region West includes all states west of the Mississippi River less the Producing Region and Iowa, Nebraska and Missouri.

Producing Region includes Texas, Oklahoma, Kansas, New Mexico, Louisiana, Arkansas, Mississippi and Alabama. In the paper I am going to use Producing Region and also total storage data.

### **Storage Differentials**

The Storage Differentials (STOR\_ch) variable is the difference between a weekly gas storage level and the 5-year running average for that date, reported in billion cubic feet (Bcf, equivalent to 1.03 million mmBtu). For more information see A.1.2

### **Shut-in Gas Production**

Shut-in production is a variable corresponding to the structural shocks (for example in Europe). The data collected are presenting the amount of natural gas shut in production in the Gulf of Mexico mostly because of Hurricanes (and also tropical storms). Since natural gas production in the Gulf of Mexico makes 10% of total U.S. gas production (according to the Energy Information Administration (EIA) of the Department of Energy), this variable might have significant impact on natural gas price. More information about this variable can be found in Appendix A.1.3.

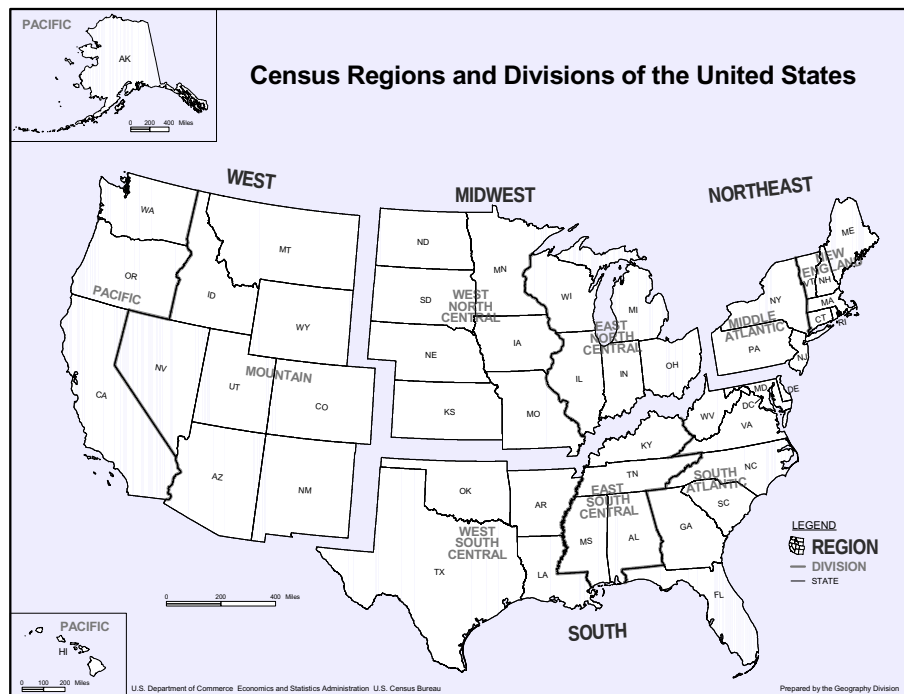


Figure 3.2: EIA Natural Gas Pipeline regions (source:U.S. Energy Information Administration (EIA))

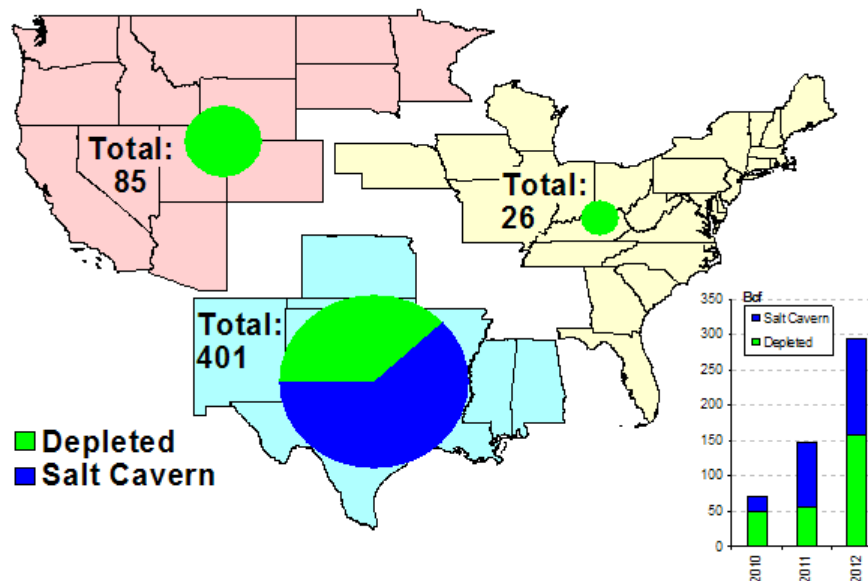


Figure 3.3: Potential natural gas storage capacity additions, 2010-2012 (source:U.S. Energy Information Administration (EIA))

# Chapter 4

## Methodology

As mentioned before, the most popular models applied for predicting natural gas price are Vector Autoregressive Models (VAR), Error Correction Model (ECM) and Autoregressive Integrated Moving Average models (ARIMA and ARIMAX). This chapter presents theoretical basics for statistical analysis that has been employed in order to reach the established objective. The main focus is on ARIMA (univariate and multivariate - ARIMAX) models. ECM models serves more as comparison models for time series (ARIMA) models.

### 4.1 Stationarity

In order to be able to conduct analysis and forecasting using time series models, data needs to be stationary (mean and variance do not depend on time). Data stationarity is crucial to avoid a spurious regression problem. Since most data in nature (and economy) has a non-stationary pattern, differencing has become the common method of bringing non-stationary time series to stationary. A time series (variable) is said to be *integrated* (4.2.1) of order one ( $I(1)$ ), if it is stationary after differencing it once, of order two ( $I(2)$ ) after differencing it twice, etc. If the data is stationary without differencing (stationary in levels), it is said to be integrated of order zero ( $I(0)$ ).

There are a few ways to determine if the data is stationary or not (i.e. has unit root). The first method is graphical data analysis. The presence of trend and volatility in the data can be verified by looking at the data plot. Also, autocorrelation function (**ACF**) as well as partial autocorrelation function (**PACF**) can deliver important information about the data stationarity. More formal tests of the integrated process (integrated process has a unit root, so the time series is not stationary) are various tests like Dickey-Fuller Test (**DF**), augmented Dickey-Fuller (**ADF**) test, Kwiatkowski-Phillips-Schmidt-Shin (**KPSS**) test, Ljung-Box test and Phillips-Perron (**PP**) Unit Root Test.

#### 4.1.1 Autocorrelation Function

The covariance function of  $X_t$  can be defined as:

$$\gamma_X(r, s) = \text{cov}(X_r, X_s) = E[(X_r - \mu_X(r)) - (X_s - \mu_X(s))] \quad (4.1)$$

where:

- $r$  and  $s$  are independent
- $\mu_X(t) = EX_t$  is the mean function of  $X_t$ .

For stationary process  $X_t$ :

- $\mu_X(t)$  is independent on  $t$  ( $\mu_X(t) = \mu$ ) and
- $\gamma(t+h, t)$  is independent on  $t$  for each  $h$  ( $\gamma_X(t+h, t) = \gamma_X(h) = \gamma_X(h+0, 0)$ ).

Let  $X_t$  be a stationary time series. The autocovariance function (**ACVF**) of  $X_t$  can be presented as:

$$\gamma_X(h) = \text{cov}(X_{t+h}, X_t) = \text{cov}(X_h, X_0) \quad (4.2)$$

where  $h = t_1 - t_2$  is the lag.

The **autocorrelation** function (**ACF**) of  $X_t$  is:

$$\rho_X(h) = \frac{\gamma_X(h)}{\gamma_X(0)} = \text{corr}(X_{t+h}, X_t) \quad (4.3)$$

Plot of  $\rho(h)$  against  $h$  is called correlogram.

#### 4.1.2 Partial Autocorrelation Function

The partial autocorrelation function (PACF) of an ARMA process  $\{X_t\}$  is the function  $\alpha(\cdot)$  defines as:

$$\alpha(0) = 1 \quad (4.4)$$

and

$$\alpha(h) = \phi_{hh}, \quad h \geq 1, \quad (4.5)$$

where  $\phi_{hh}$  is the last component of

$$\phi_h = \Gamma_h^{-1} \rho_h \quad (4.6)$$

and  $\Gamma_h = [\rho(i-j)]_{i,j=1}^h$ , and  $\rho_h = \rho(1), \rho(2), \dots, \rho(h)'$ , and  $\rho(h)$  is an autocorrelation function of  $X_t$ .

#### 4.1.3 Unit root tests

The formal method to test stationarity of a univariate time series is the unit root test. One of the most popular unit root test is **Dickey-Fuller (DF)** test.

##### Dickey-Fuller Test

Given  $AR(1)$  process:

$$y_t = u + \rho y_{t-1} + \varepsilon_t \quad (4.7)$$

where  $u$  and  $\rho$  are parameters and  $\varepsilon_t$  is assumed to be white noise,  $y$  is a **stationary** series if  $-1 < \rho < 1$ .

If  $\rho = 1$ ,  $y$  is a **nonstationary** series (a random walk with drift and intercept). It can be verified whether  $|\rho| < 1$

$H_0 : \rho = 1$  vs  $H_a : \rho < 1$

The test is carried out by estimating as equation with  $y_{t-1}$  subtracted from both sides

$$\Delta y_t = u + r y_{t-1} + \varepsilon_t \quad (4.8)$$

where  $r = \rho - 1$  and the null and alternative hypotheses are:

$$H_0 : r = 0 \text{ vs } H_a : r < 0$$

In DF test, we can include trend and intercept in the test regression:

$$y_t = u + \omega t + \rho y_{t-1} + \varepsilon_t \quad (4.9)$$

or

$$\Delta y_t = u + \omega t + r y_{t-1} + \varepsilon_t \quad (4.10)$$



### Augmented Dickey-Fuller Test

In the Augmented Dickey-Fuller (ADF) test lagged difference terms of the dependent variable are added to the right-hand side of the regression:

$$y_t = u + ry_{t-1} + b_1\Delta y_{t-1} + b_2\Delta y_{t-2} + \dots + b_p\Delta y_{t-p} + \varepsilon(t) \quad (4.11)$$

The null and alternative hypothesis are:

$$H_0 : r = 0 \text{ vs } H_a : r < 0$$

ADF test can include a constant, a constant and linear trend, or neither in the test regression

- None

$$y_t = ry_{t-1} + b_1\Delta y_{t-1} + b_2\Delta y_{t-2} + \dots + b_p\Delta y_{t-p} + \varepsilon(t) \quad (4.12)$$

- Intercept

$$y_t = u + ry_{t-1} + b_1\Delta y_{t-1} + b_2\Delta y_{t-2} + \dots + b_p\Delta y_{t-p} + \varepsilon(t) \quad (4.13)$$

- Trend and Intercept

$$y_t = u + \omega t + ry_{t-1} + b_1\Delta y_{t-1} + b_2\Delta y_{t-2} + \dots + b_p\Delta y_{t-p} + \varepsilon(t) \quad (4.14)$$

In this research, the ADF with trend and intercept is applied.

### Philips-Person Test

The **Philips-Person (PP)** evaluates the null hypothesis of a unit root in a univariate time series  $y$ . The model is

$$y_t = u + \omega t + \rho y_{t-1} + \varepsilon_t \quad (4.15)$$

where  $\varepsilon_t$  is the innovations process.

The null hypothesis restricts  $\rho = 1$ .

The test assesses the null hypothesis under the model variant appropriate for series with different growth characteristics ( $u = 0$  or  $\omega = 0$ ).

### Kwiatkowski-Phillips-Schmidt-Shin Test

The **Kwiatkowski-Phillips-Schmidt-Shin (KPSS)** test assesses the null hypothesis that a univariate time series  $y$  is trend **stationary** against the alternative that it is a nonstationary unit-root process.

**KPSS** test computes the vector of test statistics, with length equal to the number of tests, using an OLS regression of  $y$  on an intercept. If 'trend' is true, the regression includes a trend. The test statistic  $stat$  is:

$$stat = \sum_{i=1}^N (r(t))^2 / s_{NW}^2 N^2 \quad (4.16)$$

where  $r(t) = r_1 + \dots + r_t$ ,  $r$  is the vector of residuals from the regression,  $s_{NW}^2$  is the Newey-West estimator of the long-run variance, and  $N$  is the sample size.

## 4.2 Models

There are various approaches for predicting or estimating <sup>1</sup> values of a certain dependent variable. In a given model the response variable can be a result of many different factors. Certain models rely only on historical values of predicting variable, while others depend on exogenous determinants as well.

In this paper both approaches will be addressed.

### 4.2.1 Univariate (ARIMA)

The Autoregressive Integrated Moving Average (ARIMA) model is an example of a model built only upon historical values of the predicting variable. There is no assumption about the underlying economic model or structural relationships between variables [8], which makes model formulation much easier.

ARIMA model is constructed upon two models: Autoregressive (AR) model and Moving Average (MA) model.

#### Autoregressive (AR) model

In Autoregressive models the predicted (foretasted) value is a linear combination of past values of the same variable. The name ‘Autoregressive’ indicates regression of the variable against itself.

AR(p) models can be written as

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t \quad (4.17)$$

where

- $X_{t-1}, X_{t-2}, \dots, X_{t-p}$  are previous observations of variable  $X$  in time points  $t-1, t-2$  up till  $t-p$ ,
- $\varepsilon_t$  error term (or white noise) at time  $t$  is assumed to be (*i.i.d*) and  $\varepsilon_t \sim N(0, \sigma^2)$  where  $\sigma^2$  is the variance,
- $\phi_1, \phi_2, \dots, \phi_p$  are parameters to be estimated.

Popular form / presentation of Autoregressive models (as well as Moving Average and ARIMA) uses backshift (or backward shift) operator  $B$  defined as

$$B^p X_t = X_{t-p} \quad (4.18)$$

$$\nabla^d = (1 - B)^d \quad (4.19)$$

As can be seen, backshift operator  $B$  shifts the time by one unit.

$$\nabla X_t = X_t - X_{t-1} = (1 - B)X_t \quad (4.20)$$

$$\nabla^2 X_t = (1 - B)^2 X_t = X_t - 2X_{t-1} + X_{t-2} \quad (4.21)$$

$$(1 - B^2)X_t = X_t - X_{t-2} \quad (4.22)$$

Using backshift operator,  $p^{th}$  order Autoregressive model (AR(p)) can be written as:

$$\Phi(B)X_t = \varepsilon_t \quad (4.23)$$

where

$$\Phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p \quad (4.24)$$

is an autoregressive polynomial of order  $p$ .

---

<sup>1</sup>According to Ababio (2012) [8] prediction refers to predicting estimates in a distinct future while forecast refers to forecasting / projecting estimates in a near future.

### Moving Average (MA) model

The Moving Average (MA) model uses past forecast errors in a regression-like model.

MA(q) models can be written as

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_p \varepsilon_{t-q} \quad (4.25)$$

where

- $\varepsilon_t$  error term or white noise at time  $t$  is assumed to be (i.i.d) and  $\varepsilon_t \sim N(0, \sigma^2)$  where  $\sigma^2$  is the variance,
- $\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-q}$  are previous values of error terms in time points  $t-1, t-2$  up till  $t-q$ ,
- $\theta_1, \theta_2, \dots, \theta_q$  are parameters to be estimated.

Using backshift operator  $q^h$  order Moving Average (MA(q)) can be presented as:

$$X_t = \Theta(B) \varepsilon_t \quad (4.26)$$

where

$$\Theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q \quad (4.27)$$

is moving average polynomial of order  $q$ .

### Autoregressive Moving Average (ARMA) model

Autoregressive Moving Average (ARMA) model is formed by combining Autoregressive and Moving Average processes.

The general ARMA(p,q) process is defines as:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q} \quad (4.28)$$

with all the symbols defined in 4.2.1 and 4.2.1.

Using backshift operator

$$X_t = \frac{\Theta(B)}{\Phi(B)} \varepsilon_t \quad (4.29)$$

### Autoregressive Integrated Moving Average (ARIMA) model

An Autoregressive Integrated Moving Average model is a generalization of an Autoregressive Moving Average Model (4.2.1).

An ARMA(p,q) model assumption is that the investigated data is stationary (4.1). Unfortunately, the data occurring in nature (also economics) is usually nonstationary. Each observation of nonstationary time series is a result of a sum of all the past random errors. Since the present value incorporates all of the past random errors, the process is called **integrated**. A simple solution for an integrated time series is differencing (subtracting the value of the preceding period from the current period).

The first order differencing process of time series  $X_t$  is defines as:

$$\nabla X_t = (1 - B)X_t = X_t - X_{t-1} \quad (4.30)$$

The second order differencing process of time series  $X_t$  can be described:

$$\nabla^2 X_t = (1 - B)^2 X_t = X_t - 2X_{t-1} + X_{t-2} \quad (4.31)$$

Finally, the  $d^{th}$  order differencing process of time series  $X_t$ :

$$\nabla^d X_t = (1 - B)^d X_t \quad (4.32)$$

The order of integration indicates how many times data has to be differenced to restore stationarity (to be stacionary). When the stationarity in data is achieve, the ARMA(p,q) model can be employed.

Autoregressive Integrated Moving Average model with parameter (p,d,q) can be presented as:

$$\Phi(B)\nabla^d X_t = \Theta(B)\varepsilon_t \quad (4.33)$$

where  $\Phi(B)$  and  $\Theta(B)$  are autoregressive and moving average polynomials of order  $p$  and  $q$  respectively and  $\varepsilon_t$  en error term (white noise process).

#### 4.2.2 Multivariate (ARIMAX, Transfer Function)

Multivariate time series analysis includes explanatory variables other than just previous values of the response variable in the analysis process. Since natural gas price depends not only on its previous values but also on a multitude of external factors (like weather, storage and shut in production) and also other energy products (crude oil, coal) (see Chapter 3), multivariate models are widely employed (in this particular sector).

##### ARMAX models

An ARMAX models is an ARMA model with additional variable on the right hand side. It can be presented in the following form:

$$Y_t = \beta X_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q} \quad (4.34)$$

where

- $X_t$  is a covariate at the time  $t$  with its coefficient  $\beta$ ,
- $Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}$  are previous observations of variable  $Y$  in time points  $t-1, t-2$  up till  $t-p$ ,
- $\varepsilon_t$  error term or white noise at time  $t$  is assumed to be (i.i.d) and  $\varepsilon_t \sim N(0, \sigma^2)$  where  $\sigma^2$  is the variance,
- $\beta, \phi_1, \phi_2, \dots, \phi_p, \theta_1, \theta_2, \dots, \theta_q$  are parameters to be estimated,
- $p$  and  $q$  are integer orders of the model.

The difference between the ordinary least squares (OLS) regression model and ARMAX model is that the covariant coefficient is hard to interpret. The meaning of  $\beta$  represents not only the impact of  $X_t$  on  $Y_t$ . The presence of lagged values of the response variable on the right hand side of the equation indicates that coefficient  $\beta$  can only be interpreted conditional on the value of previous values of the response variable ( $Y_{t-1}, Y_{t-2}, \dots, Y_{t-p}$ ).

Using backshift operator  $B$ , the model can be written as:

$$\Phi(B)Y_t = \beta X_t + \Theta(B)\varepsilon_t \quad (4.35)$$

or

$$Y_t = \frac{\beta}{\Phi(B)} X_t + \frac{\Theta(B)}{\Phi(B)} \varepsilon_t \quad (4.36)$$

where  $\Phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$  and  $\Theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q$ .

##### ARIMAX models

When the data used in ARMAX model is not stationary, the same procedure as for ARMA and ARIMA (4.2.1) needs to be conducted. The data (all variables,  $Y_t$  and  $X_t$ ,  $X_t$  can be a matrix) needs to be differenced. In this case, the model takes the following form:

$$Y_t = \frac{\beta}{\Phi(B)\nabla^d} X_t + \frac{\Theta(B)}{\Phi(B)\nabla^d} \varepsilon_t \quad (4.37)$$

where  $\nabla^d = (1 - B)^d$  denotes the differencing operator.

## Transfer Function Models

Transfer Function model is a generalization of an ARMAX model (4.2.2). There are many various ways of presenting Transfer Function Models. One of them is:

$$Y_t = \frac{\beta(B)}{\Phi(B)} X_t + \frac{\Theta(B)}{\Phi(B)} \varepsilon_t \quad (4.38)$$

where

- $\beta(B) = 1 - \beta_1 B - \beta_2 B^2 - \dots - \beta_s B^s$ ,
- $\Phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$ ,
- and  $\Theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q$ .

An even more general form of Transfer Function model:

$$Y_t = c + \frac{\omega(B)B^b}{\gamma(B)} X_t + \frac{\Theta(B)}{\Phi(B)} \varepsilon_t \quad (4.39)$$

can be often found in the literature ([10]).

In this thesis, the Transfer Function model indicates an ARIMA model with covariants allowing for lag covariants as well. For example:

$$Y_t = \beta_0 X_t + \beta_1 X_{t-1} + \dots + \beta_s X_{t-s} + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_p \varepsilon_{t-p} \quad (4.40)$$

Transfer Function models are also referred as ‘Dynamic regression models’, although different sources use this term for various models.

### 4.2.3 Error Correction Model (ECM)

A very popular approach in modeling natural gas price is based not only on exogenous (and endogenous) variables and its values but also on the relationship between response and explanatory variables. (Vector) Error Correction Model is an example of such methodology. In order to apply a (Vector) Error Correction model, two variables need to be cointegrated.

#### Cointegration

Cointegration of variables occurs when variables are linked over a long period of time (for example, natural gas price and crude oil price or coal price). There are various statistical methods available for testing cointegration. The most popular are *Engle-Granger procedure* and *Johansen procedure* (applied in this research).

The simplest cointegration example is bivariate case  $([x_t, y_t])$ . Two variables are cointegrated if they are stationary at the same level (for example I(1) or I(2)) and error term of ordinary least square regression (OLS) is stationary at level I(0). This means it is possible to form a stationary linear combination form of two I(1) variables. Given two variables  $x_t$  and  $y_t$  that are I(1) (that means integrated of level 1, thus stationary after one differencing process), linear combination can be presented as:

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t \quad (4.41)$$

When  $\varepsilon_t$

$$\varepsilon_t = y_t - (\beta_0 + \beta_1 x_t) \quad (4.42)$$

is I(0) then variables  $y_t$  and  $x_t$  are *cointegrated*.

Cointegration in some sense resembles integration, in a way that the process of detrending the nonstationary series involves focus on the offsetting of the shared attributes of stochastic trends ([2]). Cointegration solves the problem of spurious regressions without the need of differencing data and losing the information about the level of the time series.

### Error Correction Model

Having established cointegration between two variables (for example natural gas price and crude oil price), the Error Correction model can be applied.

In a simple case, assuming the model is

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + u_t \quad (4.43)$$

where both  $y_t$  and  $x_t$  are  $I(1)$ , subtracting  $y_{t-1}$  from both sides of the equation 4.43

$$y_t - y_{t-1} = \alpha_0 + \alpha_1 y_{t-1} - y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + u_t \quad (4.44)$$

and adding  $-\beta_0 x_{t-1} + \beta_0 x_{t-1}$  to right side of the equation we get:

$$\Delta y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_0 \Delta x_t + \theta_1 x_{t-1} + \varepsilon_t \quad (4.45)$$

After finding the evidence of cointegration, the corresponding Error Correction Model can be specified. Knowing that:

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_0 x_t + u_t \quad (4.46)$$

$$u_{t-1} = y_{t-1} - \alpha_0 - \alpha_1 y_{t-1} - \beta_0 x_{t-1} \quad (4.47)$$

Combining 4.45 and 4.47, the ECM model can be written as:

$$\Delta y_t = \alpha_0 + \alpha_1 y_{t-1} + \beta_0 \Delta x_t + \rho u_{t-1} + \varepsilon_t \quad (4.48)$$

Additional lags of  $\Delta y_{t-1}$  and  $\Delta x_{t-1}$  can be added if needed.

The final version of Error Correction Model (ECM) is:

$$\Delta y_t = \sum_{i=1}^p \alpha_i y_{t-i} + \sum_{i=1}^q \beta_i \Delta x_{t-i} + \delta ec m_{t-1} + \varepsilon_t \quad (4.49)$$

where  $\delta ec m_{t-1}$  describes long run relationship between variables  $y$  and  $x$ .

### Vector Error Correction Model

In bivariate case, having vector  $y_t = [y_{1,t}, y_{2,t}]'$  (that is  $I(1)$ ) with cointegrating vector  $\beta = [1, -\beta_2]'$  and  $\beta' y_t = y_{1,t} - \beta_2 y_{2,t} \sim I(0)$ , the VECM model can be presented as:

$$\Delta y_{1,t} = \alpha_1 + \gamma_1 (y_{1,t-1} - \beta_2 y_{2,t-1}) + \sum_{i=1}^K \psi_{1,i} \Delta y_{1,t-i} + \sum_{i=1}^L \psi_{2,i} \Delta y_{2,t-i} + \varepsilon_{1,t} \quad (4.50)$$

$$\Delta y_{2,t} = \alpha_2 + \gamma_2 (y_{1,t-1} - \beta_2 y_{2,t-1}) + \sum_{i=1}^K \xi_{1,i} \Delta y_{1,t-i} + \sum_{i=1}^L \xi_{2,i} \Delta y_{2,t-i} + \varepsilon_{2,t} \quad (4.51)$$

## 4.3 Model selection criteria

Choosing an adequate model for given data is one of the most difficult tasks in time series analysis [9]. There are many model selection criteria and, as seen in Chapter 2, different researchers choose different criteria to rely on.

### 4.3.1 AIC

Akaike's Information Criterion (AIC) is one of the most commonly used model selection criteria. This criteria selects the model that minimizes the negative likelihood, including a penalty for overparameterization. AIC can be obtained by using the following formula:

$$AIC = -2\log p(L) + 2p \quad (4.52)$$

where  $L$  is the likelihood in the fitted model and  $p$  refers to the number of parameters in the model.

Akaike's Information Criterion (AIC) provides a measure of the model quality by simulating the situation where the model is tested on a different data set. After computing several different models, one can compare them using this criterion. According to Akaike's theory, the most accurate model has the smallest AIC.

### 4.3.2 BIC

Bayesian (Schwarz) Information Criterion is also commonly used model selection criterion. AIC and BIC feature the same goodness-of-fit term. The penalty term of BIC is more stringent than the penalty term of AIC. Consequently, BIC tends to favor smaller models than does AIC.

$$BIC = -2\log p(L) + p\ln(n) \quad (4.53)$$

where  $n$  is the number of observations in the data set (given variable),  $L$  is the likelihood in the fitted model and  $p$  refers to the number of parameters in the model.

### 4.3.3 RMSEin

A popular approach in model selection process is to use error metrics (like RMSE, MAE, MAPE, WMAPE) for in-sample predictions, which means comparing fitted model with the actual data. Error metrics are described in the next section (4.4).

## 4.4 Error metrics

There are many different accuracy measures to evaluate the performance of forecasting methods. Since different models have different 'forecasting power', error metrics demonstrate how risky the forecast model is. The majority of these measurements are based on differences between actual and predicted values. Naturally, a smaller difference indicates a better model. The most popular risk measures are:

- Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4.54)$$

- Mean Square Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4.55)$$

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4.56)$$

- Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \quad (4.57)$$

## 4.5 Residual Testing

After a model is fitted, it should be validated. As with standard non-linear least squares fitting, the primary tool for model diagnostic checking is residual analysis.

Error terms  $\varepsilon_t$  in the ARIMA(p,d,q) model are assumed to be (*i.i.d.*) with  $\varepsilon_t \sim N(0, \sigma^2)$ , where  $\sigma^2$  is the variance. These assumptions are crucial for further tests to check for nonlinearity not captured by the model. Residuals (error terms) not being (*i.i.d.*) can indicate a nonlinear trend left in the data, which was not found by the ARIMA model (so ARIMA did not model properly).

Residual analysis can be done in several ways, including graphical and analytical analysis. One of the most popular tests for residuals testing in Ljung-Box test.

### 4.5.1 Ljung-Box test

The Ljung-Box test is a diagnostic tool applied to time series for testing lack of fit of the model. The test examines residuals (error terms) after the data is fitted, checking its (residuals) autocorrelations. Residuals of a well fitted model should demonstrate very small autocorrelations.

The Ljung-Box test is defined as follow:

- $H_0$  : The model does not exhibit lack of fit.
- $H_a$  : The model exhibits lack of fit.

The test statistics can be written as:

$$Q = n(n+2) \sum_{k=1}^m \frac{\hat{r}_k^2}{n-k} \quad (4.58)$$

where  $n$  is number of observations (length of time series),  $\hat{r}_k$  is the estimated autocorrelation of the series at *lag* $k$ , and  $m$  is the number of lags being tested.

The null hypothesis ( $H_0$  : ) is rejected (meaning that the model has significant lack of fit) if

$$Q > \chi_{1-\alpha, h}^2 \quad (4.59)$$

where  $\chi_{1-\alpha, h}^2$  is the value of chi-square distribution with  $h$  degrees of freedom and significance level  $\alpha$

Since the test is conducted on residuals, the degrees of freedom should take into consideration parameters ( $p, q$ ) of estimated model ( $ARMA(p, q)$ ). In this case  $h = m - p - q$ <sup>2</sup>.

Graphical analysis of residuals includes autocorrelation function (ACF) plot (ACF of residuals), lag plot, histogram and normal probability plot. An example of *4 plot* is presented on Figure 4.1.

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<sup>2</sup>In some sources  $h = p + q$



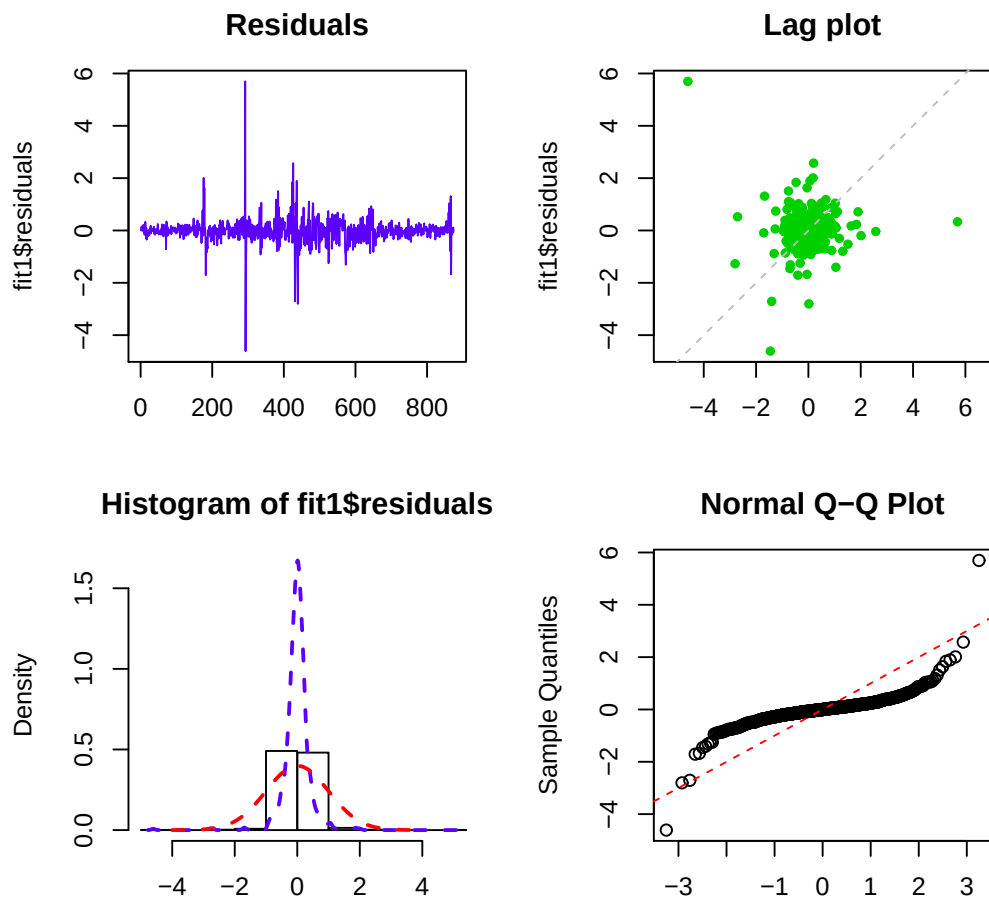


Figure 4.1: 4 plot

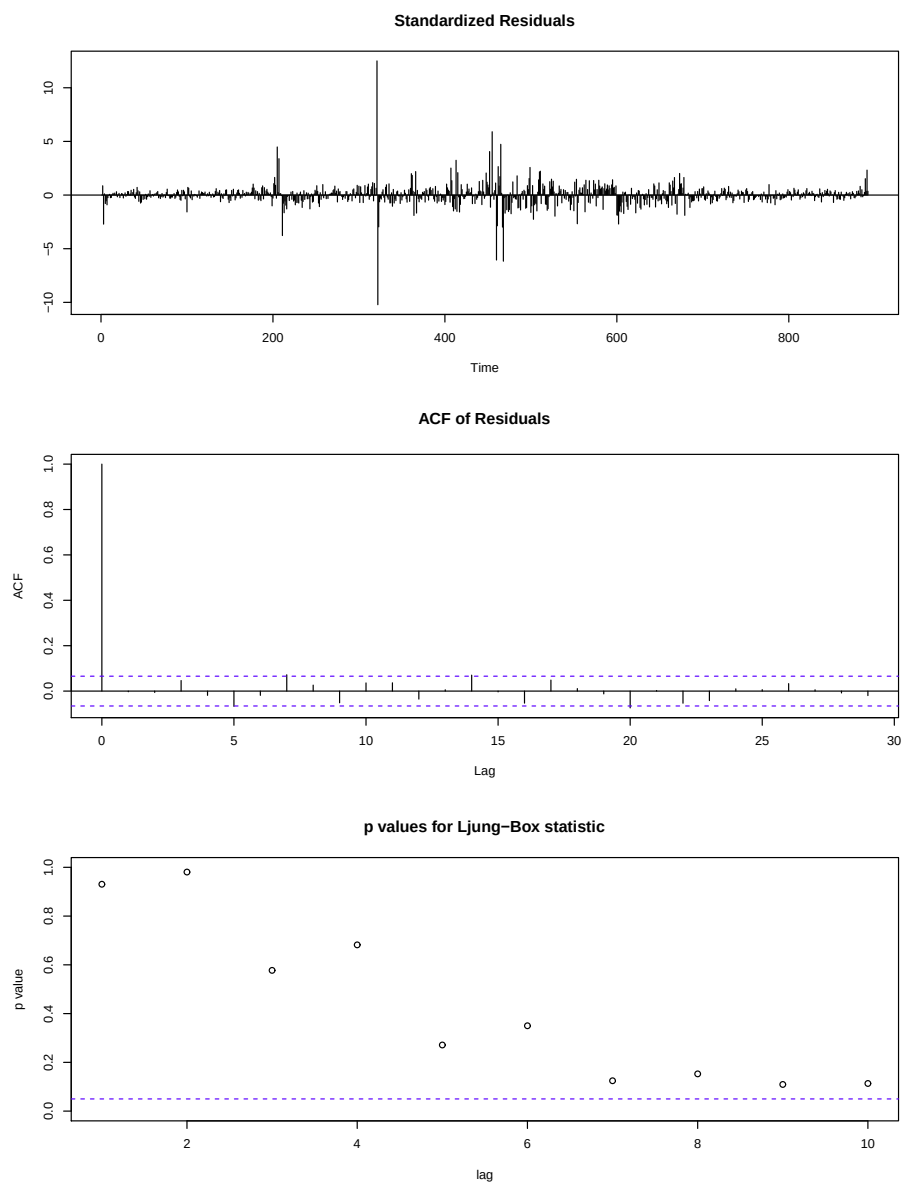


Figure 4.2: Residuals plot

## 4.6 Analysis Process

### 4.6.1 ARIMA, ARIMAX and Transfer Function models

The forecasting approach applied in this research (for ARIMA and also ARIMAX/Transfer Function) models is based on Box-Jenkins methodology. This procedure comprises following stages:

- Model identification,
- Parameter estimation,
- Diagnostic checking and
- Prediction.

The more precise description of the employed methodology can be described as following:

1. Checking the data stationarity.
2. Making various ARIMA (ARIMAX/Transfer Function) models and comparing model selection criteria (AIC and BIC).
3. Testing for residuals.
4. In case model does not satisfy the residuals normality requirement, then choosing another model.
5. Using selected model for forecast (prediction).
6. Finding error metrics (for example RMSE).
7. Graphical representation of the forecast compared with actual values.

### 4.6.2 Error Correction Model

In error correction model only two variables are used (Henry Hub Natural Gas Spot Price and West Texas Intermediate Crude Oil Spot Price). According to information in Section 4.2.3, ECM model can be applied if two variables have long run equilibrium (cointegration). In this case, the methodology contains following steps:

1. Checking the stationarity of natural gas and crude oil price.
2. Checking the cointegration:
  - Running regression

$$HH_t = \alpha + \beta WTI_t + \varepsilon_t. \quad (4.60)$$

- Storing residuals as  $ecm$ :

$$ecm = HH_t - (\alpha + \beta WTI_t + \varepsilon_t), \quad (4.61)$$

- where HH - Henry Hub Natural Gas Spot Price and WTI - West Texas Intermediate Crude Oil Spot Price.
- Checking the stationarity of  $ecm$  (residuals).

3. Constructing the ECM model:

$$\Delta HH_t = \sum_{i=1}^p \alpha_i \Delta HH_{t-i} + \sum_{i=0}^q \beta_i \Delta WTI_{t-i} + \delta ecm_{t-1} + \varepsilon_t, \quad (4.62)$$

where

- $\Delta HH$  is the first difference of the gas price,
- $\Delta WTI$  is the first difference of the crude oil price,

- $ecm_{t-1}$  is the error term of the OLS from step 2,
  - $p$  and  $q$  are parameters that can be estimated using following criteria:
    - AIC and BIC Criteria,
    - F-test,
    - T-test.
  - In this research AIC and BIC Criteria were applied.
4. Testing residuals of Error Correction Model (see Section 4.5) .
  5. Forecasting.

# Chapter 5

## Results

In this chapter weekly data (Henry Hub Natural Gas Spot Price) in Dollars per million Btu (\$\text{mmBtu})\$ from the period of *August 1, 1997* till *May 30, 2014* are analyzed. The number of observations (in one time series) equals to 878.

### 5.1 ARIMA

#### 5.1.1 Stationarity

##### ACF and PACF

As mentioned in Section 4.1, the first step in establishing stationarity of the data is to use graphical data analysis. Looking at data plot (Figure 5.1) - Henry Hub Natural Gas Spot Price on weekly base, it can be noticed that there is strong trend and high volatility, suggesting that, the considered price time series is not stationary. The next step is to plot autocorrelation (ACF) and partial autocorrelation (PACF) functions.

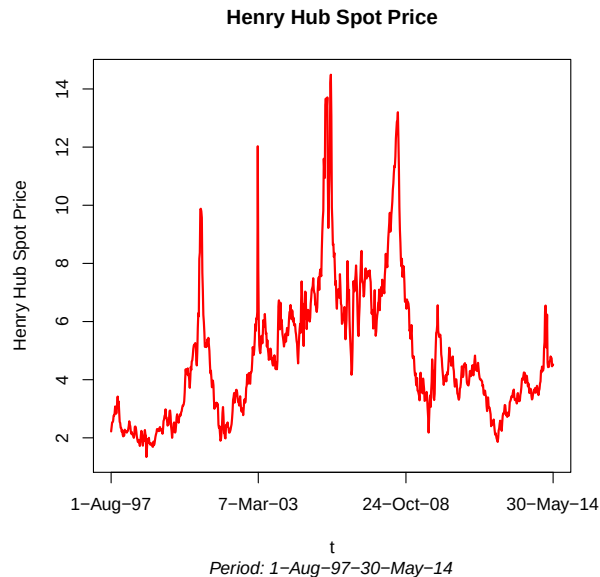


Figure 5.1: Henry Hub Natural Gas Spot Price - weekly data

The autocorrelation function (Figure 5.2) shows that several autocorrelations are persistently large and trend to zero rather slowly. This might indicate existing trend and non-stationarity.

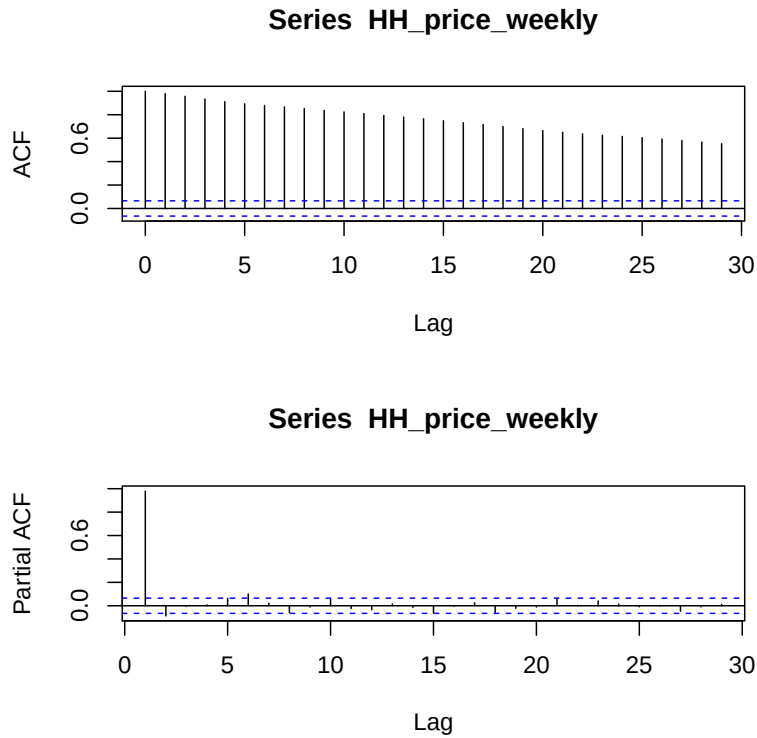


Figure 5.2: Autocorrelation and partial autocorrelation function for Henry Hub weekly spot prices

### Unit root tests

#### Unit root tests

The ADF and PP tests (Section 4.1.3) both check the stationarity in the time series. They both test if there is unit root in the time series.

$H_0$  : there is **unit root** (the time series is **nonstationary**)

$H_a$  : there is **no unit root** (the time series is **stationary**)

Significance level for all tests  $\alpha = 0.05$ .

Results presented in Table 5.1 show that  $p$  – values are greater than significance level  $\alpha$  for ADF and PP tests. This means that **Henry Hub Natural Gas weekly Prices** are **not stationary** time series and in order to conduct further analysis, we need to differentiate this data.

Results of KPSS test only confirm our presumptions. Null hypothesis in KPSS test is that the time series is stationary against the alternative hypothesis that the time series is not stationary (has unit root). The data is statistically significant at level  $\alpha$ . There is significant evidence against the null hypothesis.

Plots of standard deviations for level (original data), first and second differences also indicate that the data is I(1) (see Section 4.1). We can see huge decline in the standard deviation of level and first difference (Figure 5.3).

Table 5.1: Unit root tests for Henry Hub Natural Gas Spot Price - weekly data

| Henry Hub Natural Gas Spot Price |            |         |            |         |            |         |
|----------------------------------|------------|---------|------------|---------|------------|---------|
| Number of lag                    | ADF        |         | PP         |         | KPSS       |         |
|                                  | statistics | p-value | statistics | p-value | statistics | p-value |
| 0                                | -3.0670    | 0.1154  | -3.0670    | 0.1154  | 12.1722    | 0.01    |
| 1                                | -3.3109    | 0.0610  | -3.1979    | 0.0854  | 6.1510     | 0.01    |
| 2                                | -3.3109    | 0.0651  | -3.2423    | 0.0769  | 4.1380     | 0.01    |
| 6                                | -2.6572    | 0.2673  | -3.0797    | 0.1123  | 1.8332     | 0.01    |
| 9                                | -2.6452    | 0.2733  | -3.0051    | 0.1316  | 1.3106     | 0.01    |
| 20                               | -2.8326    | 0.1860  | -3.0740    | 0.1137  | 0.6685     | 0.01    |
| 30                               | -2.6026    | 0.2944  | -3.0722    | 0.1141  | 0.4824     | 0.01    |

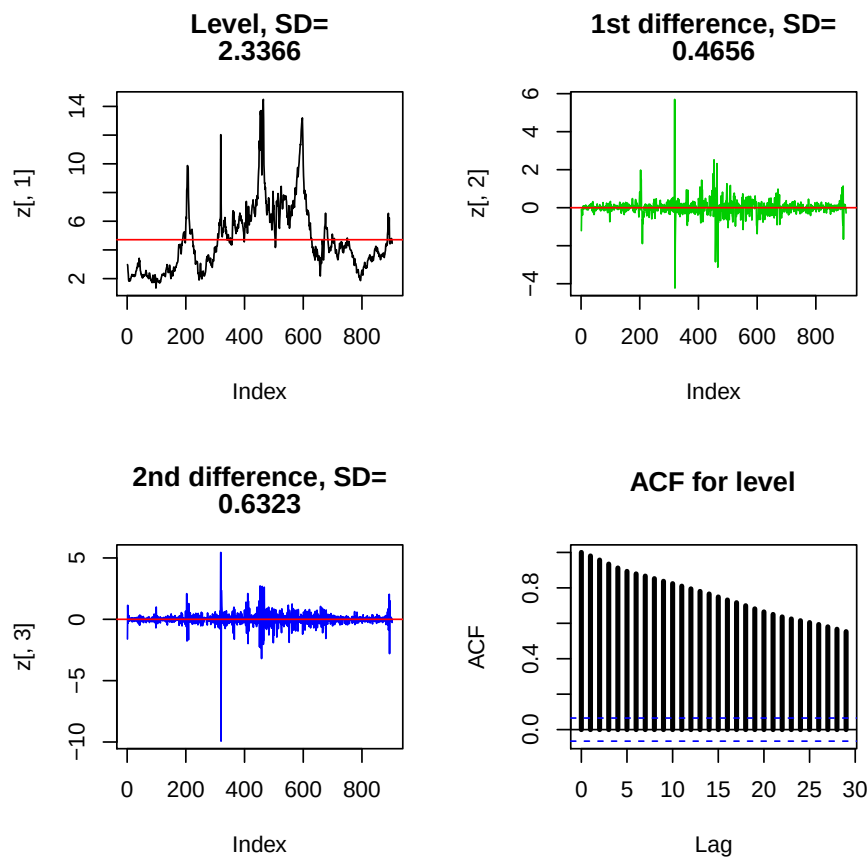


Figure 5.3: Standard deviations for the data, first and second differences, autocorrelation function

### Henry Hub Spot Price I(1)

A plot of the differenced data appears to vary about a fixed level. Also, unit roots tests show that there is significant evidence against the null hypothesis (non stationary data) in the case of the ADF and PP test. In the case of KPSS test the data is not statistically significant (so we can not reject the null hypothesis - stationarity of the time series).

The Henry Hub Natural Gas Spot Price series appears to be an I(1) process. This indicates that for further analysis Autoregressive **Integrated** Moving Average (**ARIMA**) prices will be used. The study will follow Box-Jenkins methodology, which involves the following steps: *identification, estimation, model checking and forecasting*.

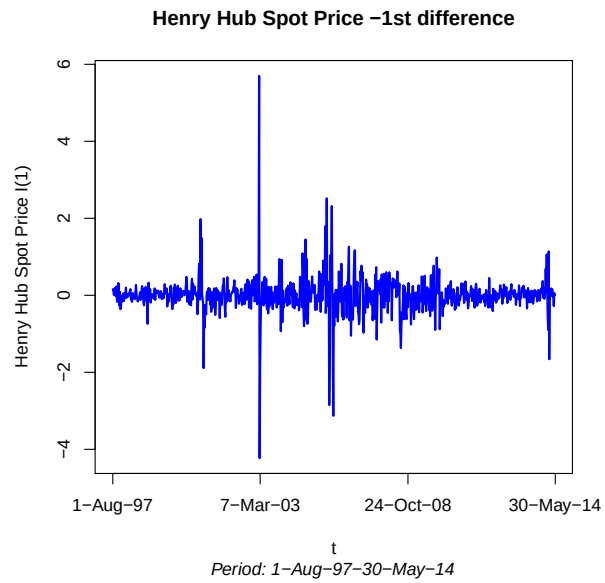


Figure 5.4: First difference (I(1)) of Henry Hub Natural Gas Spot Price - weekly data

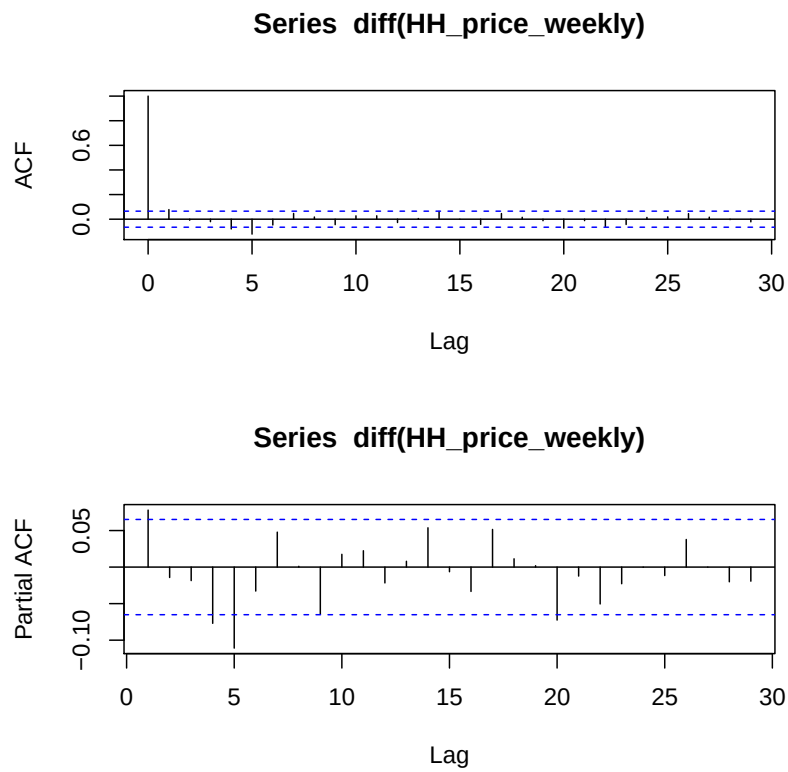


Figure 5.5: Autocorrelation and partial autocorrelation function for first difference (I(1)) Henry Hub weekly spot prices



Table 5.2: Unit root tests for first difference of Henry Hub Natural Gas Spot Price - weekly data

| First difference of Henry Hub Natural Gas Spot Price |            |         |            |         |            |         |
|------------------------------------------------------|------------|---------|------------|---------|------------|---------|
| Number of<br>lag                                     | ADF        |         | PP         |         | KPSS       |         |
|                                                      | statistics | p-value | statistics | p-value | statistics | p-value |
| 0                                                    | -27.7508   | < 0.001 | -27.7508   | < 0.001 | 0.0235     | 0.1     |
| 1                                                    | -20.6709   | < 0.001 | -27.7549   | < 0.001 | 0.0218     | 0.1     |
| 2                                                    | -17.3110   | < 0.001 | -27.7469   | < 0.001 | 0.0214     | 0.1     |
| 6                                                    | -12.4869   | < 0.001 | -27.6644   | < 0.001 | 0.0248     | 0.1     |
| 9                                                    | -10.4644   | < 0.001 | -27.7026   | < 0.001 | 0.0270     | 0.1     |
| 20                                                   | -6.8296    | < 0.001 | -27.7668   | < 0.001 | 0.0286     | 0.1     |
| 30                                                   | -6.0210    | < 0.001 | -27.9195   | < 0.001 | 0.0316     | 0.1     |

### 5.1.2 Model Identification

In order to identify the model (find the parameter  $p$  of AR and  $q$  of MA terms - the order of the model), ACF and PACF can be used. For example, if when comparing the autocorrelations with their error limits, the only significant autocorrelation is at lag 0, then MA(0) would be suggested. A similar procedure can be applied to partial autocorrelation function. If the highest PACF coefficient value is at lag 5, AR(5) would be proposed and, as the final result **ARIMA(5,1,0)** model would be fitted.

Following tables present chosen ARIMA models for each time horizon (identification and prediction) with its model selection criteria and error measures. Models chosen by R program are written in *italic* font, while models with the smallest RMSE are in **bold**.

Table 5.3: ARIMA models for Henry Hub Natural Gas Spot Price - weekly data, predicting 4 values

| Series                                                          | Model               | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE  |
|-----------------------------------------------------------------|---------------------|----------|----------|--------|--------|-------|-------|
| HH_w; Est. period 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                               | ARIMA(5,1,3)        | 1150.349 | 1198.080 | 0.462  | 5.379  | 0.258 | 5.088 |
| 2                                                               | ARIMA(5,1,2)        | 1154.888 | 1197.846 | 0.464  | 5.356  | 0.192 | 3.816 |
| 3                                                               | <i>ARIMA(2,1,2)</i> | 1156.817 | 1185.456 | 0.466  | 5.329  | 0.216 | 4.242 |
| 4                                                               | <b>ARIMA(4,1,2)</b> | 1157.009 | 1195.194 | 0.465  | 5.346  | 0.180 | 3.488 |
| HH_w; Est. period 28-Feb-03 - 2-May-14 (583 obs); Pred.4 values |                     |          |          |        |        |       |       |
| 1                                                               | ARIMA(5,1,2)        | 763.820  | 803.134  | 0.459  | 5.294  | 0.138 | 2.894 |
| 2                                                               | ARIMA(1,1,5)        | 766.924  | 801.870  | 0.461  | 5.241  | 0.134 | 2.672 |
| 3                                                               | <b>ARIMA(4,1,2)</b> | 770.240  | 805.186  | 0.462  | 5.217  | 0.083 | 1.697 |
| 4                                                               | ARIMA(4,1,5)        | 770.417  | 818.467  | 0.460  | 5.233  | 0.100 | 2.171 |
| 5                                                               | <i>ARIMA(2,1,1)</i> | 781.896  | 803.737  | 0.469  | 5.161  | 0.223 | 4.336 |
| HH_w; Est. period 9-Dec-05 - 2-May-14 (439 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                               | ARIMA(5,1,2)        | 404.315  | 441.076  | 0.373  | 4.899  | 0.115 | 2.457 |
| 2                                                               | ARIMA(5,1,4)        | 406.786  | 451.716  | 0.374  | 5.049  | 0.089 | 1.811 |
| 3                                                               | ARIMA(2,1,1)        | 410.970  | 431.392  | 0.382  | 4.997  | 0.172 | 3.270 |
| 4                                                               | <i>ARIMA(3,1,2)</i> | 414.582  | 443.173  | 0.382  | 5.007  | 0.169 | 3.204 |
| 5                                                               | <b>ARIMA(4,1,3)</b> | 415.685  | 452.446  | 0.380  | 5.004  | 0.088 | 1.566 |
| HH_w; Est. period 4-Jul-08 - 2-May-14 (305 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                               | ARIMA(3,1,2)        | 137.349  | 163.391  | 0.296  | 4.828  | 0.189 | 3.881 |
| 2                                                               | <b>ARIMA(2,1,2)</b> | 137.628  | 159.949  | 0.297  | 4.855  | 0.146 | 2.886 |
| 3                                                               | ARIMA(4,1,2)        | 138.904  | 168.666  | 0.295  | 4.844  | 0.163 | 3.302 |
| 4                                                               | ARIMA(2,1,4)        | 139.506  | 169.268  | 0.296  | 4.829  | 0.195 | 3.988 |
| 5                                                               | <i>ARIMA(0,1,1)</i> | 142.591  | 153.752  | 0.302  | 4.787  | 0.209 | 4.025 |
| HH_w; Est. period 4-Sep-09 - 2-May-14 (244 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                               | ARIMA(4,1,5)        | 67.175   | 105.644  | 0.261  | 4.518  | 0.202 | 4.156 |
| 2                                                               | ARIMA(1,1,3)        | 69.449   | 90.432   | 0.271  | 4.572  | 0.162 | 3.015 |
| 3                                                               | <i>ARIMA(2,1,2)</i> | 70.266   | 91.249   | 0.270  | 4.555  | 0.276 | 5.391 |
| 4                                                               | <b>ARIMA(5,1,3)</b> | 71.382   | 106.354  | 0.266  | 4.563  | 0.089 | 1.728 |

Table 5.4: ARIMA models for Henry Hub Natural Gas Spot Price - weekly data, predicting 12 values

| Series                                                           | Model               | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE   |
|------------------------------------------------------------------|---------------------|----------|----------|--------|--------|-------|--------|
| HH_w; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                | ARIMA(5,1,3)        | 1134.335 | 1181.974 | 0.46   | 5.356  | 1.732 | 37.75  |
| 2                                                                | ARIMA(4,1,4)        | 1134.56  | 1182.198 | 0.461  | 5.351  | 1.761 | 38.398 |
| 3                                                                | ARIMA(4,1,3)        | 1136.872 | 1179.747 | 0.462  | 5.351  | 1.683 | 36.668 |
| 4                                                                | <b>ARIMA(2,1,2)</b> | 1142.878 | 1171.462 | 0.465  | 5.331  | 1.516 | 32.883 |
| HH_w; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.12 values |                     |          |          |        |        |       |        |
| 1                                                                | ARIMA(5,1,2)        | 743.308  | 782.498  | 0.455  | 5.25   | 1.956 | 42.535 |
| 2                                                                | <b>ARIMA(2,1,3)</b> | 757.152  | 787.633  | 0.46   | 5.126  | 1.487 | 31.506 |
| 3                                                                | ARIMA(1,1,5)        | 747.967  | 782.802  | 0.457  | 5.187  | 1.807 | 39.262 |
| 4                                                                | ARIMA(4,1,5)        | 749.704  | 797.602  | 0.456  | 5.163  | 1.939 | 42.143 |
| 5                                                                | <i>ARIMA(2,1,1)</i> | 759.623  | 781.395  | 0.464  | 5.125  | 1.919 | 41.782 |
| HH_w; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                | ARIMA(5,1,2)        | 375.977  | 412.572  | 0.364  | 4.83   | 2     | 43.593 |
| 2                                                                | ARIMA(4,1,5)        | 381.345  | 426.072  | 0.367  | 4.902  | 2.007 | 43.738 |
| 3                                                                | ARIMA(2,1,1)        | 385.635  | 405.966  | 0.374  | 4.938  | 1.988 | 43.368 |
| 4                                                                | <i>ARIMA(3,1,2)</i> | 388.812  | 417.275  | 0.374  | 4.939  | 1.969 | 42.942 |
| 5                                                                | <b>ARIMA(5,1,3)</b> | 393.06   | 433.721  | 0.372  | 4.936  | 1.576 | 33.744 |
| HH_w; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                | ARIMA(2,1,3)        | 94.564   | 120.421  | 0.276  | 4.699  | 2.532 | 54.871 |
| 2                                                                | <b>ARIMA(3,1,5)</b> | 95.236   | 132.173  | 0.272  | 4.645  | 1.855 | 40.325 |
| 3                                                                | ARIMA(2,1,2)        | 102.384  | 124.546  | 0.281  | 4.726  | 2.567 | 55.611 |
| 4                                                                | <i>ARIMA(0,1,4)</i> | 104.296  | 126.459  | 0.282  | 4.756  | 1.922 | 41.915 |
| HH_w; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                | ARIMA(3,1,1)        | 17.225   | 38.008   | 0.244  | 4.466  | 2.03  | 44.239 |
| 2                                                                | ARIMA(5,1,3)        | 17.367   | 52.006   | 0.238  | 4.315  | 2.26  | 49.277 |
| 3                                                                | ARIMA(1,1,3)        | 18.04    | 38.823   | 0.244  | 4.456  | 1.949 | 42.484 |
| 4                                                                | ARIMA(2,1,3)        | 19.438   | 43.685   | 0.244  | 4.462  | 1.963 | 42.792 |
| 5                                                                | <b>ARIMA(2,1,4)</b> | 22.053   | 49.764   | 0.244  | 4.455  | 1.923 | 41.928 |
| 6                                                                | <i>ARIMA(3,1,2)</i> | 19.224   | 43.471   | 0.243  | 4.467  | 2.037 | 44.412 |

### 5.1.3 Model Estimation

In the following tables coefficients for the best models (see previous tables) are estimated. The column  $1:n$  in each table indicates the intercept of the model. Models without intercept were also evaluated, but did not bring satisfactory results.

Table 5.5: ARIMA (4,1,2) model for Henry Hub Natural Gas Spot Price - weekly data (est 1 – 874, pred 4)

| Coefficients:                    | AR(1)  | AR(2)   | AR(3)  | AR(4)                  | MA(1)   | MA(2)  | 1:n    |
|----------------------------------|--------|---------|--------|------------------------|---------|--------|--------|
|                                  | 0.8228 | -0.4576 | 0.0433 | -0.1042                | -0.7519 | 0.3857 | 0.0085 |
| s.e.                             | 0.2186 | 0.2516  | 0.0549 | 0.0412                 | 0.2203  | 0.2521 | 0.0144 |
| $\sigma^2$ estimated as 0.2163 : |        |         |        | loglikelihood = -570.5 |         |        |        |

Table 5.6: ARIMA (4,1,2) model for Henry Hub Natural Gas Spot Price - weekly data (est 292 – 874, pred 4)

| Coefficients:                    | AR(1)  | AR(2)   | AR(3)  | AR(4)                   | MA(1)   | MA(2)  | 1:n     |
|----------------------------------|--------|---------|--------|-------------------------|---------|--------|---------|
|                                  | 0.8701 | -0.8500 | 0.2388 | -0.1637                 | -0.6098 | 0.6775 | -0.0111 |
| s.e.                             | 0.1252 | 0.1305  | 0.0652 | 0.0476                  | 0.1228  | 0.1111 | 0.0226  |
| $\sigma^2$ estimated as 0.2139 : |        |         |        | loglikelihood = -377.12 |         |        |         |

Table 5.7: ARIMA (5,1,4) model for Henry Hub Natural Gas Spot Price - weekly data (est 436 – 874, pred 4)

| Coefficients:                    | AR(1)   | AR(2)  | AR(3)   | AR(4)   | AR(5)                   | MA(1)  | MA(2)   | MA(3)  | MA(4)  | 1:n     |
|----------------------------------|---------|--------|---------|---------|-------------------------|--------|---------|--------|--------|---------|
|                                  | -0.2941 | 0.8028 | -0.4718 | -0.8015 | 0.1491                  | 0.5357 | -0.6218 | 0.3493 | 0.8859 | -0.0163 |
| s.e.                             | 0.0591  | 0.0781 | 0.0832  | 0.0558  | 0.0571                  | 0.0463 | 0.0847  | 0.1061 | 0.0743 | 0.0238  |
| $\sigma^2$ estimated as 0.1401 : |         |        |         |         | loglikelihood = -192.39 |        |         |        |        |         |

Table 5.8: ARIMA (2,1,2) model for Henry Hub Natural Gas Spot Price - weekly data (est 570 – 874, pred 4)

| Coefficients:                     | AR(1)   | AR(2)  | MA(1)                  | MA(2)   | 1:n     |
|-----------------------------------|---------|--------|------------------------|---------|---------|
|                                   | -0.5485 | 0.4214 | 0.7239                 | -0.2074 | -0.0242 |
| s.e.                              | 0.2752  | 0.2716 | 0.2912                 | 0.2836  | 0.0229  |
| $\sigma^2$ estimated as 0.08843 : |         |        | loglikelihood = -62.81 |         |         |

Table 5.9: ARIMA (5,1,3) model for Henry Hub Natural Gas Spot Price - weekly data (est 631 – 874, pred 4)

| Coefficients:                     | AR(1)  | AR(2)  | AR(3)   | AR(4)  | AR(5)                  | MA(1)   | MA(2)   | MA(3)  | 1:n    |
|-----------------------------------|--------|--------|---------|--------|------------------------|---------|---------|--------|--------|
|                                   | 0.6112 | 0.0928 | -0.8468 | 0.1276 | -0.0845                | -0.5794 | -0.0619 | 0.7807 | 0.0113 |
| s.e.                              | 0.1841 | 0.2507 | 0.1501  | 0.0878 | 0.0762                 | 0.1773  | 0.2409  | 0.1760 | 0.0177 |
| $\sigma^2$ estimated as 0.07103 : |        |        |         |        | loglikelihood = -25.69 |         |         |        |        |

Since results for forecasting horizon equal to 12 are not satisfactory, another identification and estimation period was employed. The start dates are the same, but the period end with observation number 863 (Feb 14, 2014).

Table 5.10: ARIMA (2,1,2) model for Henry Hub Natural Gas Spot Price - weekly data (est 1 – 866, pred 12)

| Coefficients:                    | AR(1)  | AR(2)   | MA(1)                          | MA(2)   | 1:n    |
|----------------------------------|--------|---------|--------------------------------|---------|--------|
|                                  | 1.0039 | -0.0496 | -0.9200                        | -0.0620 | 0.0077 |
| s.e.                             | 0.3285 | 0.3188  | 0.3287                         | 0.3252  | 0.0081 |
| $\sigma^2$ estimated as 0.2164 : |        |         | <i>loglikelihood</i> = -565.44 |         |        |

Table 5.11: ARIMA (2,1,3) model for Henry Hub Natural Gas Spot Price - weekly data (est 292 – 866, pred 12)

| Coefficients:                    | AR(1)  | AR(2)  | MA(1)  | MA(2)                          | MA(3)   | 1:n     |
|----------------------------------|--------|--------|--------|--------------------------------|---------|---------|
|                                  | 0.2144 | 0.7126 | 0.0660 | -0.7780                        | -0.2877 | -0.0069 |
| s.e.                             | 0.2260 | 0.2179 | 0.2262 | 0.1638                         | 0.0742  | 0.0034  |
| $\sigma^2$ estimated as 0.2127 : |        |        |        | <i>loglikelihood</i> = -371.58 |         |         |

Table 5.12: ARIMA (5,1,3) model for Henry Hub Natural Gas Spot Price - weekly data (est 436 – 866, pred 12)

| Coefficients:                    | AR(1)  | AR(2)  | AR(3)   | AR(4)  | AR(5)                          | MA(1)   | MA(2)   | MA(3)  | 1:n     |
|----------------------------------|--------|--------|---------|--------|--------------------------------|---------|---------|--------|---------|
|                                  | 0.8226 | 0.8139 | -0.8137 | 0.1226 | 0.0399                         | -0.5592 | -0.9341 | 0.4933 | -0.0144 |
| s.e.                             | 0.6540 | 0.2697 | 0.5873  | 0.2042 | 0.0601                         | 0.6545  | 0.1329  | 0.6103 | 0.0070  |
| $\sigma^2$ estimated as 0.1388 : |        |        |         |        | <i>loglikelihood</i> = -186.53 |         |         |        |         |

Table 5.13: ARIMA (3,1,5) model for Henry Hub Natural Gas Spot Price - weekly data (est 570 – 866, pred 12)

| Coefficients:                    | AR(1)  | AR(2)  | AR(3)   | MA(1)   | MA(2)                         | MA(3)  | MA(4)  | MA(5)  | 1:n     |
|----------------------------------|--------|--------|---------|---------|-------------------------------|--------|--------|--------|---------|
|                                  | 0.8422 | 0.8595 | -0.8784 | -0.5918 | -1.0378                       | 0.5619 | 0.3134 | 0.0562 | -0.0239 |
| s.e.                             | 0.0602 | 0.0826 | 0.0502  | 0.0922  | 0.0945                        | 0.1072 | 0.0792 | 0.0760 | 0.0270  |
| $\sigma^2$ estimated as 0.0742 : |        |        |         |         | <i>loglikelihood</i> = -37.62 |        |        |        |         |

Table 5.14: ARIMA (1,1,3) model for Henry Hub Natural Gas Spot Price - weekly data (est 631 – 866, pred 12)

| Coefficients:                     | AR(1)   | MA(1)  | MA(2)                        | MA(3)   | 1:n    |
|-----------------------------------|---------|--------|------------------------------|---------|--------|
|                                   | -0.8508 | 1.0727 | 0.1763                       | -0.1484 | 0.0156 |
| s.e.                              | 0.1000  | 0.1159 | 0.1119                       | 0.0813  | 0.0181 |
| $\sigma^2$ estimated as 0.05996 : |         |        | <i>loglikelihood</i> = -3.02 |         |        |

Table 5.15: ARIMA (5,1,3) model for Henry Hub Natural Gas Spot Price - weekly data (est 1 – 863, Aug 19,2005-Feb 14,2014, pred 12)

| Coefficients:                    | AR(1)  | AR(2)   | AR(3)                 | AR(4)   | AR(5)                          | MA(1)               | MA(2)  | MA(3)   | 1:n    |
|----------------------------------|--------|---------|-----------------------|---------|--------------------------------|---------------------|--------|---------|--------|
|                                  | 0.4449 | -0.7250 | 0.5242                | -0.1014 | -0.1024                        | -0.3582             | 0.6853 | -0.4925 | 0.0087 |
| s.e.                             | 0.1712 | 0.1036  | 0.1616                | 0.0441  | 0.0427                         | 0.1702              | 0.0952 | 0.1485  | 0.0136 |
| $\sigma^2$ estimated as 0.2101 : |        |         |                       |         | <i>loglikelihood</i> = -550.86 |                     |        |         |        |
| <i>AIC</i> = 1121.71             |        |         | <i>AICc</i> = 1121.97 |         |                                | <i>BIC</i> = 1169.3 |        |         |        |

Table 5.16: ARIMA (5,1,3) model for Henry Hub Natural Gas Spot Price - weekly data (est 421 – 863 Jan 10,1997 - Feb 14,2014, pred 12)

| Coefficients:                   | AR(1)  | AR(2)   | AR(3)  | AR(4)          | AR(5)                      | MA(1)   | MA(2)         | MA(3)   | 1:n     |
|---------------------------------|--------|---------|--------|----------------|----------------------------|---------|---------------|---------|---------|
|                                 | 0.3041 | -0.6876 | 0.4039 | -0.1563        | -0.2352                    | -0.0358 | 0.6386        | -0.2579 | -0.0101 |
| s.e.                            | 0.1714 | 0.1051  | 0.1722 | 0.0788         | 0.0627                     | 0.1729  | 0.0902        | 0.1559  | 0.0195  |
| $\sigma^2$ estimated as 0.174 : |        |         |        |                | $\loglikelihood = -241.29$ |         |               |         |         |
| $AIC = 502.59$                  |        |         |        | $AICc = 503.1$ |                            |         | $BIC = 543.5$ |         |         |

Table 5.17: ARIMA (1,1,5) model for Henry Hub Natural Gas Spot Price - weekly data (est 421 – 863 Jan 10,1997 - Feb 14,2014, pred 12)

| Coefficients:                    | AR(1)  | MA(1)  | MA(2)        | MA(3)                      | MA(4)   | MA(5)         | 1:n     |
|----------------------------------|--------|--------|--------------|----------------------------|---------|---------------|---------|
|                                  | 0.1987 | 0.0603 | -0.0138      | 0.0249                     | -0.0726 | -0.2492       | -0.0097 |
| s.e.                             | 0.1471 | 0.1411 | 0.0550       | 0.0495                     | 0.0452  | 0.0449        | 0.0190  |
| $\sigma^2$ estimated as 0.1804 : |        |        |              | $\loglikelihood = -248.84$ |         |               |         |
| $AIC = 513.67$                   |        |        | $AICc = 514$ |                            |         | $BIC = 546.4$ |         |

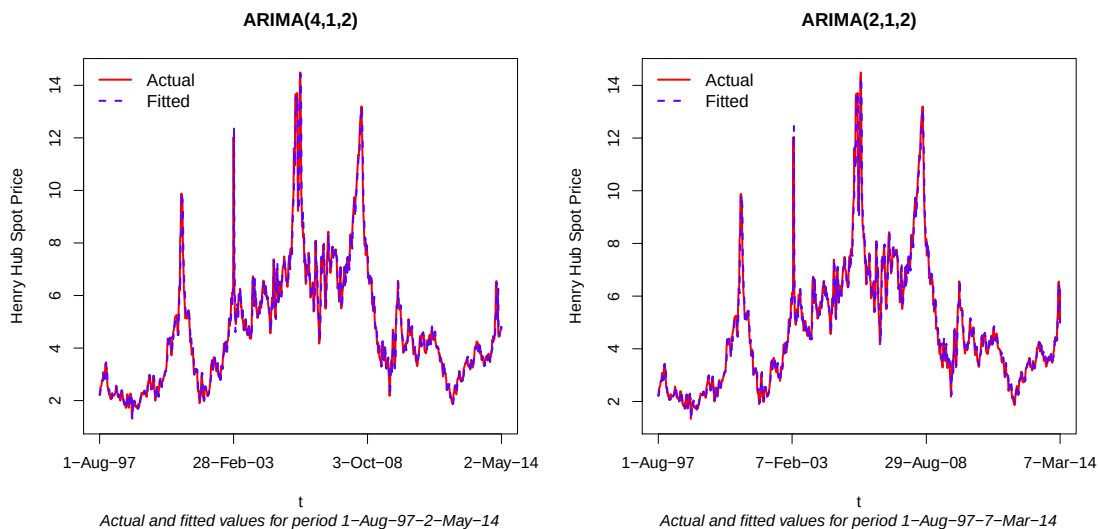


Figure 5.6: Fitted and actual values for Henry Hub Natural Gas Spot Price - weekly

### 5.1.4 Model Verification

#### Residuals analysis

In case of ARIMA models, model verification builds upon residuals analysis (see Section 4.5). For each model presented in tables 5.3 and 5.4, the Ljung-Box test with degrees of freedom equal to  $24 - p - q$  (where  $p$  and  $q$  are AR and MA parameters for  $ARIMA(p, d, q)$  model) was applied. For clarity of the thesis, results are not presented. Residuals of all of the proposed models passed normality test ( $p$ -value of Ljung-Box test  $>$  significance level  $\alpha = 0.05$ ). Following graphics are another example of residuals analysis in this paper.

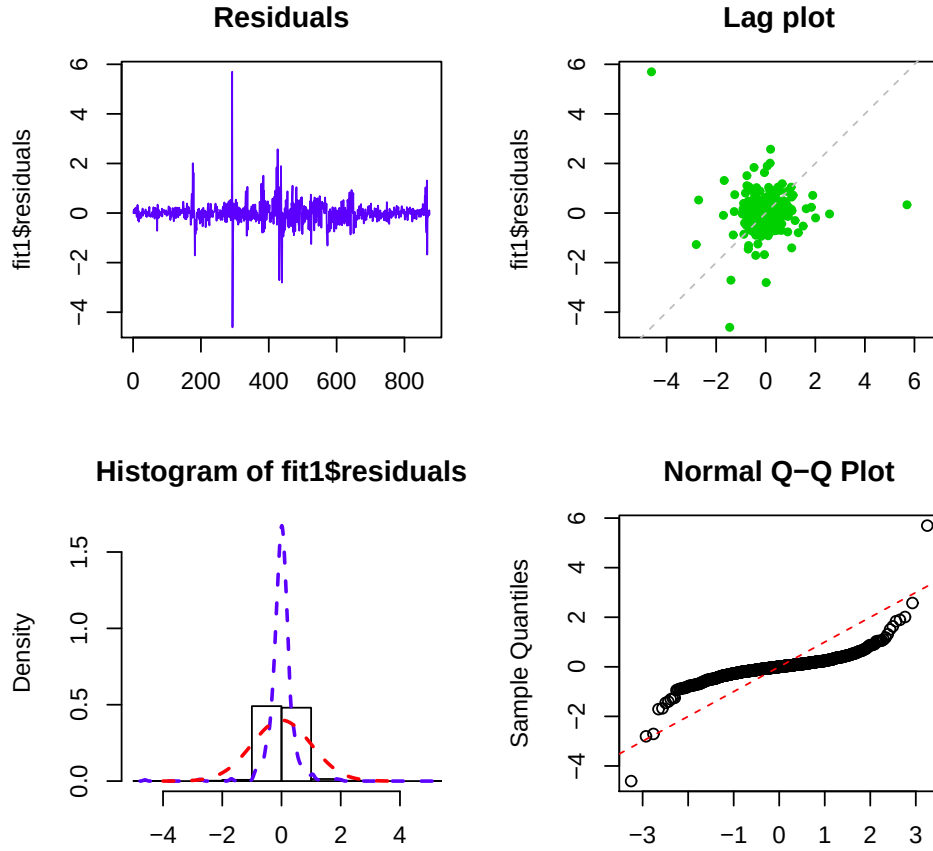


Figure 5.7: 4 plot for ARIMA(4,1,2) model - Henry Hub Natural Gas Spot Price, weekly data, 874 observations, 4 predictions

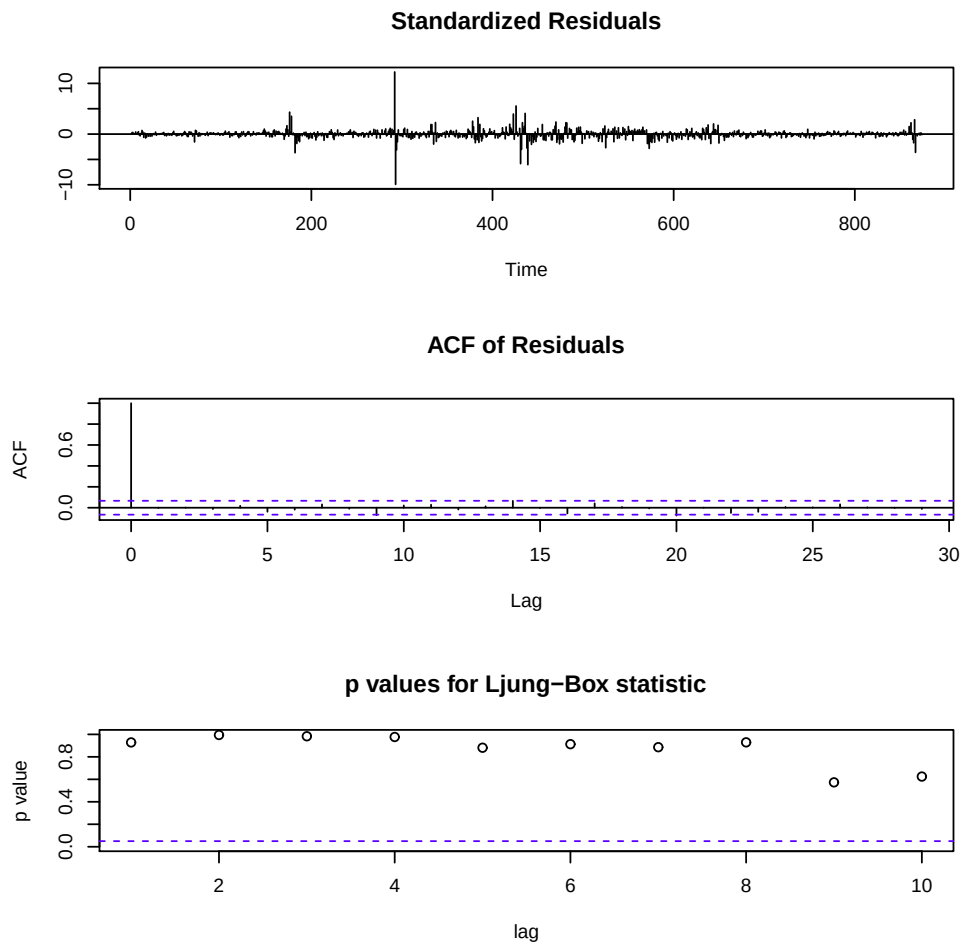


Figure 5.8: Residuals of ARIMA(4,1,2) model -Henry Hub Natural Gas Spot Price, weekly data, 874 observations, 4 predictions

### 5.1.5 Model Forecast

The last step of the Box-Jenkins method is model forecast. After model verification (residuals testing), only models with normal, i.i.d. residuals are taken into consideration.

#### Confidence Intervals

Confidence Intervals and predicted values for chosen models are depicted on the graphics presented below (Figure 5.11, Figure 5.12). It can be easily noticed, that ARIMA models predict much better for short time horizons.



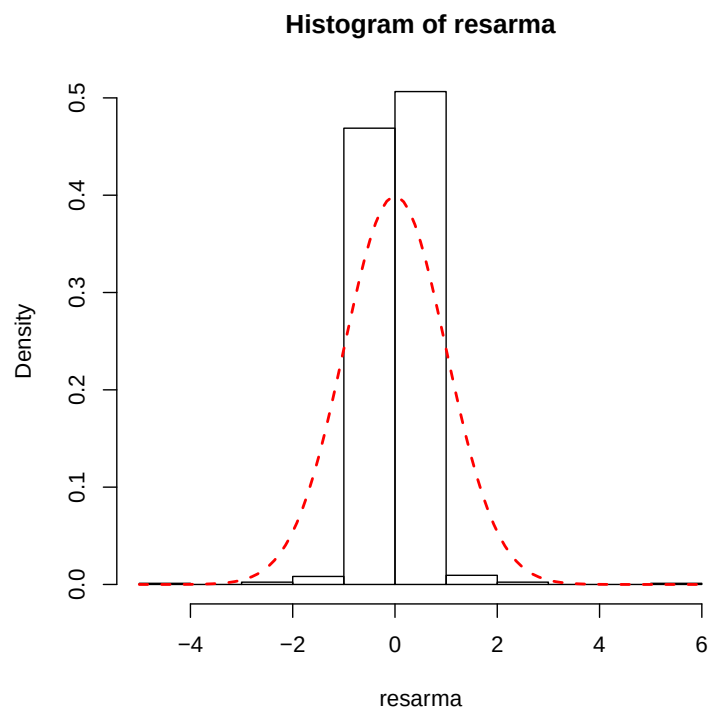


Figure 5.9: Residuals histogram - ARIMA(4,1,1) model for Henry Hub Natural Gas Spot Price - weekly data

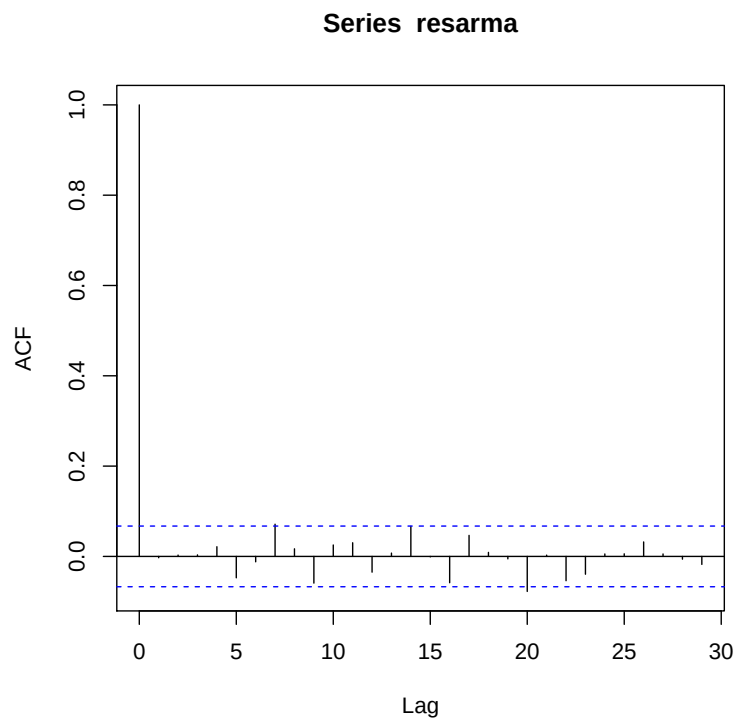


Figure 5.10: Residuals ACF - ARIMA(4,1,1) model for Henry Hub Natural Gas Spot Price - weekly

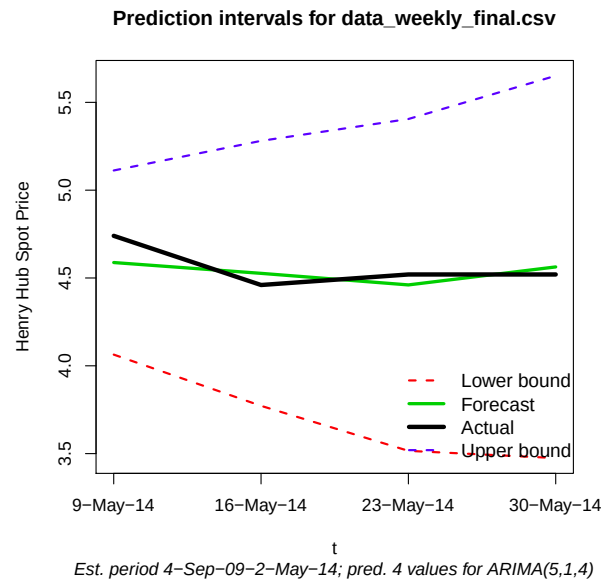
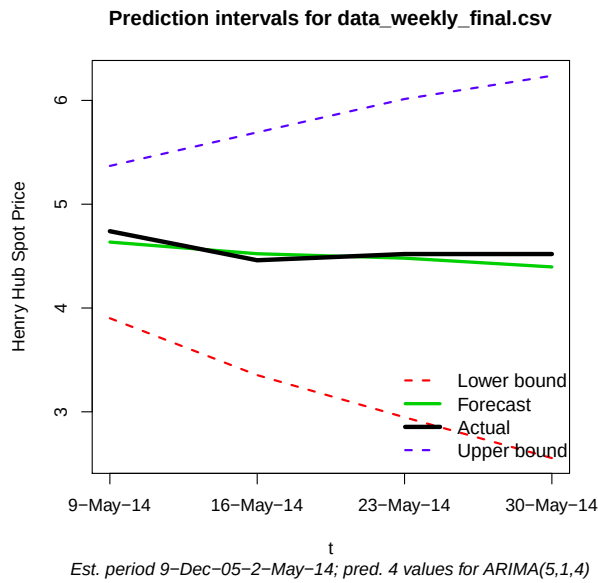
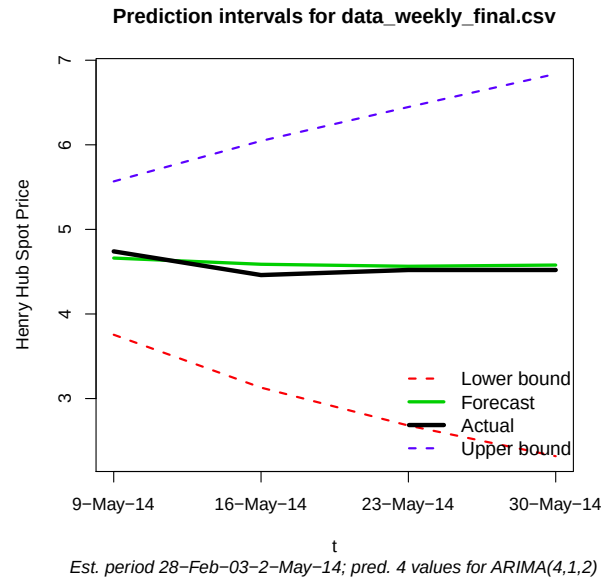
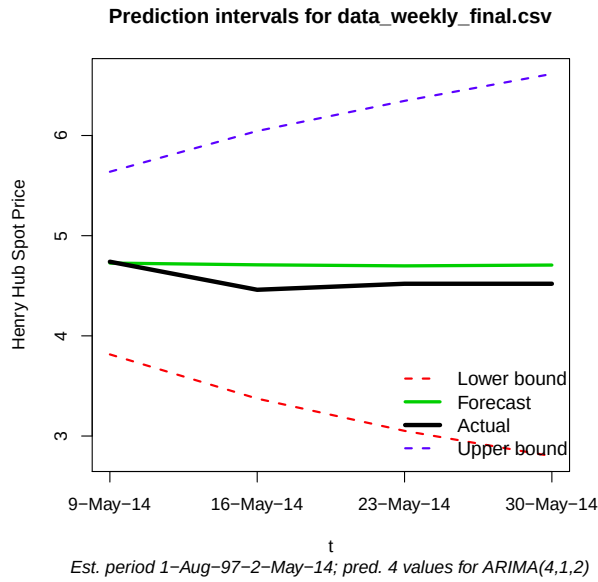


Figure 5.11: Forecast (n=4) for Henry Hub Natural Gas Spot Price - weekly price

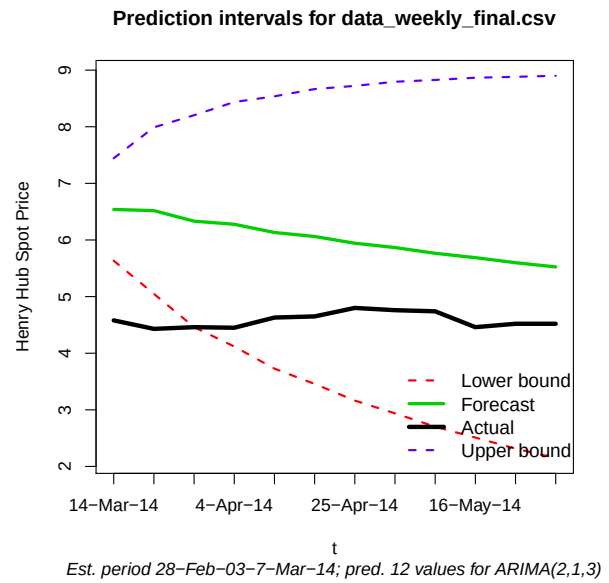
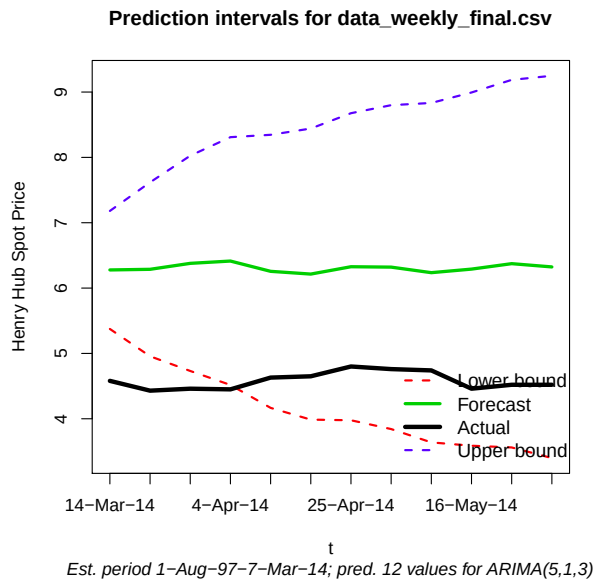
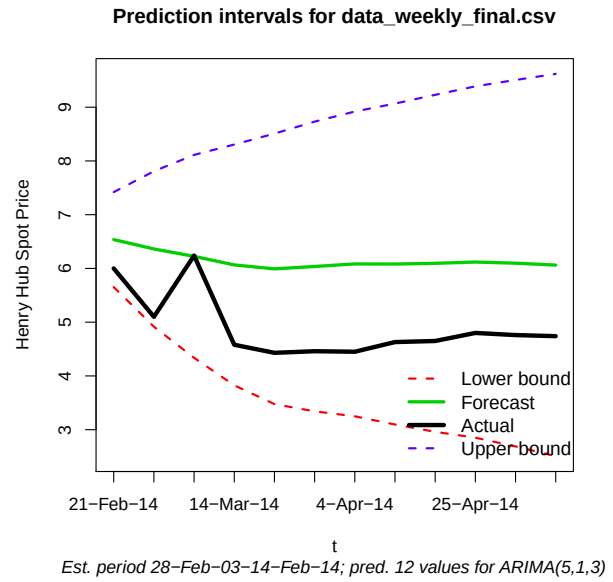
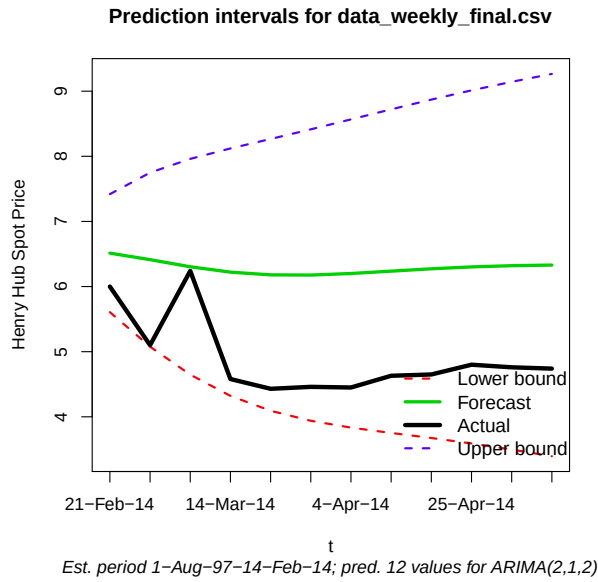


Figure 5.12: Forecast (n=12) for Henry Hub Natural Gas Spot Price - weekly price

## 5.2 ARIMAX

As previously stated in Section 4.2.2, natural gas price depends not only on its historical values, but also on multiple other (exogenous and endogenous) variables. Different research use various factors, like crude oil price, coal price, and also gold price and nominal effective exchange rate (NEER) of the U.S. dollar as well as the U.S. interest rate (for details see Chapter 2).

In order to establish which variables are significant for Henry Hub Natural Gas Sport Price, correlations between suspected variables were calculated. Table 5.19 presents results of this analysis.

Table 5.18: Variables description- weekly data

| Variable description                                       | Name used in dataset | Number used in data set |
|------------------------------------------------------------|----------------------|-------------------------|
| Henry Hub Natural Gas Spot Price weekly                    | g_HH_w               | 2                       |
| West Texas Intermediate Crude Oil Price weekly             | o_WTI_w              | 3                       |
| Heating Degree Days West South Central weekly              | HDD_WSC_w            | 4                       |
| Heating Degree Days deviation West South Central weekly    | HDDdev_WSC_w         | 5                       |
| Heating Degree Days total (United States) weekly           | HDD_USA_w            | 6                       |
| Heating Degree Days deviation total (United States) weekly | HDDdev_USA_w         | 7                       |
| Cooling Degree Days West South Central weekly              | CDD_WSC_w            | 8                       |
| Cooling Degree Days deviation West South Central weekly    | CDDdev_WSC_w         | 9                       |
| Cooling Degree Days total (United States) weekly           | CDD_USA_w            | 10                      |
| Cooling Degree Days deviation total (United States) weekly | CDDdev_USA_w         | 11                      |
| Storage Production region weekly                           | STOR_PROD_w          | 12                      |
| Storage total (United States) weekly                       | STOR_USA_w           | 13                      |
| Storage Differentials production region weekly             | STOR_PROD_ch_w       | 14                      |
| Storage Differentials total (United States) weekly         | STOR_USA_ch_w        | 15                      |
| Storage East region weekly                                 | STORAGE.EAST_w       | 16                      |
| Storage West region weekly                                 | STORAGE.WEST_w       | 17                      |
| Shut in natural gas production in Gold of Mexico weekly    | HH_shutin_w          | 18                      |
| Shut in crude oil production in Gold of Mexico weekly      | WTI_shutin_w         | 19                      |

As discussed earlier (Chapter 3), because of the local character of natural gas price, it is reasonable to use certain data only from the local area (West South Central Census Region for weather variables <sup>1</sup> and Producing Region for Storage variable <sup>2</sup>).

Tables 5.20 and 5.21 present correlations between variables either from the local area or the total sum from the whole United States. Since the data from the whole United States is strongly dependent on the local values, it is wise to make two different data sets - local and global.

Besides the division for local and global date, the analysis is also based on different time horizons (different identification and estimation set as well as predicting different number of values). Various time horizons and prediction intervals can be found in Section 5.1 (Table 3.2).

The stationarity of Henry Hub Natural Gas spot price was already discussed in the previous section. This section will focus on adding additional variables to ARIMA models.

Analyzing different factors affecting Henry Hub Natural Gas spot price, the best results were obtained for full data set (all the variables for a certain region - either local or global). Since the variables Storage and Storage Differential were also highly correlated, only one (i.e. Storage Differential) was used as covariant.

<sup>1</sup>West South Central Census Region includes Texas, Oklahoma, Louisiana and Arkansas

<sup>2</sup>Producing Region includes Texas, Oklahoma, Kansas, New Mexico, Louisiana, Arkansas, Mississippi and Alabama

Local data set contains following variables:

- g\_HH\_w (as dependent variable)
- o\_WTI\_w
- HDD\_WSC\_w
- HDDdev\_WSC\_w
- CDD\_WSC\_w
- CDDdev\_WSC\_w
- STOR\_PROD\_ch\_w
- HH\_shutin\_w

Similarly, the global data set consists of variables:

- g\_HH\_w (as dependent variable)
- o\_WTI\_w
- HDD\_USA\_w
- HDDdev\_USA\_w
- CDD\_USA\_w
- CDDdev\_USA\_w
- STOR\_USA\_ch\_w
- HH\_shutin\_w

The following tables present chosen ARIMAX models for local data and each time horizon (identification and prediction) with its model selection criteria and error measures.

Similarly results for global data were obtained.

As can be seen, local and global data give fairly similar results. Local data displays slightly better forecasting abilities (smaller forecasting errors).

Below few examples of estimated models are presented.

Table 5.19: Correlation matrix - weekly data

|    | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    | 11    | 12    | 13    | 14    | 15    | 16    | 17    | 18    | 19    |
|----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 2  | 1.00  | 0.34  | 0.10  | 0.12  | 0.09  | 0.13  | -0.04 | -0.01 | -0.02 | 0.03  | -0.14 | -0.11 | -0.28 | -0.18 | -0.08 | -0.08 | 0.40  | 0.37  |
| 3  | 0.34  | 1.00  | -0.02 | 0.06  | -0.02 | 0.04  | 0.06  | 0.08  | 0.07  | 0.13  | 0.50  | 0.26  | -0.07 | -0.01 | 0.08  | 0.40  | 0.06  | 0.08  |
| 4  | 0.10  | -0.02 | 1.00  | 0.25  | 0.94  | 0.18  | -0.75 | -0.36 | -0.66 | -0.34 | -0.04 | -0.01 | -0.08 | -0.09 | 0.02  | -0.07 | -0.06 | -0.08 |
| 5  | 0.12  | 0.06  | 0.25  | 1.00  | 0.02  | 0.76  | 0.05  | -0.21 | 0.07  | -0.06 | -0.05 | -0.04 | -0.20 | -0.19 | -0.05 | -0.01 | -0.01 | -0.02 |
| 6  | 0.09  | -0.02 | 0.94  | 0.02  | 1.00  | 0.11  | -0.86 | -0.32 | -0.80 | -0.39 | -0.10 | -0.09 | -0.03 | -0.06 | -0.07 | -0.15 | -0.07 | -0.09 |
| 7  | 0.13  | 0.04  | 0.18  | 0.76  | 0.11  | 1.00  | 0.07  | -0.20 | 0.08  | -0.15 | -0.07 | -0.07 | -0.20 | -0.22 | -0.08 | -0.04 | -0.03 | -0.04 |
| 8  | -0.04 | 0.06  | -0.75 | 0.05  | -0.86 | 0.07  | 1.00  | 0.48  | 0.96  | 0.49  | 0.10  | 0.09  | 0.00  | 0.02  | 0.06  | 0.16  | 0.03  | 0.06  |
| 9  | -0.01 | 0.08  | -0.36 | -0.21 | -0.32 | -0.20 | 0.48  | 1.00  | 0.31  | 0.59  | 0.08  | 0.04  | 0.05  | 0.07  | 0.01  | 0.06  | 0.05  | 0.08  |
| 10 | -0.02 | 0.07  | -0.66 | 0.07  | -0.80 | 0.08  | 0.96  | 0.31  | 1.00  | 0.55  | 0.12  | 0.12  | 0.00  | 0.01  | 0.09  | 0.19  | 0.02  | 0.05  |
| 11 | 0.03  | 0.13  | -0.34 | -0.06 | -0.39 | -0.15 | 0.49  | 0.59  | 0.55  | 1.00  | 0.13  | 0.08  | 0.05  | 0.07  | 0.04  | 0.13  | 0.09  | 0.13  |
| 12 | -0.14 | 0.50  | -0.04 | -0.05 | -0.10 | -0.07 | 0.10  | 0.08  | 0.12  | 0.13  | 1.00  | 0.91  | 0.35  | 0.35  | 0.78  | 0.93  | 0.07  | 0.08  |
| 13 | -0.11 | 0.26  | -0.01 | -0.04 | -0.09 | -0.07 | 0.09  | 0.04  | 0.12  | 0.08  | 0.91  | 1.00  | 0.21  | 0.24  | 0.97  | 0.95  | 0.17  | 0.18  |
| 14 | -0.28 | -0.07 | -0.08 | -0.20 | -0.03 | -0.20 | 0.00  | 0.05  | 0.00  | 0.05  | 0.35  | 0.21  | 1.00  | 0.94  | 0.14  | 0.14  | -0.06 | -0.05 |
| 15 | -0.18 | -0.01 | -0.09 | -0.19 | -0.06 | -0.22 | 0.02  | 0.07  | 0.01  | 0.07  | 0.35  | 0.24  | 0.94  | 1.00  | 0.17  | 0.18  | 0.01  | 0.02  |
| 16 | -0.08 | 0.08  | 0.02  | -0.05 | -0.07 | -0.08 | 0.06  | 0.01  | 0.09  | 0.04  | 0.78  | 0.97  | 0.14  | 0.17  | 1.00  | 0.86  | 0.21  | 0.21  |
| 17 | -0.08 | 0.40  | -0.07 | -0.01 | -0.15 | -0.04 | 0.16  | 0.06  | 0.19  | 0.13  | 0.93  | 0.95  | 0.14  | 0.18  | 0.86  | 1.00  | 0.17  | 0.17  |
| 18 | 0.40  | 0.06  | -0.06 | -0.01 | -0.07 | -0.03 | 0.03  | 0.05  | 0.02  | 0.09  | 0.07  | 0.17  | -0.06 | 0.01  | 0.21  | 0.17  | 1.00  | 0.97  |
| 19 | 0.37  | 0.08  | -0.08 | -0.02 | -0.09 | -0.04 | 0.06  | 0.08  | 0.05  | 0.13  | 0.08  | 0.18  | -0.05 | 0.02  | 0.21  | 0.17  | 0.97  | 1.00  |

Table 5.20: Correlation matrix for local variables - weekly data

|    | 2      | 3      | 4      | 5      | 8      | 9      | 12     | 14     | 18     |
|----|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 2  | 1      | 0.340  | 0.104  | 0.122  | -0.037 | -0.011 | -0.144 | -0.282 | 0.399  |
| 3  | 0.340  | 1      | -0.018 | 0.060  | 0.056  | 0.081  | 0.503  | -0.074 | 0.065  |
| 4  | 0.104  | -0.018 | 1      | 0.250  | -0.747 | -0.357 | -0.042 | -0.076 | -0.064 |
| 5  | 0.122  | 0.060  | 0.250  | 1      | 0.047  | -0.215 | -0.046 | -0.201 | -0.009 |
| 8  | -0.037 | 0.056  | -0.747 | 0.047  | 1      | 0.479  | 0.099  | 0.003  | 0.026  |
| 9  | -0.011 | 0.081  | -0.357 | -0.215 | 0.479  | 1      | 0.080  | 0.049  | 0.048  |
| 12 | -0.144 | 0.503  | -0.042 | -0.046 | 0.099  | 0.080  | 1      | 0.347  | 0.074  |
| 14 | -0.282 | -0.074 | -0.076 | -0.201 | 0.003  | 0.049  | 0.347  | 1      | -0.061 |
| 18 | 0.399  | 0.065  | -0.064 | -0.009 | 0.026  | 0.048  | 0.074  | -0.061 | 1      |

Table 5.21: Correlation matrix for global variables - weekly data

|    | 2      | 3      | 6      | 7      | 10     | 11     | 13     | 15     | 18     |
|----|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 2  | 1      | 0.340  | 0.090  | 0.135  | -0.018 | 0.033  | -0.105 | -0.179 | 0.399  |
| 3  | 0.340  | 1      | -0.023 | 0.040  | 0.068  | 0.128  | 0.259  | -0.010 | 0.065  |
| 6  | 0.090  | -0.023 | 1      | 0.114  | -0.800 | -0.387 | -0.094 | -0.059 | -0.070 |
| 7  | 0.135  | 0.040  | 0.114  | 1      | 0.079  | -0.148 | -0.075 | -0.220 | -0.033 |
| 10 | -0.018 | 0.068  | -0.800 | 0.079  | 1      | 0.554  | 0.120  | 0.013  | 0.022  |
| 11 | 0.033  | 0.128  | -0.387 | -0.148 | 0.554  | 1      | 0.083  | 0.069  | 0.089  |
| 13 | -0.105 | 0.259  | -0.094 | -0.075 | 0.120  | 0.083  | 1      | 0.238  | 0.175  |
| 15 | -0.179 | -0.010 | -0.059 | -0.220 | 0.013  | 0.069  | 0.238  | 1      | 0.009  |
| 18 | 0.399  | 0.065  | -0.070 | -0.033 | 0.022  | 0.089  | 0.175  | 0.009  | 1      |

Table 5.22: ARIMAX models for Henry Hub Natural Gas Spot Price - local variables, weekly data, predicting 4 values

| Series                                                                  | Model         | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE  |
|-------------------------------------------------------------------------|---------------|----------|----------|--------|--------|-------|-------|
| ARIMAX local; Est. period 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                       | ARIMAX(5,1,3) | 1023.303 | 1104.446 | 0.426  | 5.517  | 0.163 | 3.288 |
| 2                                                                       | ARIMAX(5,1,2) | 1031.185 | 1107.554 | 0.429  | 5.517  | 0.098 | 1.937 |
| ARIMAX local; Est. period 28-Feb-03 - 2-May-14 (583 obs); Pred.4 values |               |          |          |        |        |       |       |
| 1                                                                       | ARIMAX(5,1,2) | 681.044  | 750.935  | 0.422  | 5.113  | 0.091 | 1.871 |
| 2                                                                       | ARIMAX(4,1,2) | 691.606  | 757.129  | 0.427  | 5.089  | 0.143 | 2.943 |
| ARIMAX local; Est. period 9-Dec-05 - 2-May-14 (439 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                       | ARIMAX(5,1,4) | 365.837  | 439.358  | 0.350  | 4.761  | 0.137 | 2.273 |
| 2                                                                       | ARIMAX(1,1,1) | 368.821  | 413.751  | 0.359  | 4.794  | 0.177 | 3.455 |
| 3                                                                       | ARIMAX(5,1,2) | 371.681  | 437.033  | 0.356  | 4.824  | 0.167 | 3.060 |
| ARIMAX local; Est. period 4-Jul-08 - 2-May-14 (305 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                       | ARIMAX(3,1,0) | 96.378   | 141.022  | 0.272  | 4.690  | 0.184 | 3.530 |
| 2                                                                       | ARIMAX(2,1,2) | 96.765   | 145.129  | 0.270  | 4.645  | 0.215 | 4.314 |
| 3                                                                       | ARIMAX(5,1,0) | 98.781   | 150.865  | 0.271  | 4.699  | 0.151 | 2.854 |
| ARIMAX local; Est. period 4-Sep-09 - 2-May-14 (244 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                       | ARIMAX(2,1,3) | 23.904   | 72.864   | 0.238  | 4.525  | 0.117 | 2.133 |
| 2                                                                       | ARIMAX(5,1,0) | 26.592   | 75.552   | 0.241  | 4.518  | 0.119 | 2.173 |
| 3                                                                       | ARIMAX(5,1,3) | 34.575   | 94.027   | 0.240  | 4.502  | 0.162 | 3.342 |



Table 5.23: ARIMAX models for Henry Hub Natural Gas Spot Price - local variables, weekly data, predicting 12 values

| Series                                                                   | Model         | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE   |
|--------------------------------------------------------------------------|---------------|----------|----------|--------|--------|-------|--------|
| ARIMAX local; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                        | ARIMAX(5,1,3) | 1013.904 | 1094.890 | 0.426  | 5.489  | 1.242 | 26.772 |
| 2                                                                        | ARIMAX(5,1,2) | 1020.819 | 1097.041 | 0.428  | 5.492  | 1.337 | 28.785 |
| ARIMAX local; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.12 values |               |          |          |        |        |       |        |
| 1                                                                        | ARIMAX(5,1,2) | 668.818  | 738.488  | 0.421  | 5.060  | 1.304 | 27.762 |
| 2                                                                        | ARIMAX(4,1,2) | 678.326  | 743.642  | 0.425  | 5.044  | 1.070 | 21.293 |
| ARIMAX local; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                        | ARIMAX(5,1,2) | 343.566  | 408.624  | 0.347  | 4.716  | 1.772 | 38.421 |
| 2                                                                        | ARIMAX(1,1,1) | 350.051  | 394.779  | 0.354  | 4.757  | 1.711 | 37.194 |
| 3                                                                        | ARIMAX(5,1,4) | 352.563  | 425.753  | 0.348  | 4.772  | 1.721 | 37.325 |
| ARIMAX local; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                        | ARIMAX(1,1,1) | 71.758   | 112.389  | 0.263  | 4.545  | 1.428 | 30.824 |
| 2                                                                        | ARIMAX(3,1,0) | 72.577   | 116.901  | 0.262  | 4.553  | 1.458 | 31.443 |
| 3                                                                        | ARIMAX(2,1,2) | 73.231   | 121.249  | 0.261  | 4.521  | 1.501 | 32.509 |
| ARIMAX local; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                        | ARIMAX(5,1,3) | -16.786  | 42.100   | 0.214  | 4.138  | 1.606 | 34.863 |
| 2                                                                        | ARIMAX(3,1,3) | -15.044  | 36.914   | 0.216  | 4.156  | 1.341 | 28.923 |
| 3                                                                        | ARIMAX(2,1,3) | -4.962   | 43.532   | 0.225  | 4.284  | 1.573 | 34.139 |
| 4                                                                        | ARIMAX(5,1,0) | -3.910   | 44.584   | 0.225  | 4.293  | 1.531 | 33.167 |

Table 5.24: ARIMAX models for Henry Hub Natural Gas Spot Price - global variables, weekly data, predicting 4 values

| Series                                                                   | Model         | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE  |
|--------------------------------------------------------------------------|---------------|----------|----------|--------|--------|-------|-------|
| ARIMAX global; Est. period 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                        | ARIMAX(5,1,3) | 1033.690 | 1114.832 | 0.429  | 5.445  | 0.140 | 2.595 |
| 2                                                                        | ARIMAX(5,1,2) | 1038.629 | 1114.998 | 0.430  | 5.463  | 0.234 | 4.469 |
| ARIMAX global; Est. period 28-Feb-03 - 2-May-14 (583 obs); Pred.4 values |               |          |          |        |        |       |       |
| 1                                                                        | ARIMAX(5,1,2) | 693.800  | 763.691  | 0.427  | 5.105  | 0.210 | 4.194 |
| 2                                                                        | ARIMAX(4,1,2) | 705.737  | 771.260  | 0.432  | 5.093  | 0.108 | 2.021 |
| ARIMAX global; Est. period 9-Dec-05 - 2-May-14 (439 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                        | ARIMAX(1,1,1) | 370.217  | 415.146  | 0.360  | 4.845  | 0.100 | 1.808 |
| 2                                                                        | ARIMAX(5,1,4) | 370.549  | 444.070  | 0.353  | 4.917  | 0.116 | 1.680 |
| 3                                                                        | ARIMAX(5,1,2) | 372.547  | 437.899  | 0.356  | 4.884  | 0.152 | 3.067 |
| ARIMAX global; Est. period 4-Jul-08 - 2-May-14 (305 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                        | ARIMAX(2,1,0) | 107.704  | 148.628  | 0.278  | 4.708  | 0.107 | 1.836 |
| 2                                                                        | ARIMAX(2,1,2) | 108.793  | 157.157  | 0.277  | 4.642  | 0.125 | 2.629 |
| 3                                                                        | ARIMAX(3,1,0) | 109.066  | 153.710  | 0.278  | 4.707  | 0.100 | 1.826 |
| 4                                                                        | ARIMAX(2,1,3) | 110.431  | 162.515  | 0.277  | 4.678  | 0.118 | 2.465 |
| 5                                                                        | ARIMAX(5,1,0) | 110.616  | 162.700  | 0.277  | 4.702  | 0.122 | 2.480 |
| 6                                                                        | ARIMAX(5,1,3) | 115.750  | 178.995  | 0.275  | 4.666  | 0.121 | 1.981 |
| ARIMAX global; Est. period 4-Sep-09 - 2-May-14 (244 obs); Pred.4 values  |               |          |          |        |        |       |       |
| 1                                                                        | ARIMAX(2,1,3) | 43.049   | 92.009   | 0.249  | 4.542  | 0.095 | 1.667 |
| 2                                                                        | ARIMAX(5,1,2) | 43.133   | 99.088   | 0.246  | 4.452  | 0.094 | 1.764 |
| 3                                                                        | ARIMAX(5,1,0) | 44.079   | 93.040   | 0.250  | 4.552  | 0.134 | 2.641 |
| 4                                                                        | ARIMAX(3,1,3) | 44.547   | 97.005   | 0.248  | 4.526  | 0.091 | 1.695 |
| 5                                                                        | ARIMAX(3,1,0) | 45.600   | 87.566   | 0.252  | 4.542  | 0.086 | 1.563 |

Table 5.25: ARIMAX models for Henry Hub Natural Gas Spot Price - global variables, weekly data, predicting 12 values

| Series                                                                    | Model         | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE   |
|---------------------------------------------------------------------------|---------------|----------|----------|--------|--------|-------|--------|
| ARIMAX global; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                         | ARIMAX(5,1,3) | 1022.523 | 1103.509 | 0.428  | 5.408  | 1.352 | 28.886 |
| 2                                                                         | ARIMAX(5,1,2) | 1027.209 | 1103.431 | 0.430  | 5.433  | 1.427 | 30.463 |
| ARIMAX global; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.12 values |               |          |          |        |        |       |        |
| 1                                                                         | ARIMAX(5,1,2) | 680.482  | 750.152  | 0.425  | 5.061  | 1.537 | 32.740 |
| 2                                                                         | ARIMAX(4,1,2) | 689.873  | 755.189  | 0.429  | 5.029  | 1.323 | 27.155 |
| ARIMAX global; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                         | ARIMAX(5,1,2) | 345.024  | 410.082  | 0.347  | 4.771  | 1.920 | 41.507 |
| 2                                                                         | ARIMAX(1,1,1) | 350.584  | 395.311  | 0.354  | 4.801  | 1.809 | 39.206 |
| 3                                                                         | ARIMAX(5,1,4) | 352.163  | 425.353  | 0.347  | 4.857  | 1.840 | 39.697 |
| ARIMAX global; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                         | ARIMAX(1,1,1) | 80.779   | 121.410  | 0.267  | 4.575  | 1.621 | 34.962 |
| 2                                                                         | ARIMAX(3,1,2) | 81.320   | 133.032  | 0.264  | 4.534  | 1.698 | 36.766 |
| 3                                                                         | ARIMAX(3,1,0) | 82.244   | 126.569  | 0.267  | 4.592  | 1.768 | 38.258 |
| ARIMAX global; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred.12 values  |               |          |          |        |        |       |        |
| 1                                                                         | ARIMAX(2,1,3) | 5.404    | 53.897   | 0.230  | 4.317  | 1.622 | 34.943 |
| 2                                                                         | ARIMAX(5,1,2) | 6.648    | 62.069   | 0.227  | 4.238  | 1.362 | 28.128 |
| 3                                                                         | ARIMAX(3,1,3) | 7.094    | 59.052   | 0.230  | 4.333  | 1.522 | 32.657 |
| 4                                                                         | ARIMAX(5,1,0) | 7.342    | 55.836   | 0.231  | 4.328  | 1.664 | 35.866 |

### 5.2.1 Model Estimation

Table 5.26 presents estimated parameters for ARIMAX(5,1,2) model with all exogenous variables (given above) for local data. Column *1:nobs* indicates the intercept of the model. Similar to the ARIMA models, the ARIMAX models with intercept predicted much better than models without it.

Table 5.26: ARIMAX (5,1,2) model for Henry Hub Natural Gas Spot Price - local variables weekly data (est 1 – 874, pred 4)

| Coefficients:                  | AR(1)         | AR(2)            | AR(3)         | AR(4)                          | AR(5)              | MA(1)           | MA(2)  |
|--------------------------------|---------------|------------------|---------------|--------------------------------|--------------------|-----------------|--------|
|                                | 0.4385        | -0.4224          | -0.0246       | -0.0460                        | -0.1163            | -0.4298         | 0.4092 |
| s.e.                           | 0.2108        | 0.2227           | 0.0405        | 0.0379                         | 0.0410             | 0.2111          | 0.2233 |
| WTI_w                          | HDD_<br>WSC_w | HDDdev_<br>WSC_w | CDD_<br>WSC_w | CDDdev_<br>WSC_w               | STOR_<br>PROD_ch_w | HH_<br>shutin_w | 1:nobs |
| 0.0359                         | 0.0028        | 0.0016           | -0.0017       | 0.0045                         | -2.4281            | 0.0001          | 0.0041 |
| 0.0060                         | 0.0015        | 0.0015           | 0.0021        | 0.0023                         | 0.4618             | 0               | 0.0008 |
| $\sigma^2$ estimated as 0.1839 |               |                  |               | <i>loglikelihood</i> = -499.59 |                    |                 |        |

Table 5.27: ARIMAX (5,1,3) model for Henry Hub Natural Gas Spot Price - global variables weekly data (est 1 – 874, pred 4)

| Coefficients:                   | AR(1)  | AR(2)         | AR(3)            | AR(4)         | AR(5)                          | MA(1)             | MA(2)           | MA(3)   |
|---------------------------------|--------|---------------|------------------|---------------|--------------------------------|-------------------|-----------------|---------|
|                                 | 0.5257 | -0.5960       | 0.6316           | -0.0599       | -0.1055                        | -0.5176           | 0.5532          | -0.6505 |
| s.e.                            | 0.1439 | 0.1262        | 0.1413           | 0.0444        | 0.0449                         | 0.1428            | 0.1215          | 0.1275  |
| Coefficients:                   | WTI_w  | HDD_<br>USA_w | HDDdev_<br>USA_w | CDD_<br>USA_w | CDDdev_<br>USA_w               | STOR_<br>USA_ch_w | HH_<br>shutin_w | 1:nobs  |
|                                 | 0.0373 | 0.0017        | 0.0031           | -0.0037       | 0.0098                         | -3.1563           | 0.001           | 0.0034  |
| s.e.                            | 0.0059 | 0.0010        | 0.0012           | 0.0035        | 0.0041                         | 0.5873            | 0               | 0.0008  |
| $\sigma^2$ estimated as 0.184 : |        |               |                  |               | <i>loglikelihood</i> = -499.84 |                   |                 |         |

## 5.2.2 Model Verification

### Residuals Analysis

Similar to the ARIMA models, ARIMAX model verification builds upon residuals analysis (see Section 4.5).

For each model presented in tables 5.22, 5.23, 5.24, 5.25, the Ljung-Box test with  $24 - p - q$  degrees of freedom (where  $p$  and  $q$  are AR and MA parameters for  $ARIMAX(p, d, q)$  model) was applied. For clarity of the thesis, results are not presented. Residuals of all of the proposed models passed normality test ( $p$ -value of Ljung-Box test  $>$  significance level  $\alpha = 0.05$ ). Following graphs are another example of residuals analysis in this paper.

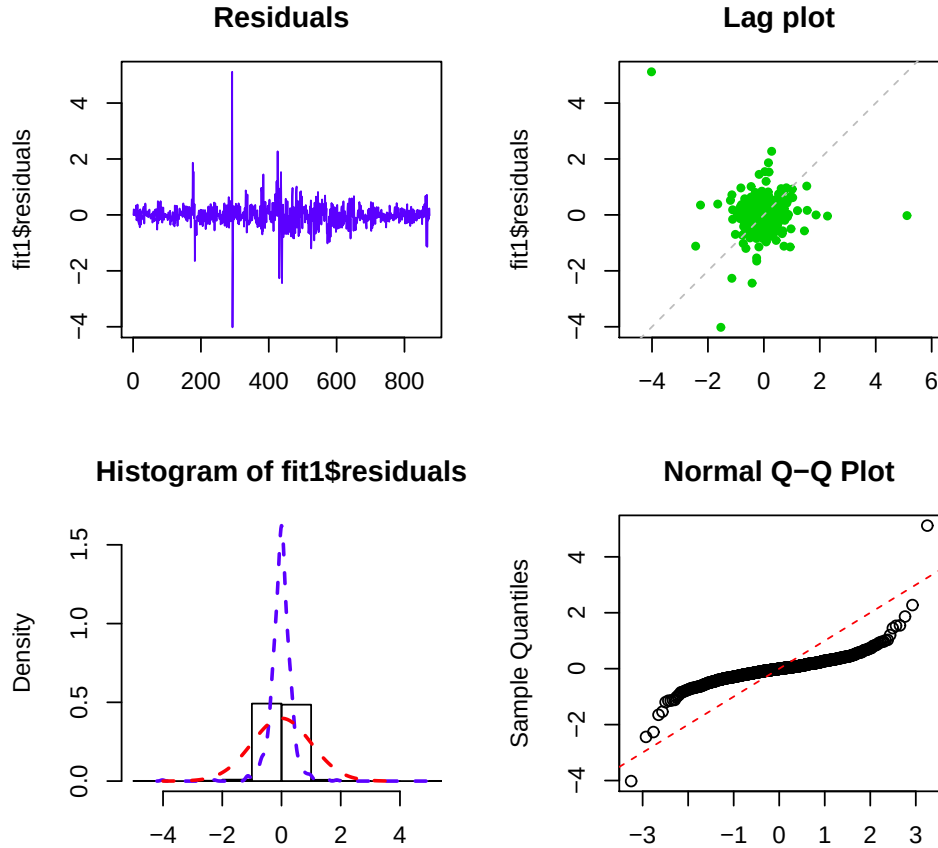


Figure 5.13: 4 plot for ARIMAX(5,1,3) with local variables model -Henry Hub Natural Gas Spot Price, weekly data, 874 observations, 4 predictions

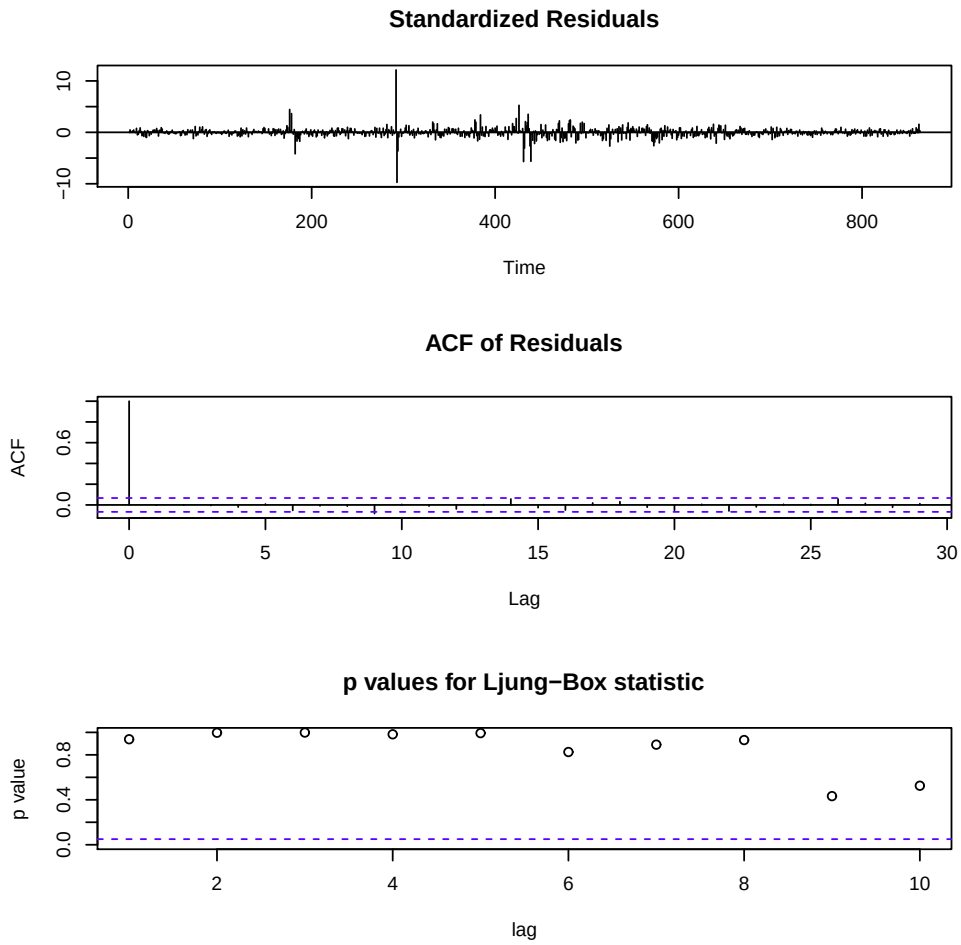


Figure 5.14: Residuals of ARIMAX(5,1,2) with local variables model -Henry Hub Natural Gas Spot Price, weekly data, 874 observations, 4 predictions

### 5.2.3 Model Forecast

The last step of Box-Jenkins method is model forecast. After model verification (residuals testing), only models with normal, i.i.d. residuals are taken into consideration. Plots on the Figure 5.15 depict predictions and confidence intervals for Henry Hub Natural Gas Spot Price (weekly data). Upper plots present 4 predicted values, while lower plots present prediction for 12 values. Graphics on the left hand side use local data as exogenous variables, and graphics on the right hand side illustrate prediction for ARIMAX models using global variables.

#### Confidence Intervals

Confidence intervals were calculated and are depicted on the following graphs.

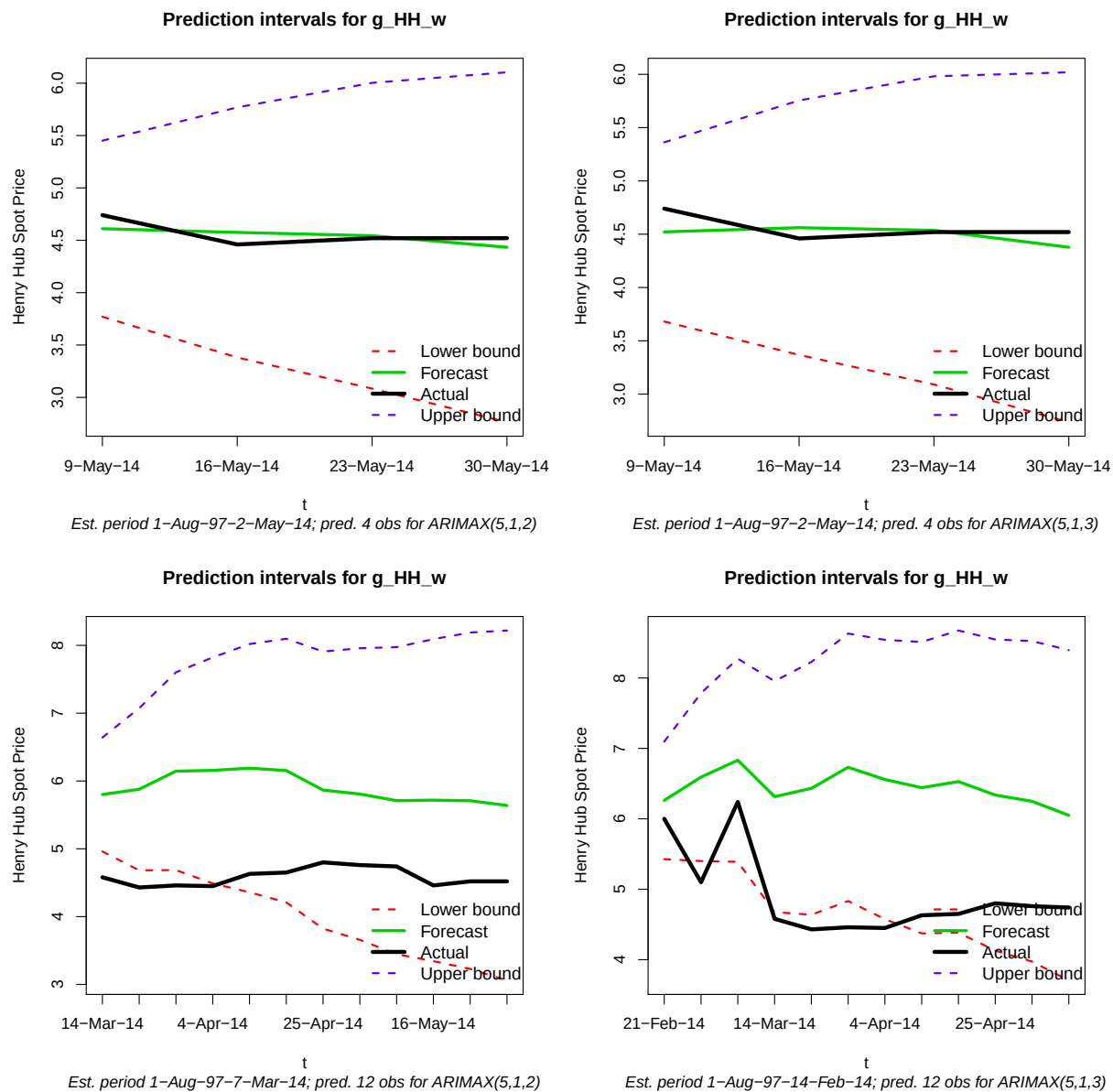


Figure 5.15: Forecast (n=4 and n=12) for Henry Hub Natural Gas Spot Price - weekly price, ARIMAX models for local data (left hand side) and global data(right hand side)

## 5.3 Transfer Function

As mentioned in Section 4.2.2 Transfer Function models in this paper are treated as ARIMAX models allowing lag of covariates as additional variables. Since the most significant factor for natural gas price is crude oil price (Table 5.19), for model simplicity, only lags of this variable (crude oil price - o\_WTI\_w) are allowed in the Transfer Function model.

Since lags of crude oil price variable (o\_WTI\_w) are allowed in the model, the numbers of variables (names of variables used in data set) are different than in previous section (5.2 - ARIMAX) and are presented in the Table 5.28.

Table 5.28: Variables description for Transfer Function Model

| Variable description                                       | Name used in dataset | Number used in data set |
|------------------------------------------------------------|----------------------|-------------------------|
| Henry Hub Natural Gas Spot Price weekly                    | 2                    | g_HH_w                  |
| West Texas Intermediate Crude Oil Price weekly             | 3                    | o_WTI_w                 |
| West Texas Intermediate Crude Oil Price weekly lag 1       | 4                    | WTI-1_w                 |
| West Texas Intermediate Crude Oil Price weekly lag 2       | 5                    | WTI-2_w                 |
| West Texas Intermediate Crude Oil Price weekly lag 3       | 6                    | WTI-3_w                 |
| Heating Degree Days West South Central weekly              | 7                    | HDD_WSC_w               |
| Heating Degree Days deviation West South Central weekly    | 8                    | HDDdev_WSC_w            |
| Heating Degree Days total (United States) weekly           | 9                    | HDD_USA_w               |
| Heating Degree Days deviation total (United States) weekly | 10                   | HDDdev_USA_w            |
| Cooling Degree Days West South Central weekly              | 11                   | CDD_WSC_w               |
| Cooling Degree Days deviation West South Central weekly    | 12                   | CDDdev_WSC_w            |
| Cooling Degree Days total (United States) weekly           | 13                   | CDD_USA_w               |
| Cooling Degree Days deviation total (United States) weekly | 14                   | CDDdev_USA_w            |
| Storage Production region weekly                           | 15                   | STOR_PROD_w             |
| Storage total (United States) weekly                       | 16                   | STOR_USA_w              |
| Storage Differentials production region weekly             | 17                   | STOR_PROD_ch_w          |
| Storage Differentials total (United States) weekly         | 18                   | STOR_USA_ch_w           |
| Shut in natural gas production in Gold of Mexico weekly    | 19                   | HH_shutin_w             |

### 5.3.1 Model Identification

Below, model identifications for different time horizons and different data (local and global are presented). The number in column lag indicates lags of crude oil variable (o\_WTI\_w) included in the model. Similar to the ARIMA and ARIMAX forecasts, the best predicting power expose models predicting into short-time horizon (4 values versus 12 values).

Similarly results for global data were obtained (Table 5.31).



Table 5.29: Transfer function models for Henry Hub Natural Gas Spot Price - local variables, weekly data, predicting 4 values

| Series                                                                                    | Model            | lag | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE  |
|-------------------------------------------------------------------------------------------|------------------|-----|----------|----------|--------|--------|-------|-------|
| Transfer function ARIMAX local; Est. period 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                         | TF ARIMAX(5,1,3) | 1   | 1024.451 | 1110.367 | 0.426  | 5.512  | 0.158 | 3.208 |
| 2                                                                                         | TF ARIMAX(5,1,2) | 1   | 1032.866 | 1114.009 | 0.428  | 5.517  | 0.096 | 1.886 |
| Transfer function ARIMAX local; Est. period 28-Feb-03 - 2-May-14 (583 obs); Pred.4 values |                  |     |          |          |        |        |       |       |
| 1                                                                                         | TF ARIMAX(5,1,2) | 1   | 681.626  | 755.885  | 0.422  | 5.114  | 0.094 | 1.766 |
| 2                                                                                         | TF ARIMAX(4,1,2) | 1   | 691.342  | 761.233  | 0.426  | 5.069  | 0.098 | 1.832 |
| Transfer function ARIMAX local; Est. period 9-Dec-05 - 2-May-14 (439 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                         | TF ARIMAX(5,1,4) | 1   | 366.524  | 444.130  | 0.350  | 4.742  | 0.162 | 2.868 |
| 2                                                                                         | TF ARIMAX(1,1,1) | 1   | 370.304  | 419.318  | 0.359  | 4.790  | 0.168 | 3.321 |
| 3                                                                                         | TF ARIMAX(5,1,2) | 1   | 371.352  | 440.788  | 0.355  | 4.816  | 0.120 | 2.018 |
| Transfer function ARIMAX local; Est. period 4-Jul-08 - 2-May-14 (305 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                         | TF ARIMAX(3,1,0) | 1   | 97.801   | 146.165  | 0.272  | 4.692  | 0.185 | 3.582 |
| 2                                                                                         | TF ARIMAX(2,1,2) | 2   | 98.642   | 154.447  | 0.270  | 4.667  | 0.206 | 4.044 |
| 3                                                                                         | TF ARIMAX(5,1,3) | 2   | 99.636   | 170.322  | 0.266  | 4.577  | 0.179 | 3.625 |
| 4                                                                                         | TF ARIMAX(5,1,0) | 1   | 99.684   | 155.489  | 0.271  | 4.708  | 0.142 | 2.733 |
| 5                                                                                         | TF ARIMAX(2,1,3) | 2   | 100.824  | 160.349  | 0.271  | 4.676  | 0.212 | 4.187 |
| Transfer function ARIMAX local; Est. period 4-Sep-09 - 2-May-14 (244 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                         | TF ARIMAX(3,1,2) | 1   | 24.302   | 76.760   | 0.237  | 4.551  | 0.130 | 2.253 |
| 2                                                                                         | TF ARIMAX(2,1,3) | 2   | 29.669   | 85.624   | 0.240  | 4.589  | 0.187 | 3.620 |
| 3                                                                                         | TF ARIMAX(5,1,0) | 2   | 29.915   | 85.870   | 0.240  | 4.537  | 0.115 | 1.977 |
| 4                                                                                         | TF ARIMAX(5,1,3) | 2   | 37.885   | 104.331  | 0.240  | 4.516  | 0.146 | 3.029 |

Table 5.30: Transfer function models for Henry Hub Natural Gas Spot Price - local variables, weekly data, predicting 12 values

| Series                                                                                     | Model            | lag | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE   |
|--------------------------------------------------------------------------------------------|------------------|-----|----------|----------|--------|--------|-------|--------|
| Transfer function ARIMAX local; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                          | TF ARIMAX(5,1,3) | 1   | 1014.962 | 1100.712 | 0.426  | 5.491  | 1.219 | 26.257 |
| 2                                                                                          | TF ARIMAX(5,1,2) | 1   | 1022.485 | 1103.471 | 0.428  | 5.490  | 1.330 | 28.633 |
| Transfer function ARIMAX local; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.12 values |                  |     |          |          |        |        |       |        |
| 1                                                                                          | TF ARIMAX(5,1,2) | 1   | 669.730  | 743.754  | 0.420  | 5.058  | 1.306 | 27.824 |
| 2                                                                                          | TF ARIMAX(4,1,2) | 1   | 677.964  | 747.634  | 0.423  | 5.040  | 1.035 | 20.235 |
| Transfer function ARIMAX local; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                          | TF ARIMAX(5,1,4) | 1   | 345.153  | 422.409  | 0.343  | 4.723  | 1.621 | 35.132 |
| 2                                                                                          | TF ARIMAX(1,1,1) | 1   | 351.443  | 400.237  | 0.354  | 4.753  | 1.684 | 36.573 |
| 3                                                                                          | TF ARIMAX(5,1,2) | 2   | 352.335  | 425.525  | 0.348  | 4.751  | 1.158 | 22.193 |
| Transfer function ARIMAX local; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                          | TF ARIMAX(3,1,0) | 2   | 72.029   | 123.741  | 0.260  | 4.519  | 1.522 | 32.864 |
| 2                                                                                          | TF ARIMAX(1,1,1) | 1   | 72.809   | 117.134  | 0.262  | 4.542  | 1.463 | 31.664 |
| 3                                                                                          | TF ARIMAX(2,1,0) | 2   | 73.125   | 121.144  | 0.262  | 4.545  | 1.591 | 34.604 |
| 4                                                                                          | TF ARIMAX(2,1,2) | 3   | 73.244   | 132.344  | 0.259  | 4.493  | 1.608 | 34.939 |
| 5                                                                                          | TF ARIMAX(3,1,2) | 1   | 74.135   | 129.541  | 0.260  | 4.517  | 1.448 | 31.218 |
| Transfer function ARIMAX local; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                          | TF ARIMAX(5,1,3) | 1   | -14.060  | 48.289   | 0.214  | 4.151  | 1.543 | 33.464 |
| 2                                                                                          | TF ARIMAX(3,1,3) | 1   | -12.853  | 42.568   | 0.217  | 4.146  | 1.373 | 29.696 |
| 3                                                                                          | TF ARIMAX(5,1,0) | 1   | -3.209   | 48.748   | 0.225  | 4.296  | 1.548 | 33.528 |

Table 5.31: Transfer function models for Henry Hub Natural Gas Spot Price - global variables, weekly data, predicting 4 values

| Series                                                                                     | Model            | lag | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE  |
|--------------------------------------------------------------------------------------------|------------------|-----|----------|----------|--------|--------|-------|-------|
| Transfer function ARIMAX global; Est. period 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                          | TF ARIMAX(5,1,3) | 3   | 1038.349 | 1133.811 | 0.428  | 5.448  | 0.134 | 2.490 |
| 2                                                                                          | TF ARIMAX(5,1,2) | 2   | 1042.326 | 1128.241 | 0.430  | 5.451  | 0.210 | 3.968 |
| 3                                                                                          | TF ARIMAX(1,1,1) | 2   | 1057.281 | 1119.331 | 0.437  | 5.388  | 0.117 | 2.394 |
| Transfer function ARIMAX global; Est. period 28-Feb-03 - 2-May-14 (583 obs); Pred.4 values |                  |     |          |          |        |        |       |       |
| 1                                                                                          | TF ARIMAX(5,1,2) | 2   | 696.356  | 774.983  | 0.426  | 5.101  | 0.196 | 3.875 |
| 2                                                                                          | TF ARIMAX(4,1,2) | 1   | 706.657  | 776.548  | 0.431  | 5.097  | 0.102 | 1.861 |
| 3                                                                                          | TF ARIMAX(1,1,1) | 1   | 715.342  | 767.760  | 0.438  | 5.034  | 0.103 | 1.903 |
| Transfer function ARIMAX global; Est. period 9-Dec-05 - 2-May-14 (439 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                          | TF ARIMAX(5,1,4) | 2   | 364.923  | 446.613  | 0.348  | 4.878  | 0.109 | 2.260 |
| 2                                                                                          | TF ARIMAX(1,1,1) | 1   | 371.520  | 420.534  | 0.359  | 4.836  | 0.098 | 1.838 |
| 3                                                                                          | TF ARIMAX(5,1,2) | 3   | 375.192  | 452.798  | 0.355  | 4.889  | 0.220 | 3.991 |
| Transfer function ARIMAX global; Est. period 4-Jul-08 - 2-May-14 (305 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                          | TF ARIMAX(1,1,1) | 1   | 109.441  | 154.085  | 0.278  | 4.648  | 0.096 | 1.779 |
| 2                                                                                          | TF ARIMAX(3,1,0) | 1   | 110.533  | 158.897  | 0.278  | 4.725  | 0.095 | 1.739 |
| 3                                                                                          | TF ARIMAX(2,1,3) | 2   | 110.866  | 170.391  | 0.275  | 4.657  | 0.118 | 2.221 |
| 4                                                                                          | TF ARIMAX(5,1,0) | 2   | 111.045  | 170.570  | 0.275  | 4.688  | 0.109 | 2.146 |
| 5                                                                                          | TF ARIMAX(3,1,2) | 2   | 113.001  | 172.526  | 0.276  | 4.681  | 0.116 | 1.945 |
| Transfer function ARIMAX global; Est. period 4-Sep-09 - 2-May-14 (244 obs); Pred.4 values  |                  |     |          |          |        |        |       |       |
| 1                                                                                          | TF ARIMAX(3,1,2) | 1   | 43.527   | 95.985   | 0.247  | 4.503  | 0.089 | 1.697 |
| 2                                                                                          | TF ARIMAX(2,1,3) | 1   | 44.361   | 96.818   | 0.248  | 4.529  | 0.091 | 1.695 |
| 3                                                                                          | TF ARIMAX(5,1,2) | 1   | 44.451   | 103.903  | 0.245  | 4.447  | 0.098 | 1.825 |
| 4                                                                                          | TF ARIMAX(5,1,0) | 3   | 47.903   | 107.355  | 0.248  | 4.539  | 0.118 | 2.404 |

Table 5.32: Transfer function models for Henry Hub Natural Gas Spot Price - global variables, weekly data, predicting 12 values

| Series                                                                                      | Model            | lag | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE   |
|---------------------------------------------------------------------------------------------|------------------|-----|----------|----------|--------|--------|-------|--------|
| Transfer function ARIMAX global; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                           | TF ARIMAX(5,1,3) | 1   | 1023.544 | 1109.294 | 0.428  | 5.420  | 1.336 | 28.523 |
| 2                                                                                           | TF ARIMAX(5,1,2) | 1   | 1028.782 | 1109.768 | 0.430  | 5.431  | 1.422 | 30.343 |
| 3                                                                                           | TF ARIMAX(1,1,1) | 1   | 1043.156 | 1100.323 | 0.436  | 5.359  | 1.376 | 29.479 |
| Transfer function ARIMAX global; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.12 values |                  |     |          |          |        |        |       |        |
| 1                                                                                           | TF ARIMAX(5,1,2) | 1   | 680.724  | 754.749  | 0.425  | 5.056  | 1.490 | 31.622 |
| 2                                                                                           | TF ARIMAX(4,1,2) | 1   | 687.635  | 757.305  | 0.427  | 5.026  | 1.111 | 20.875 |
| Transfer function ARIMAX global; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                           | TF ARIMAX(5,1,4) | 3   | 347.325  | 432.713  | 0.343  | 4.805  | 1.754 | 37.866 |
| 2                                                                                           | TF ARIMAX(5,1,2) | 1   | 354.124  | 423.248  | 0.351  | 4.818  | 1.841 | 39.702 |
| Transfer function ARIMAX global; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                           | TF ARIMAX(2,1,2) | 1   | 77.678   | 129.390  | 0.263  | 4.525  | 1.570 | 33.637 |
| 2                                                                                           | TF ARIMAX(3,1,2) | 1   | 79.918   | 135.324  | 0.263  | 4.532  | 1.540 | 32.873 |
| 3                                                                                           | TF ARIMAX(1,1,1) | 1   | 80.429   | 124.754  | 0.266  | 4.570  | 1.571 | 33.704 |
| Transfer function ARIMAX global; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred.12 values  |                  |     |          |          |        |        |       |        |
| 1                                                                                           | TF ARIMAX(2,1,3) | 2   | 7.123    | 62.544   | 0.227  | 4.229  | 1.274 | 25.646 |
| 2                                                                                           | TF ARIMAX(5,1,2) | 1   | 7.443    | 66.329   | 0.227  | 4.248  | 1.378 | 28.477 |
| 3                                                                                           | TF ARIMAX(5,1,0) | 2   | 9.940    | 65.361   | 0.230  | 4.322  | 1.682 | 36.269 |
| 4                                                                                           | TF ARIMAX(3,1,3) | 2   | 9.948    | 68.833   | 0.229  | 4.318  | 1.598 | 34.412 |

### 5.3.2 Model Estimation

Table 5.33 presents Transfer Function model for local data (all variables) including 2 lags of West Texas Intermediate Crude Oil Sport Price (o\_WTI\_w) with 2 lags. Estimation period covers time horizon from August 1, 1997 to May 2, 2014 (1 - 874 observations).

Table 5.33: Transfer function model for Henry Hub Natural Gas Spot Price - local variables, 2 lags of o\_WTI\_w, weekly data

|                                 |           |              |                      |             |           |
|---------------------------------|-----------|--------------|----------------------|-------------|-----------|
| Coefficients:                   | AR(1)     | AR(2)        | AR(3)                | AR(4)       | MA(1)     |
|                                 | 0.742     | 0.171        | -0.034               | -0.020      | -0.743    |
| s.e.                            | 1.163     | 1.118        | 0.065                | 0.048       | 1.163     |
| Coefficients:                   | MA(2)     | o_WTI_w      | WTI.1_w              | WTI.2_w     | HDD_WSC_w |
|                                 | -0.203    | 0.039        | 0.005                | -0.002      | 0.002     |
| s.e.                            | 1.118     | 0.006        | 0.006                | 0.006       | 0.002     |
| HDDdev_WSC_w                    | CDD_WSC_w | CDDdev_WSC_w | STOR_PROD_ch_w       | HH_shutin_w | 1:nobs    |
| 0.002                           | -0.002    | 0.005        | -2.591               | 0.0001      | -0.003    |
| 0.0016                          | 0.002     | 0.003        | 0.443                | 0.0000      | 0.001     |
| $\sigma^2$ estimated as 0.184 : |           |              | loglikelihood = -500 |             |           |

Table 5.34: Transfer function model for Henry Hub Natural Gas Spot Price - global variables, 2 lags of o\_WTI\_w, weekly data

|                                  |           |              |                         |             |           |
|----------------------------------|-----------|--------------|-------------------------|-------------|-----------|
| Coefficients:                    | AR(1)     | AR(2)        | AR(3)                   | AR(4)       | MA(1)     |
|                                  | 0.882     | -0.133       | 0.017                   | -0.076      | -0.874    |
| s.e.                             | 0.270     | 0.336        | 0.052                   | 0.053       | 0.269     |
| Coefficients:                    | MA(2)     | o_WTI_w      | WTI.1_w                 | WTI.2_w     | HDD_USA_w |
|                                  | 0.077     | 0.038        | 0.003                   | 0.000       | 0.002     |
| s.e.                             | 0.333     | 0.006        | 0.006                   | 0.006       | 0.001     |
| HDDdev_USA_w                     | CDD_USA_w | CDDdev_USA_w | STOR_USA_ch_w           | HH_shutin_w | 1:nobs    |
| 0.0027                           | -0.003    | 0.010        | -3.104                  | 0.0001      | 0.001     |
| 0.0012                           | 0.004     | 0.004        | 0.620                   | 0.0000      | 0.001     |
| $\sigma^2$ estimated as 0.1865 : |           |              | loglikelihood = -505.89 |             |           |

## 5.4 ECM

The Error Correction Model is usually applied whenever long lasting relationship (cointegration) between explanatory and response variables is discovered. In the case of natural gas price, a good candidate for such a factor is crude oil price. In fact, West Texas Intermediate Spot Price is cointegrated with Henry Hub Natural Gas Spot Price.

In order to be cointegrated, two variables must be integrated at the same level and residuals of their linear combination must be stationary (residuals in level should be  $I(0)$ ).

Knowing from the previous section that Henry Hub Natural Gas Spot Price (weekly data) is  $I(1)$ , West Texas Intermediate should also be  $I(1)$ .

The methodology of verifying the stationarity is exactly the same as for natural gas variable (Section 5.1).

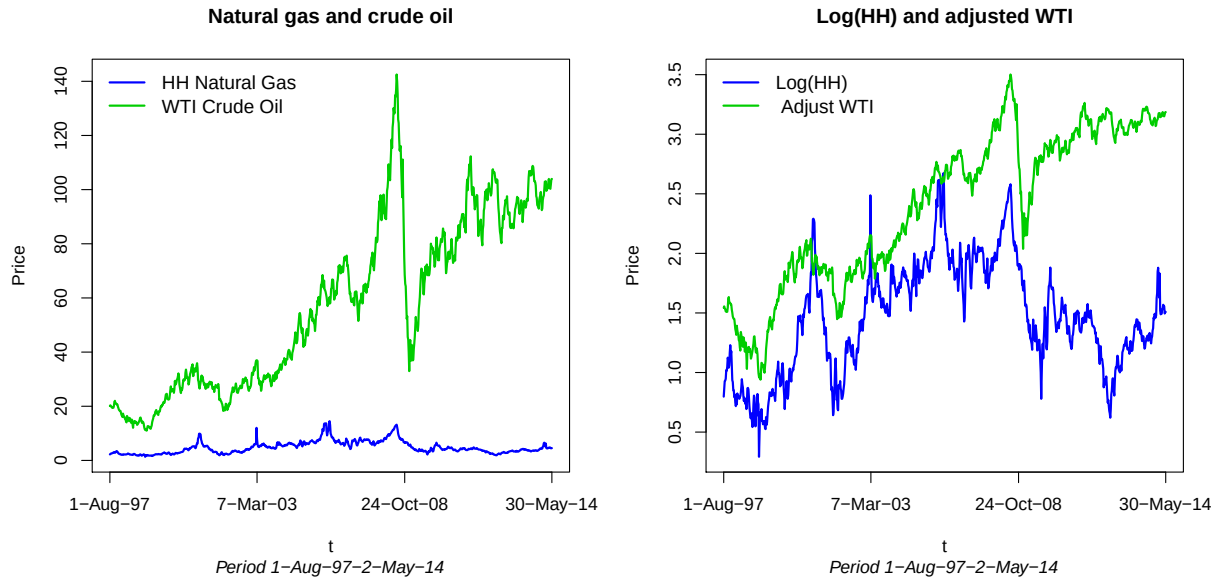


Figure 5.16: Henry Hub Natural Gas Spot Price and West Texas Intermediate Crude Oil Spot Price - weekly data

Adjusted crude oil price:  $AdjustedCrudeOilPrice = \log(WTICrudeOil) - \text{mean}(\log(HenryHubNaturalGasPrice))$ .

### ACF and PACF

As mentioned above, the first step is to use graphical data analysis. Looking at our data plot (Figure 5.17) - West Texas Intermediate Crude Oil Spot Price on a monthly base, one can see that there is strong trend and high volatility, suggesting that our price time series is not stationary.

The next step is to plot autocorrelation (ACF) and partial autocorrelation (PACF) functions.

The autocorrelation function (Figure 5.18) shows that several autocorrelations are persistently large and trend to zero rather slowly. This might indicate existing trend and non-stationarity.

### Unit root tests

Plots of standard deviations for level (original data), first and second differences also indicate that the data is  $I(1)$ . A huge decline can be seen in the standard deviation of level and first difference (Figure 5.19).

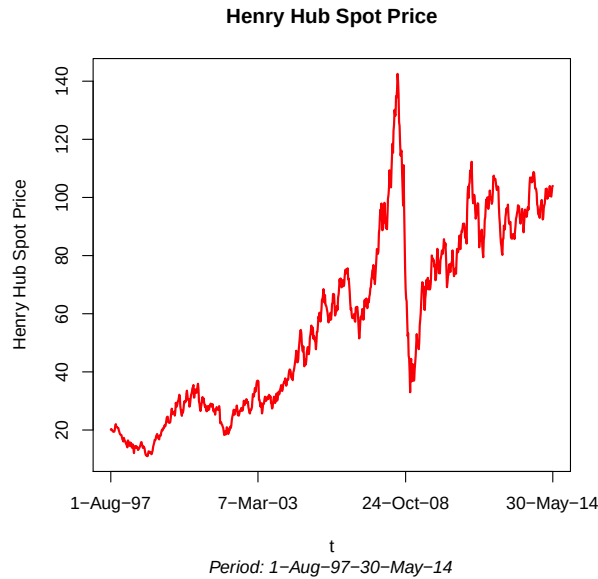


Figure 5.17: West Texas Intermediate Crude Oil Spot Price - weekly data

Table 5.35: Unit root tests for West Texas Intermediate Crude Oil - weekly data

| West Texas Intermediate Crude Oil |            |         |            |         |            |         |
|-----------------------------------|------------|---------|------------|---------|------------|---------|
| Number of lag                     | ADF        |         | PP         |         | KPSS       |         |
|                                   | statistics | p-value | statistics | p-value | statistics | p-value |
| 0                                 | -2.756     | 0.218   | -2.756     | 0.218   | 2.147      | 0.010   |
| 1                                 | -3.298     | 0.067   | -2.986     | 0.137   | 1.083      | 0.010   |
| 2                                 | -3.289     | 0.069   | -3.086     | 0.111   | 0.727      | 0.010   |
| 6                                 | -3.783     | 0.018   | -3.345     | 0.060   | 0.322      | 0.010   |
| 9                                 | -4.612     | 0.001   | -3.546     | 0.035   | 0.232      | 0.010   |
| 20                                | -4.457     | 0.003   | -3.886     | 0.013   | 0.126      | 0.087   |
| 30                                | -3.629     | 0.028   | -3.810     | 0.017   | 0.099      | 0.100   |

### Henry Hub Spot Price I(1)

A plot of the differenced data appears to vary about a fixed level. Also, unit roots tests show that there is significant evidence against the null hypothesis (non-stationary data) in the case of the ADF and PP test. In the case of KPSS test the data is not statistically significant (so we cannot reject the null hypothesis - stationarity of the time series).

The West Texas Intermediate Crude Oil Spot Price series appears to be an I(1) process. Further analysis proved that the residuals of regression are I(0). Those facts indicate that Henry Hub Natural Gas Spot Price and West Texas Intermediate Crude Oil Spot Price are cointegrated. In this case, the Error Correction Model can be applied.

Confidence Intervals for different ECM models are presented on the Figure

Results of the Error Correction Model are presented in the tables below ( Table 5.38 and Table 5.39).

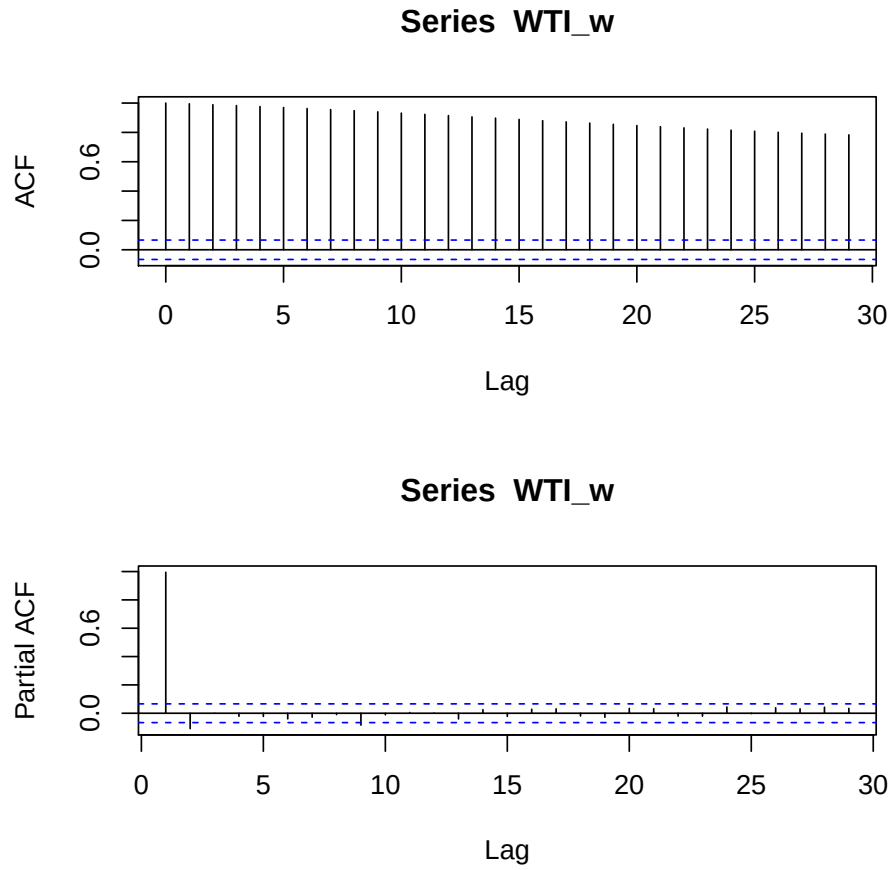


Figure 5.18: Autocorrelation and partial autocorrelation function for West Texas Intermediate Crude Oil Spot Price

Table 5.36: Unit root tests for first difference of West Texas Intermediate Crude Oil Spot Price - weekly data

| First difference of Henry Hub Natural Gas Spot Price |            |         |            |         |            |         |
|------------------------------------------------------|------------|---------|------------|---------|------------|---------|
| Number of<br>lag                                     | ADF        |         | PP         |         | KPSS       |         |
|                                                      | statistics | p-value | statistics | p-value | statistics | p-value |
| 0                                                    | -24.625    | 0.001   | -24.625    | 0.001   | 0.040      | 0.100   |
| 1                                                    | -19.006    | 0.001   | -24.627    | 0.001   | 0.034      | 0.100   |
| 2                                                    | -15.301    | 0.001   | -24.608    | 0.001   | 0.032      | 0.100   |
| 6                                                    | -9.771     | 0.001   | -24.789    | 0.001   | 0.027      | 0.100   |
| 9                                                    | -7.516     | 0.001   | -25.072    | 0.001   | 0.024      | 0.100   |
| 20                                                   | -6.341     | 0.001   | -25.623    | 0.001   | 0.021      | 0.100   |
| 30                                                   | -6.552     | 0.001   | -25.354    | 0.001   | 0.022      | 0.100   |

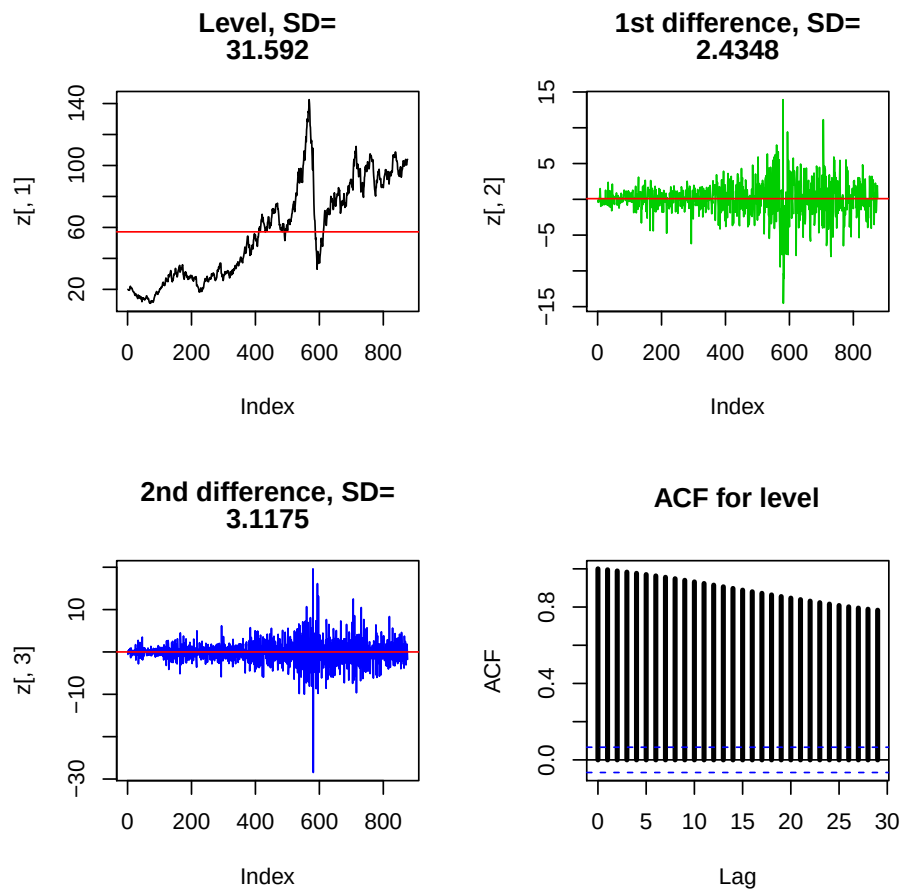


Figure 5.19: Standard deviations for the data, first and second differences, autocorrelation function



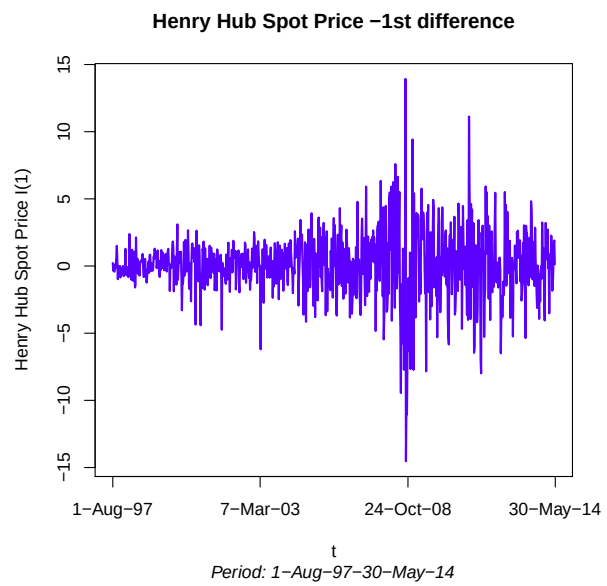


Figure 5.20: First difference ( $I(1)$ ) of West Texas Intermediate Crude Oil Spot Price - monthly data

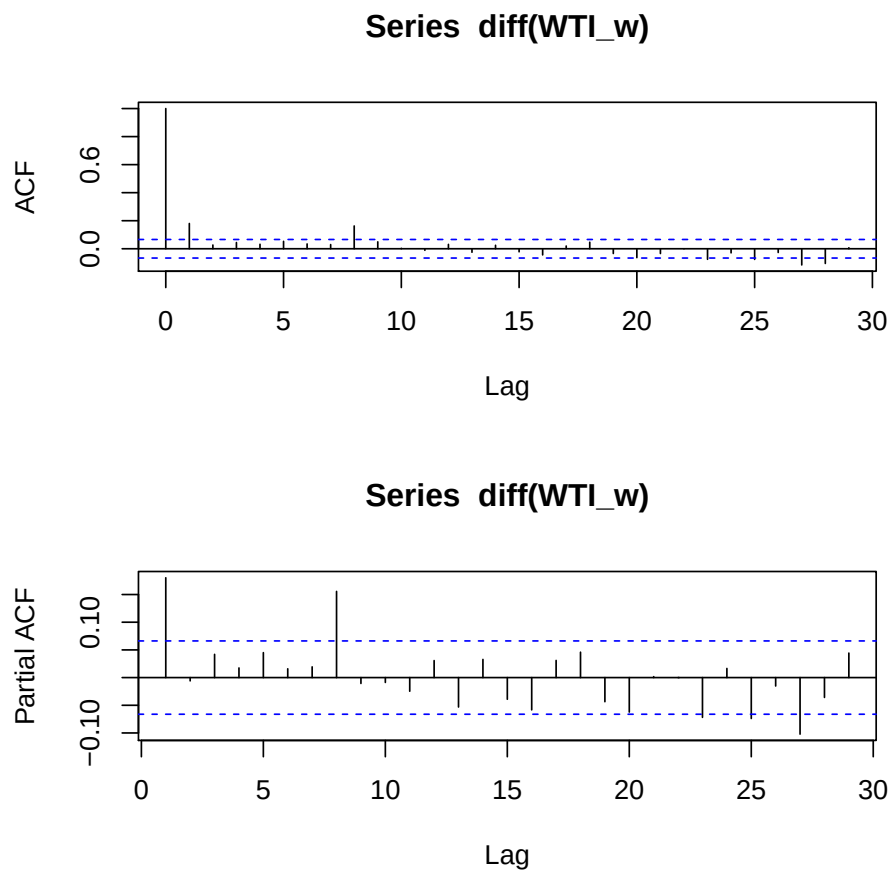


Figure 5.21: Autocorrelation and partial autocorrelation function for first difference ( $I(1)$ ) Henry Hub monthly spot prices

Table 5.37: ECM model for Henry Hub Natural Gas Spot Price and West Texas Intermediate Crude Oil Spot Price - weekly data

| Residuals:                                                      |            |            |         |                 |
|-----------------------------------------------------------------|------------|------------|---------|-----------------|
| Min                                                             | 1Q         | Median     | 3Q      | Max             |
| -4.5058                                                         | -0.1775    | -0.0026    | 0.1510  | 5.7264          |
| Coefficients:                                                   | Estimate   | Std. Error | t-value | Pr(> t )        |
| (Intercept)                                                     | -0.0015405 | 0.0157017  | -0.098  | 0.92187         |
| eHH[, 2:p]1                                                     | 0.0708761  | 0.0346167  | 2.047   | 0.04092         |
| eHH[, 2:p]2                                                     | -0.0104922 | 0.0347981  | -0.302  | 0.76309         |
| eHH[, 2:p]3                                                     | -0.002225  | 0.0347553  | -0.064  | 0.94897         |
| eHH[, 2:p]4                                                     | -0.064063  | 0.034769   | -1.843  | 0.06575         |
| eHH[, 2:p]5                                                     | -0.0981233 | 0.0348146  | -2.818  | 0.00494         |
| eHH[, 2:p]6                                                     | -0.0313814 | 0.0349253  | -0.899  | 0.36916         |
| eHH[, 2:p]7                                                     | 0.0459461  | 0.0346864  | 1.325   | 0.18566         |
| eHH[, 2:p]8                                                     | 0.0016542  | 0.034565   | 0.048   | 0.96184         |
| eHH[, 2:p]9                                                     | -0.0453263 | 0.0346433  | -1.308  | 0.19111         |
| eHH[, 2:p]10                                                    | 0.0144482  | 0.0346273  | 0.417   | 0.6766          |
| eHH[, 2:p]11                                                    | 0.0223517  | 0.0346209  | 0.646   | 0.51871         |
| eWTI[, 1:p]1                                                    | 0.0320907  | 0.0066573  | 4.82    | $1.7 * 10^{-6}$ |
| eWTI[, 1:p]2                                                    | 0.0057842  | 0.0068456  | 0.845   | 0.39838         |
| eWTI[, 1:p]3                                                    | -0.0083583 | 0.0068517  | -1.22   | 0.22285         |
| eWTI[, 1:p]4                                                    | 0.0013541  | 0.0068518  | 0.198   | 0.84338         |
| eWTI[, 1:p]5                                                    | -0.0038577 | 0.0067784  | -0.569  | 0.56942         |
| eWTI[, 1:p]6                                                    | -0.0075744 | 0.0067814  | -1.117  | 0.26434         |
| eWTI[, 1:p]7                                                    | 0.0087057  | 0.0067888  | 1.282   | 0.20007         |
| eWTI[, 1:p]8                                                    | 0.0023364  | 0.00679    | 0.344   | 0.73086         |
| eWTI[, 1:p]9                                                    | 0.0068782  | 0.006857   | 1.003   | 0.3161          |
| eWTI[, 1:p]10                                                   | -0.0003433 | 0.0068455  | -0.05   | 0.96002         |
| eWTI[, 1:p]11                                                   | 0.0079252  | 0.0068229  | 1.162   | 0.24574         |
| eWTI[, 1:p]12                                                   | -0.0034969 | 0.0067362  | -0.519  | 0.60382         |
| ecm                                                             | -0.0198736 | 0.0075228  | -2.642  | 0.0084          |
| Residual standard error: 0.4597 on 841 degrees of freedom       |            |            |         |                 |
| Multiple R-squared: 0.0811, Adjusted R-squared: 0.05488         |            |            |         |                 |
| F-statistic: 3.093 on 24 and 841 DF, p-value: $1.064 * 10^{-6}$ |            |            |         |                 |

Table 5.38: ECM models for Henry Hub Natural Gas Spot Price - weekly data, predicting 4 values

| Series                                                               | Model               | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE  |
|----------------------------------------------------------------------|---------------------|----------|----------|--------|--------|-------|-------|
| ECM model; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                                    | ECM $r = 1$ $k = 2$ | 6864.359 | 6897.142 | 0.466  | 5.368  | 0.258 | 5.056 |
| ECM model; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.4 values |                     |          |          |        |        |       |       |
| 1                                                                    | ECM $r = 1$ $k = 2$ | 4948.152 | 4978.500 | 0.436  | 5.088  | 0.211 | 4.064 |
| ECM model; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                                    | ECM $r = 1$ $k = 2$ | 3881.184 | 3909.823 | 0.376  | 5.068  | 0.253 | 4.959 |
| ECM model; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                                    | ECM $r = 1$ $k = 2$ | 2704.045 | 2730.486 | 0.279  | 4.800  | 0.084 | 1.667 |
| ECM model; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred.4 values  |                     |          |          |        |        |       |       |
| 1                                                                    | ECM $r = 1$ $k = 2$ | 1963.502 | 1988.594 | 0.268  | 4.523  | 0.081 | 1.458 |

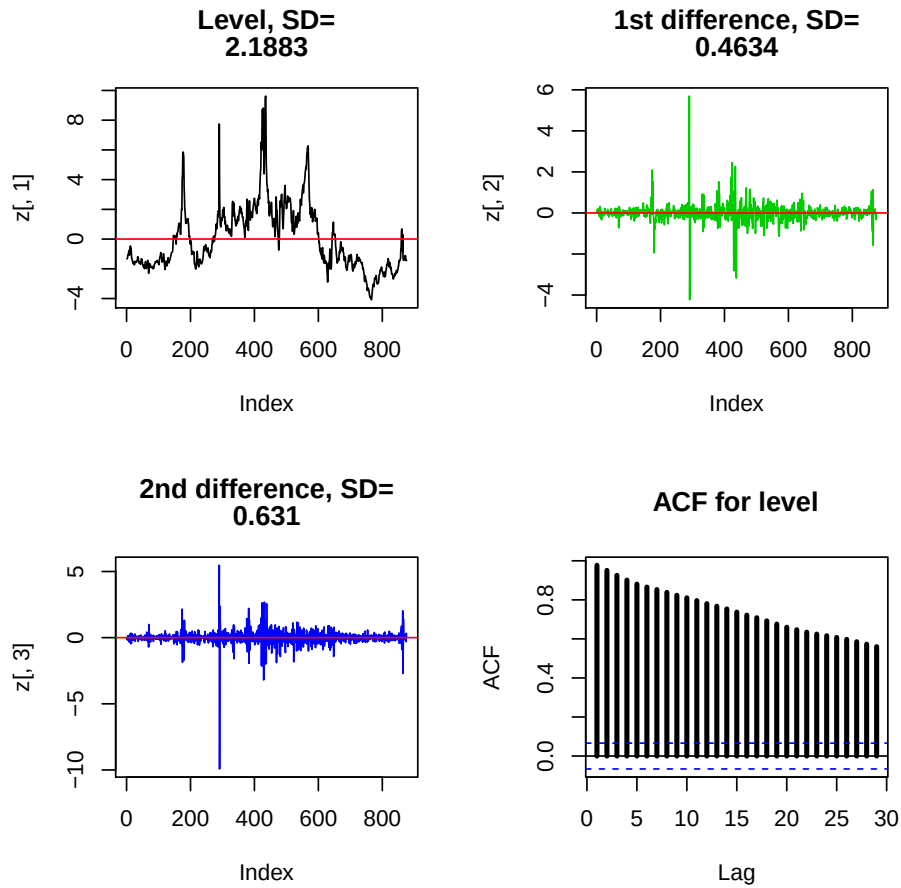


Figure 5.22: Standard deviations for the data, first and second differences, autocorrelation function for residuals of regression (Henry Hub Natural Gas Spot Price on West Texas Intermediate Crude Oil Spot Price)

Table 5.39: ECM models for Henry Hub Natural Gas Spot Price - weekly data, predicting 12 values

| Series                                                                | Model               | AIC      | BIC      | RMSEin | MAPEin | RMSE  | MAPE   |
|-----------------------------------------------------------------------|---------------------|----------|----------|--------|--------|-------|--------|
| ECM model; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                     | ECM $r = 1$ $k = 2$ | 6807.784 | 6840.512 | 0.464  | 5.357  | 1.701 | 37.096 |
| ECM model; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.12 values |                     |          |          |        |        |       |        |
| 1                                                                     | ECM $r = 1$ $k = 2$ | 4889.095 | 4919.359 | 0.432  | 5.067  | 1.770 | 38.472 |
| ECM model; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                     | ECM $r = 1$ $k = 2$ | 3820.445 | 3848.972 | 0.367  | 5.045  | 1.815 | 39.493 |
| ECM model; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                     | ECM $r = 1$ $k = 2$ | 2644.362 | 2670.642 | 0.265  | 4.705  | 1.021 | 19.426 |
| ECM model; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred.12 values  |                     |          |          |        |        |       |        |
| 1                                                                     | ECM $r = 1$ $k = 2$ | 1908.672 | 1933.563 | 0.249  | 4.346  | 1.448 | 31.151 |

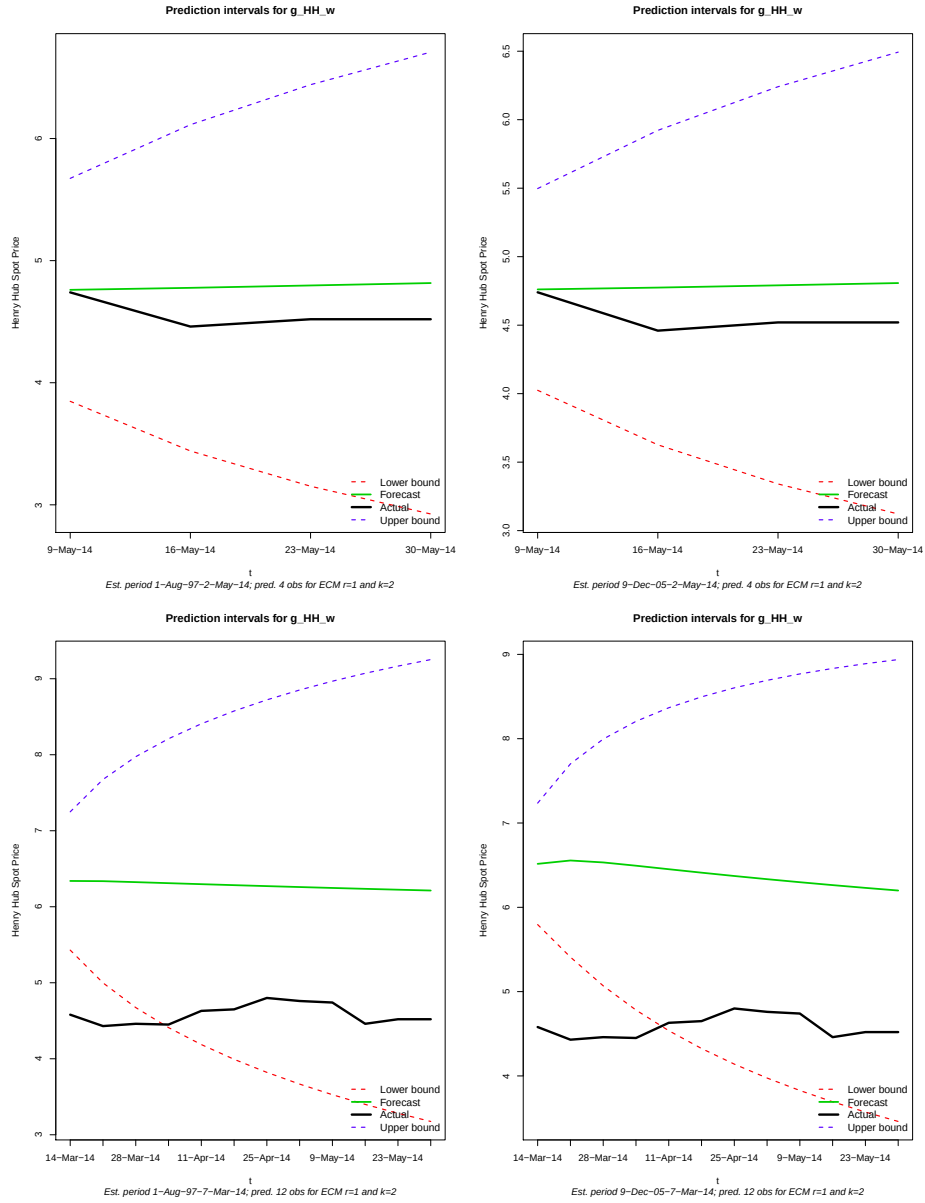


Figure 5.23: Fitted and actual values for Henry Hub Natural Gas Spot Price - weekly

## 5.5 Model comparison

There are many different methods of choosing the best criterion for model comparison (see Section 4.3). Since the main focus of this thesis is natural gas price prediction, the most important feature of the model is its predictive ability. For this reason, error measurements (RMSE and MAPE) were chosen for the model selection.

Tables below present forecasting results (forecasting errors) for different models, organized by time horizons (for identification and forecasting sets). As mentioned above, time series have the best predictive power for short-time horizons. It can be easily seen by looking at listed tables. Surprisingly, for midpoint identification and estimation data set (half of the available time horizon), the best models occur to be ARIMA models. For the longer predictions (12 values, quarter-ahead prediction) ARIMAX/Transfer Function models and ECM exhibit better predicting abilities. The underlying reason can be that the gas price depends also on other factors (like crude oil price, weather and storage) and longer prediction can capture this relationship (especially ECM model). Data presented in following tables are sorted by RMSE (Root Mean Square Error) since this was my model selection criterion.

Table 5.40: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 1-874, predicting 4 values

| Series                                                        | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE  |
|---------------------------------------------------------------|--------------------|---------------|--------|--------|-------|-------|
| Estimation comp; 1-Aug-97 - 2-May-14 (874 obs); Pred.4 values |                    |               |        |        |       |       |
| 1                                                             | Trans Fun local    | ARIMAX(5,1,2) | 0.428  | 5.517  | 0.096 | 1.886 |
| 2                                                             | ARIMAX local       | ARIMAX(5,1,2) | 0.429  | 5.517  | 0.098 | 1.937 |
| 3                                                             | Trans fun global 2 | ARIMAX(1,1,1) | 0.437  | 5.388  | 0.117 | 2.394 |
| 4                                                             | Trans fun global 3 | ARIMAX(5,1,3) | 0.428  | 5.448  | 0.134 | 2.490 |
| 5                                                             | ARIMAX globabl     | ARIMAX(5,1,3) | 0.429  | 5.445  | 0.140 | 2.595 |
| 6                                                             | Trans Fun local    | ARIMAX(5,1,3) | 0.426  | 5.512  | 0.158 | 3.208 |
| 7                                                             | ARIMAX local       | ARIMAX(5,1,3) | 0.426  | 5.517  | 0.163 | 3.288 |
| 8                                                             | ARIMA              | ARIMA(4,1,2)  | 0.465  | 5.346  | 0.180 | 3.488 |
| 9                                                             | ARIMA              | ARIMA(5,1,2)  | 0.464  | 5.356  | 0.192 | 3.816 |
| 10                                                            | Trans fun global 2 | ARIMAX(5,1,2) | 0.430  | 5.451  | 0.210 | 3.968 |
| 11                                                            | ARIMA              | ARIMA(2,1,2)  | 0.466  | 5.329  | 0.216 | 4.242 |
| 12                                                            | ARIMAX globabl     | ARIMAX(5,1,2) | 0.430  | 5.463  | 0.234 | 4.469 |
| 13                                                            | ECM                | ECM r=1 k=2   | 0.466  | 5.368  | 0.258 | 5.056 |
| 14                                                            | ECM                | ECM r=1 k=2   | 0.466  | 5.368  | 0.258 | 5.056 |
| 15                                                            | ARIMA              | ARIMA(5,1,3)  | 0.462  | 5.379  | 0.258 | 5.088 |

Table 5.41: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 292-874, predicting 4 values

| Series                                                         | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE  |
|----------------------------------------------------------------|--------------------|---------------|--------|--------|-------|-------|
| Estimation comp; 28-Feb-03 - 2-May-14 (583 obs); Pred.4 values |                    |               |        |        |       |       |
| 1                                                              | ARIMA              | ARIMA(4,1,2)  | 0.462  | 5.217  | 0.083 | 1.697 |
| 2                                                              | ARIMAX local       | ARIMAX(5,1,2) | 0.422  | 5.113  | 0.091 | 1.871 |
| 3                                                              | Trans Fun local 1  | ARIMAX(5,1,2) | 0.422  | 5.114  | 0.094 | 1.766 |
| 4                                                              | Trans Fun local 1  | ARIMAX(4,1,2) | 0.426  | 5.069  | 0.098 | 1.832 |
| 5                                                              | Trans fun global 1 | ARIMAX(4,1,2) | 0.431  | 5.097  | 0.102 | 1.861 |
| 6                                                              | Trans fun global 1 | ARIMAX(1,1,1) | 0.438  | 5.034  | 0.103 | 1.903 |
| 7                                                              | ARIMAX globabl     | ARIMAX(4,1,2) | 0.432  | 5.093  | 0.108 | 2.021 |
| 8                                                              | ARIMA              | ARIMA(1,1,5)  | 0.461  | 5.241  | 0.134 | 2.672 |
| 9                                                              | ARIMA              | ARIMA(5,1,2)  | 0.459  | 5.294  | 0.138 | 2.894 |
| 10                                                             | ARIMAX local       | ARIMAX(4,1,2) | 0.427  | 5.089  | 0.143 | 2.943 |
| 11                                                             | Trans fun global 2 | ARIMAX(5,1,2) | 0.426  | 5.101  | 0.196 | 3.875 |
| 12                                                             | ARIMAX globabl     | ARIMAX(5,1,2) | 0.427  | 5.105  | 0.210 | 4.194 |
| 13                                                             | ECM                | ECM r=1 k=2   | 0.436  | 5.088  | 0.211 | 4.064 |

Table 5.42: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 436-874, predicting 4 values

| Series                                                        | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE  |
|---------------------------------------------------------------|--------------------|---------------|--------|--------|-------|-------|
| Estimation comp; 9-Dec-05 - 2-May-14 (439 obs); Pred.4 values |                    |               |        |        |       |       |
| 1                                                             | ARIMA              | ARIMA(4,1,3)  | 0.380  | 5.004  | 0.088 | 1.566 |
| 2                                                             | ARIMA              | ARIMA(5,1,4)  | 0.374  | 5.049  | 0.089 | 1.811 |
| 3                                                             | Trans fun global 1 | ARIMAX(1,1,1) | 0.359  | 4.836  | 0.098 | 1.838 |
| 4                                                             | ARIMAX globabl     | ARIMAX(1,1,1) | 0.360  | 4.845  | 0.100 | 1.808 |
| 5                                                             | Trans fun global 2 | ARIMAX(5,1,4) | 0.348  | 4.878  | 0.109 | 2.260 |
| 6                                                             | ARIMAX globabl     | ARIMAX(5,1,4) | 0.353  | 4.917  | 0.116 | 1.680 |
| 7                                                             | Trans Fun local 1  | ARIMAX(5,1,2) | 0.355  | 4.816  | 0.120 | 2.018 |
| 8                                                             | ARIMAX local       | ARIMAX(5,1,4) | 0.350  | 4.761  | 0.137 | 2.273 |
| 9                                                             | ARIMAX globabl     | ARIMAX(5,1,2) | 0.356  | 4.884  | 0.152 | 3.067 |
| 10                                                            | Trans Fun local 1  | ARIMAX(5,1,4) | 0.350  | 4.742  | 0.162 | 2.868 |
| 11                                                            | ARIMAX local       | ARIMAX(5,1,2) | 0.356  | 4.824  | 0.167 | 3.060 |
| 12                                                            | Trans Fun local 1  | ARIMAX(1,1,1) | 0.359  | 4.790  | 0.168 | 3.321 |
| 13                                                            | ARIMA              | ARIMA(3,1,2)  | 0.382  | 5.007  | 0.169 | 3.204 |
| 14                                                            | ARIMA              | ARIMA(2,1,1)  | 0.382  | 4.997  | 0.172 | 3.270 |
| 15                                                            | ARIMAX local       | ARIMAX(1,1,1) | 0.359  | 4.794  | 0.177 | 3.455 |
| 16                                                            | Trans fun global 3 | ARIMAX(5,1,2) | 0.355  | 4.889  | 0.220 | 3.991 |
| 17                                                            | ECM                | ECM r=1 k=2   | 0.376  | 5.068  | 0.253 | 4.959 |

Table 5.43: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 570-874, predicting 4 values

| Series                                                                    | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE  |
|---------------------------------------------------------------------------|--------------------|---------------|--------|--------|-------|-------|
| Estimation comp; Est. period 4-Sep-09 - 2-May-14 (305 obs); Pred.4 values |                    |               |        |        |       |       |
| 1                                                                         | ECM                | ECM r=1 k=2   | 0.279  | 4.800  | 0.084 | 1.667 |
| 2                                                                         | Trans fun global 1 | ARIMAX(3,1,0) | 0.278  | 4.725  | 0.095 | 1.739 |
| 3                                                                         | Trans fun global 1 | ARIMAX(1,1,1) | 0.278  | 4.648  | 0.096 | 1.779 |
| 4                                                                         | ARIMAX globabl     | ARIMAX(3,1,0) | 0.278  | 4.707  | 0.100 | 1.826 |
| 5                                                                         | ARIMAX globabl     | ARIMAX(2,1,0) | 0.278  | 4.708  | 0.107 | 1.836 |
| 6                                                                         | Trans fun global 2 | ARIMAX(5,1,0) | 0.275  | 4.688  | 0.109 | 2.146 |
| 7                                                                         | Trans fun global 2 | ARIMAX(3,1,2) | 0.276  | 4.681  | 0.116 | 1.945 |
| 8                                                                         | Trans fun global 2 | ARIMAX(2,1,3) | 0.275  | 4.657  | 0.118 | 2.221 |
| 9                                                                         | ARIMAX globabl     | ARIMAX(2,1,3) | 0.277  | 4.678  | 0.118 | 2.465 |
| 10                                                                        | ARIMAX globabl     | ARIMAX(5,1,3) | 0.275  | 4.666  | 0.121 | 1.981 |
| 11                                                                        | ARIMAX globabl     | ARIMAX(5,1,0) | 0.277  | 4.702  | 0.122 | 2.480 |
| 12                                                                        | ARIMAX globabl     | ARIMAX(2,1,2) | 0.277  | 4.642  | 0.125 | 2.629 |
| 13                                                                        | Trans Fun local 1  | ARIMAX(5,1,0) | 0.271  | 4.708  | 0.142 | 2.733 |
| 14                                                                        | ARIMA              | ARIMA(2,1,2)  | 0.297  | 4.855  | 0.146 | 2.886 |
| 15                                                                        | ARIMAX local       | ARIMAX(5,1,0) | 0.271  | 4.699  | 0.151 | 2.854 |
| 16                                                                        | ARIMA              | ARIMA(4,1,2)  | 0.295  | 4.844  | 0.163 | 3.302 |
| 17                                                                        | Trans Fun local 2  | ARIMAX(5,1,3) | 0.266  | 4.577  | 0.179 | 3.625 |
| 18                                                                        | ARIMAX local       | ARIMAX(3,1,0) | 0.272  | 4.690  | 0.184 | 3.530 |
| 19                                                                        | Trans Fun local 1  | ARIMAX(3,1,0) | 0.272  | 4.692  | 0.185 | 3.582 |
| 20                                                                        | ARIMA              | ARIMA(3,1,2)  | 0.296  | 4.828  | 0.189 | 3.881 |
| 21                                                                        | Trans Fun local 2  | ARIMAX(2,1,2) | 0.270  | 4.667  | 0.206 | 4.044 |
| 22                                                                        | Trans Fun local 2  | ARIMAX(2,1,3) | 0.271  | 4.676  | 0.212 | 4.187 |
| 23                                                                        | ARIMAX local       | ARIMAX(2,1,2) | 0.270  | 4.645  | 0.215 | 4.314 |



Table 5.44: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 631-874, predicting 4 values

| Series                                                                    | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE  |
|---------------------------------------------------------------------------|--------------------|---------------|--------|--------|-------|-------|
| Estimation comp; Est. period 4-Sep-09 - 2-May-14 (244 obs); Pred.4 values |                    |               |        |        |       |       |
| 1                                                                         | ECM                | ECM r=1 k=2   | 0.268  | 4.523  | 0.081 | 1.458 |
| 2                                                                         | ARIMAX globabl     | ARIMAX(3,1,0) | 0.252  | 4.542  | 0.086 | 1.563 |
| 3                                                                         | Trans fun global 1 | ARIMAX(3,1,2) | 0.247  | 4.503  | 0.089 | 1.697 |
| 4                                                                         | ARIMA              | ARIMA(5,1,3)  | 0.266  | 4.563  | 0.089 | 1.728 |
| 5                                                                         | ARIMAX globabl     | ARIMAX(3,1,3) | 0.248  | 4.526  | 0.091 | 1.695 |
| 6                                                                         | Trans fun global 1 | ARIMAX(2,1,3) | 0.248  | 4.529  | 0.091 | 1.695 |
| 7                                                                         | ARIMAX globabl     | ARIMAX(5,1,2) | 0.246  | 4.452  | 0.094 | 1.764 |
| 8                                                                         | ARIMAX globabl     | ARIMAX(2,1,3) | 0.249  | 4.542  | 0.095 | 1.667 |
| 9                                                                         | Trans fun global 1 | ARIMAX(5,1,2) | 0.245  | 4.447  | 0.098 | 1.825 |
| 10                                                                        | Trans Fun local 2  | ARIMAX(5,1,0) | 0.240  | 4.537  | 0.115 | 1.977 |
| 11                                                                        | ARIMAX local       | ARIMAX(2,1,3) | 0.238  | 4.525  | 0.117 | 2.133 |
| 12                                                                        | Trans fun global 3 | ARIMAX(5,1,0) | 0.248  | 4.539  | 0.118 | 2.404 |
| 13                                                                        | ARIMAX local       | ARIMAX(5,1,0) | 0.241  | 4.518  | 0.119 | 2.173 |
| 14                                                                        | Trans Fun local 1  | ARIMAX(3,1,2) | 0.237  | 4.551  | 0.130 | 2.253 |
| 15                                                                        | ARIMAX globabl     | ARIMAX(5,1,0) | 0.250  | 4.552  | 0.134 | 2.641 |
| 16                                                                        | Trans Fun local 2  | ARIMAX(5,1,3) | 0.240  | 4.516  | 0.146 | 3.029 |
| 17                                                                        | ARIMA              | ARIMA(1,1,3)  | 0.271  | 4.572  | 0.162 | 3.015 |
| 18                                                                        | ARIMAX local       | ARIMAX(5,1,3) | 0.240  | 4.502  | 0.162 | 3.342 |
| 19                                                                        | Trans Fun local 2  | ARIMAX(2,1,3) | 0.240  | 4.589  | 0.187 | 3.620 |
| 20                                                                        | ARIMA              | ARIMA(4,1,5)  | 0.261  | 4.518  | 0.202 | 4.156 |
| 21                                                                        | ARIMA              | ARIMA(2,1,2)  | 0.270  | 4.555  | 0.276 | 5.391 |

Table 5.45: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 1-866, predicting 12 values

| Series                                                                     | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE   |
|----------------------------------------------------------------------------|--------------------|---------------|--------|--------|-------|--------|
| Estimation comp; Est. period 1-Aug-97 - 7-Mar-14 (866 obs); Pred.12 values |                    |               |        |        |       |        |
| 1                                                                          | Trans Fun local 1  | ARIMAX(5,1,3) | 0.426  | 5.491  | 1.219 | 26.257 |
| 2                                                                          | ARIMAX local       | ARIMAX(5,1,3) | 0.426  | 5.489  | 1.242 | 26.772 |
| 3                                                                          | Trans Fun local 1  | ARIMAX(5,1,2) | 0.428  | 5.490  | 1.330 | 28.633 |
| 4                                                                          | Trans fun global 1 | ARIMAX(5,1,3) | 0.428  | 5.420  | 1.336 | 28.523 |
| 5                                                                          | ARIMAX local       | ARIMAX(5,1,2) | 0.428  | 5.492  | 1.337 | 28.785 |
| 6                                                                          | ARIMAX globabl     | ARIMAX(5,1,3) | 0.428  | 5.408  | 1.352 | 28.886 |
| 7                                                                          | Trans fun global 1 | ARIMAX(1,1,1) | 0.436  | 5.359  | 1.376 | 29.479 |
| 8                                                                          | Trans fun global 1 | ARIMAX(5,1,2) | 0.430  | 5.431  | 1.422 | 30.343 |
| 9                                                                          | ARIMAX globabl     | ARIMAX(5,1,2) | 0.430  | 5.433  | 1.427 | 30.463 |
| 10                                                                         | ARIMA              | ARIMA(2,1,2)  | 0.465  | 5.331  | 1.516 | 32.883 |
| 11                                                                         | ARIMA              | ARIMA(4,1,3)  | 0.462  | 5.351  | 1.683 | 36.668 |
| 12                                                                         | ECM                | ECM r=1 k=2   | 0.464  | 5.357  | 1.701 | 37.096 |

Table 5.46: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 292-866, predicting 12 values

| Series                                                                      | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE   |
|-----------------------------------------------------------------------------|--------------------|---------------|--------|--------|-------|--------|
| Estimation comp; Est. period 28-Feb-03 - 7-Mar-14 (575 obs); Pred.12 values |                    |               |        |        |       |        |
| 1                                                                           | Trans Fun local 1  | ARIMAX(4,1,2) | 0.423  | 5.040  | 1.035 | 20.235 |
| 2                                                                           | ARIMAX local       | ARIMAX(4,1,2) | 0.425  | 5.044  | 1.070 | 21.293 |
| 3                                                                           | Trans fun global 1 | ARIMAX(4,1,2) | 0.427  | 5.026  | 1.111 | 20.875 |
| 4                                                                           | ARIMAX local       | ARIMAX(5,1,2) | 0.421  | 5.060  | 1.304 | 27.762 |
| 5                                                                           | Trans Fun local 1  | ARIMAX(5,1,2) | 0.420  | 5.058  | 1.306 | 27.824 |
| 6                                                                           | ARIMAX globabl     | ARIMAX(4,1,2) | 0.429  | 5.029  | 1.323 | 27.155 |
| 7                                                                           | Trans fun global 1 | ARIMAX(5,1,2) | 0.425  | 5.056  | 1.490 | 31.622 |
| 8                                                                           | ARIMAX globabl     | ARIMAX(5,1,2) | 0.425  | 5.061  | 1.537 | 32.740 |
| 9                                                                           | ECM                | ECM r=1 k=2   | 0.432  | 5.067  | 1.770 | 38.472 |
| 10                                                                          | ARIMA              | ARIMA(1,1,5)  | 0.457  | 5.187  | 1.807 | 39.262 |
| 11                                                                          | ARIMA              | ARIMA(2,1,1)  | 0.464  | 5.125  | 1.919 | 41.782 |
| 12                                                                          | ARIMA              | ARIMA(4,1,5)  | 0.456  | 5.163  | 1.939 | 42.143 |
| 13                                                                          | ARIMA              | ARIMA(5,1,2)  | 0.455  | 5.250  | 1.956 | 42.535 |

Table 5.47: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 436-866, predicting 12 values

| Series                                                                     | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE   |
|----------------------------------------------------------------------------|--------------------|---------------|--------|--------|-------|--------|
| Estimation comp; Est. period 9-Dec-05 - 7-Mar-14 (431 obs); Pred.12 values |                    |               |        |        |       |        |
| 1                                                                          | Trans Fun local 2  | ARIMAX(5,1,2) | 0.348  | 4.751  | 1.158 | 22.193 |
| 2                                                                          | ARIMA              | ARIMA(5,1,3)  | 0.372  | 4.936  | 1.576 | 33.744 |
| 3                                                                          | Trans Fun local 1  | ARIMAX(5,1,4) | 0.343  | 4.723  | 1.621 | 35.132 |
| 4                                                                          | Trans Fun local 1  | ARIMAX(1,1,1) | 0.354  | 4.753  | 1.684 | 36.573 |
| 5                                                                          | ARIMAX local       | ARIMAX(1,1,1) | 0.354  | 4.757  | 1.711 | 37.194 |
| 6                                                                          | ARIMAX local       | ARIMAX(5,1,4) | 0.348  | 4.772  | 1.721 | 37.325 |
| 7                                                                          | Trans fun global 3 | ARIMAX(5,1,4) | 0.343  | 4.805  | 1.754 | 37.866 |
| 8                                                                          | ARIMAX local       | ARIMAX(5,1,2) | 0.347  | 4.716  | 1.772 | 38.421 |
| 9                                                                          | ARIMAX globabl     | ARIMAX(1,1,1) | 0.354  | 4.801  | 1.809 | 39.206 |
| 10                                                                         | ECM                | ECM r=1 k=2   | 0.367  | 5.045  | 1.815 | 39.493 |
| 11                                                                         | ARIMAX globabl     | ARIMAX(5,1,4) | 0.347  | 4.857  | 1.840 | 39.697 |
| 12                                                                         | Trans fun global 1 | ARIMAX(5,1,2) | 0.351  | 4.818  | 1.841 | 39.702 |
| 13                                                                         | ARIMAX globabl     | ARIMAX(5,1,2) | 0.347  | 4.771  | 1.920 | 41.507 |
| 14                                                                         | ARIMA              | ARIMA(3,1,2)  | 0.374  | 4.939  | 1.969 | 42.942 |
| 15                                                                         | ARIMA              | ARIMA(2,1,1)  | 0.374  | 4.938  | 1.988 | 43.368 |

Table 5.48: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 570-866, predicting 12 values

| Series                                                                      | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE   |
|-----------------------------------------------------------------------------|--------------------|---------------|--------|--------|-------|--------|
| Estimation comp; Est. period 4-Sep-09 - 7-Mar-14 (297 obs); Pred. 12 values |                    |               |        |        |       |        |
| 1                                                                           | ECM                | ECM r=1 k=2   | 0.265  | 4.705  | 1.021 | 19.426 |
| 2                                                                           | ARIMAX local       | ARIMAX(1,1,1) | 0.263  | 4.545  | 1.428 | 30.824 |
| 3                                                                           | Trans Fun local 1  | ARIMAX(3,1,2) | 0.260  | 4.517  | 1.448 | 31.218 |
| 4                                                                           | ARIMAX local       | ARIMAX(3,1,0) | 0.262  | 4.553  | 1.458 | 31.443 |
| 5                                                                           | Trans Fun local 1  | ARIMAX(1,1,1) | 0.262  | 4.542  | 1.463 | 31.664 |
| 6                                                                           | ARIMAX local       | ARIMAX(2,1,2) | 0.261  | 4.521  | 1.501 | 32.509 |
| 7                                                                           | Trans Fun local 2  | ARIMAX(3,1,0) | 0.260  | 4.519  | 1.522 | 32.864 |
| 8                                                                           | Trans fun global 1 | ARIMAX(3,1,2) | 0.263  | 4.532  | 1.540 | 32.873 |
| 9                                                                           | Trans fun global 1 | ARIMAX(2,1,2) | 0.263  | 4.525  | 1.570 | 33.637 |
| 10                                                                          | Trans fun global 1 | ARIMAX(1,1,1) | 0.266  | 4.570  | 1.571 | 33.704 |
| 11                                                                          | Trans Fun local 2  | ARIMAX(2,1,0) | 0.262  | 4.545  | 1.591 | 34.604 |
| 12                                                                          | Trans Fun local 3  | ARIMAX(2,1,2) | 0.259  | 4.493  | 1.608 | 34.939 |
| 13                                                                          | ARIMAX globabl     | ARIMAX(1,1,1) | 0.267  | 4.575  | 1.621 | 34.962 |
| 14                                                                          | ARIMAX globabl     | ARIMAX(3,1,2) | 0.264  | 4.534  | 1.698 | 36.766 |
| 15                                                                          | ARIMAX globabl     | ARIMAX(3,1,0) | 0.267  | 4.592  | 1.768 | 38.258 |
| 16                                                                          | ARIMA              | ARIMA(3,1,5)  | 0.272  | 4.645  | 1.855 | 40.325 |
| 17                                                                          | ARIMA              | ARIMA(0,1,4)  | 0.282  | 4.756  | 1.922 | 41.915 |
| 18                                                                          | ARIMA              | ARIMA(2,1,2)  | 0.281  | 4.726  | 2.567 | 55.611 |

Table 5.49: Model comparison for Henry Hub Natural Gas Spot Price - weekly data, 631-866, predicting 12 values

| Series                                                                      | Model              | Param         | RMSEin | MAPEin | RMSE  | MAPE   |
|-----------------------------------------------------------------------------|--------------------|---------------|--------|--------|-------|--------|
| Estimation comp; Est. period 4-Sep-09 - 7-Mar-14 (236 obs); Pred. 12 values |                    |               |        |        |       |        |
| 1                                                                           | Trans fun global 2 | ARIMAX(2,1,3) | 0.227  | 4.229  | 1.274 | 25.646 |
| 2                                                                           | ARIMAX local       | ARIMAX(3,1,3) | 0.216  | 4.156  | 1.341 | 28.923 |
| 3                                                                           | ARIMAX globabl     | ARIMAX(5,1,2) | 0.227  | 4.238  | 1.362 | 28.128 |
| 4                                                                           | Trans Fun local 1  | ARIMAX(3,1,3) | 0.217  | 4.146  | 1.373 | 29.696 |
| 5                                                                           | Trans fun global 1 | ARIMAX(5,1,2) | 0.227  | 4.248  | 1.378 | 28.477 |
| 6                                                                           | ECM                | ECM r=1 k=2   | 0.249  | 4.346  | 1.448 | 31.151 |
| 7                                                                           | ARIMAX globabl     | ARIMAX(3,1,3) | 0.230  | 4.333  | 1.522 | 32.657 |
| 8                                                                           | ARIMAX local       | ARIMAX(5,1,0) | 0.225  | 4.293  | 1.531 | 33.167 |
| 9                                                                           | Trans Fun local 1  | ARIMAX(5,1,3) | 0.214  | 4.151  | 1.543 | 33.464 |
| 10                                                                          | Trans Fun local 1  | ARIMAX(5,1,0) | 0.225  | 4.296  | 1.548 | 33.528 |
| 11                                                                          | ARIMAX local       | ARIMAX(2,1,3) | 0.225  | 4.284  | 1.573 | 34.139 |
| 12                                                                          | Trans fun global 2 | ARIMAX(3,1,3) | 0.229  | 4.318  | 1.598 | 34.412 |
| 13                                                                          | ARIMAX local       | ARIMAX(5,1,3) | 0.214  | 4.138  | 1.606 | 34.863 |
| 14                                                                          | ARIMAX globabl     | ARIMAX(2,1,3) | 0.230  | 4.317  | 1.622 | 34.943 |
| 15                                                                          | ARIMAX globabl     | ARIMAX(5,1,0) | 0.231  | 4.328  | 1.664 | 35.866 |
| 16                                                                          | Trans fun global 2 | ARIMAX(5,1,0) | 0.230  | 4.322  | 1.682 | 36.269 |
| 17                                                                          | ARIMA              | ARIMA(1,1,3)  | 0.244  | 4.456  | 1.949 | 42.484 |
| 18                                                                          | ARIMA              | ARIMA(2,1,3)  | 0.244  | 4.462  | 1.963 | 42.792 |
| 19                                                                          | ARIMA              | ARIMA(3,1,1)  | 0.244  | 4.466  | 2.030 | 44.239 |

Plots on the Figure ?? depict predicted values and its errors for all of the considered models. It is noticeable that ARIMA and ECM models forecast similarly, while ARIMAX and Transfer Function models' forecasts are following also a common pattern. This fact could be explained again by the input of exogenous variables. ARIMAX and Transfer Function models use all of the available variables, while ARIMA is based only on its previous values. Surprisingly, the ECM model follows ARIMA in prediction, although it is linked with crude oil price fairly strongly. Expected outcome would be that ECM model predictions are much better than ARIMA since crude oil price is cointegrated with natural gas price and additionally, its correlation coefficient is significant.

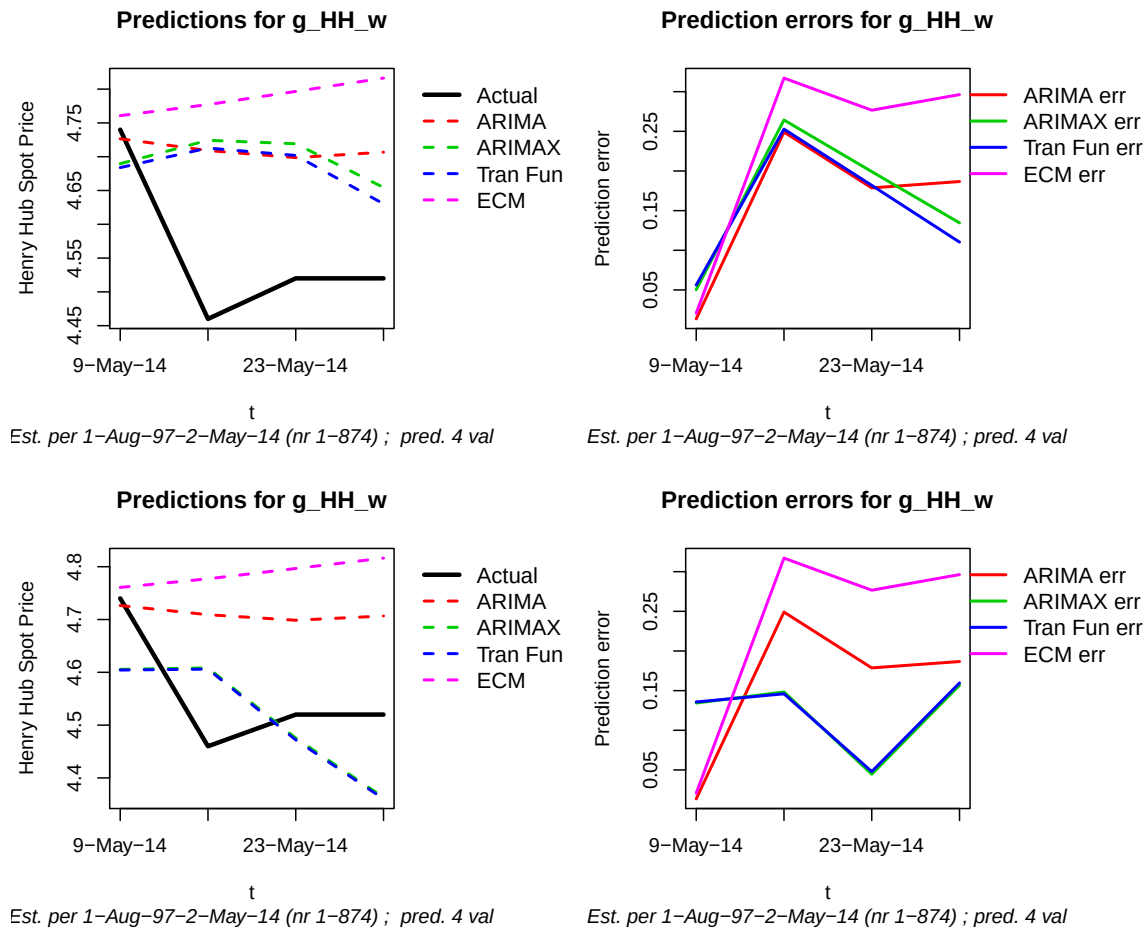


Figure 5.24: Forecast (n=4) for Henry Hub Natural Gas Spot Price all models for local data (upper figures) and global data (lower figures) and their errors

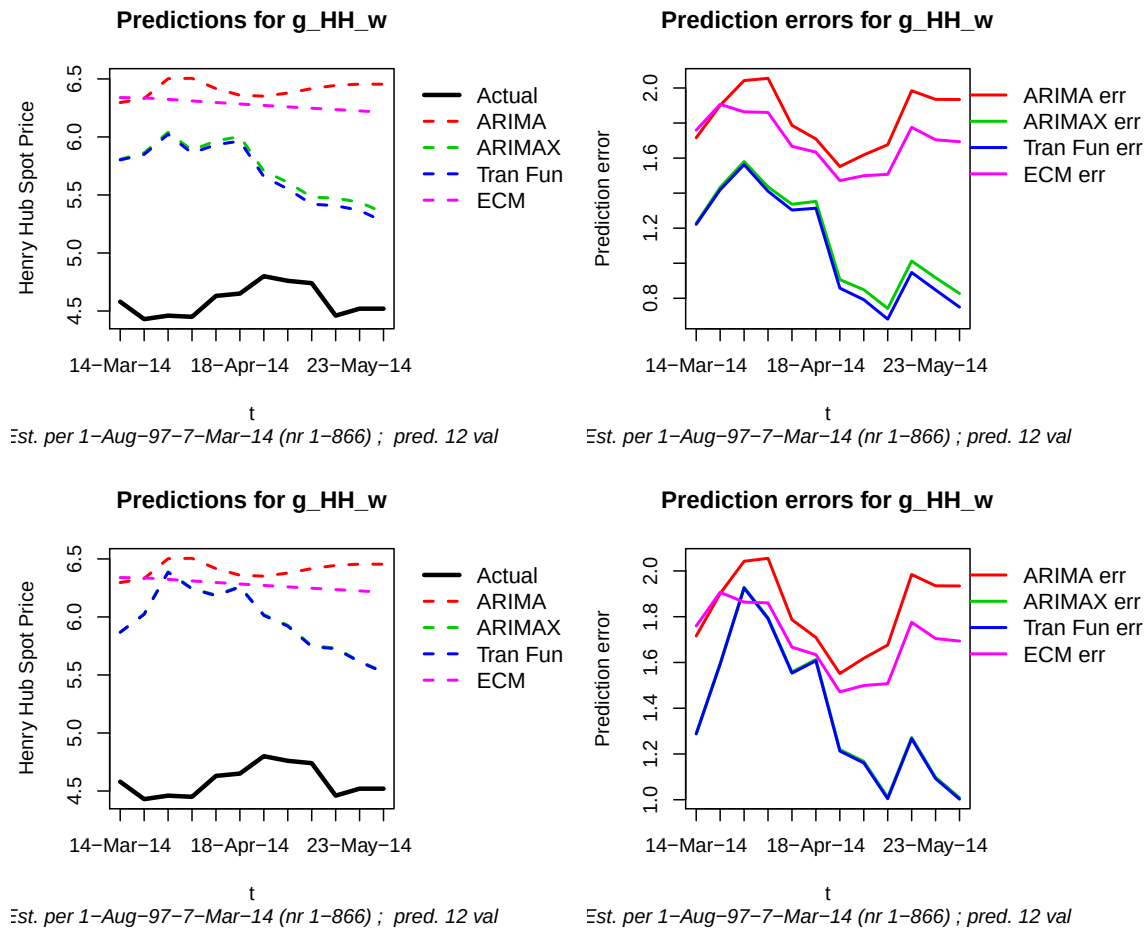


Figure 5.25: Forecast (n=12) for Henry Hub Natural Gas Spot Price - weekly price, all models for local data (upper figures) and global data (lower figures) and their errors

Table 5.50: Prediction comparison for Henry Hub Natural Gas Spot Price - weekly local data, 1-874, predicting 4 values

| Model                                                                                        | ACTUAL | ARIMA        | ARIMAX                         | Transfer Function        | ECM         |
|----------------------------------------------------------------------------------------------|--------|--------------|--------------------------------|--------------------------|-------------|
| Prediction intervals for g_HH_w Est. period 1-Aug-97-2-May-14 (obs nr 1-874) ; pred.4 values |        |              |                                |                          |             |
| Pred nr                                                                                      |        | ARIMA(4,1,2) | variables<br>3,7,8,11,12,17,19 | variabl + lags<br>2      | ECM r=1 k=2 |
| 1                                                                                            | 4.74   | 4.73         | 4.69                           | 4.68                     | 4.76        |
| 2                                                                                            | 4.46   | 4.71         | 4.72                           | 4.71                     | 4.78        |
| 3                                                                                            | 4.52   | 4.70         | 4.72                           | 4.70                     | 4.80        |
| 4                                                                                            | 4.52   | 4.71         | 4.65                           | 4.63                     | 4.82        |
| Err nr                                                                                       | ACTUAL | ARIMA<br>err | ARIMAX<br>err                  | Transfer Function<br>err | ECM<br>err  |
| 1                                                                                            |        | 0.01         | 0.05                           | 0.06                     | 0.02        |
| 2                                                                                            |        | 0.25         | 0.26                           | 0.25                     | 0.32        |
| 3                                                                                            |        | 0.18         | 0.20                           | 0.18                     | 0.28        |
| 4                                                                                            |        | 0.19         | 0.13                           | 0.11                     | 0.30        |

Table 5.51: Prediction comparison for Henry Hub Natural Gas Spot Price - global data, 1-874, predicting 4 values

| Model                                                                                        | ACTUAL | ARIMA        | ARIMAX                              | Transfer Function        | ECM         |
|----------------------------------------------------------------------------------------------|--------|--------------|-------------------------------------|--------------------------|-------------|
| Prediction intervals for g_HH_w Est. period 1-Aug-97-2-May-14 (obs nr 1-874) ; pred.4 values |        |              |                                     |                          |             |
| Pred<br>nr                                                                                   |        | ARIMA(4,1,2) | variabl<br>3, 9, 10, 13, 14, 18, 19 | lags<br>2                | ECM r=1 k=2 |
| 1                                                                                            | 4.74   | 4.73         | 4.61                                | 4.60                     | 4.76        |
| 2                                                                                            | 4.46   | 4.71         | 4.61                                | 4.61                     | 4.78        |
| 3                                                                                            | 4.52   | 4.70         | 4.48                                | 4.47                     | 4.80        |
| 4                                                                                            | 4.52   | 4.71         | 4.36                                | 4.36                     | 4.82        |
| Err<br>nr                                                                                    |        | ARIMA<br>err | ARIMAX<br>err                       | Transfer function<br>err | ECM<br>err  |
| 1                                                                                            |        | 0.01         | 0.13                                | 0.14                     | 0.02        |
| 2                                                                                            |        | 0.25         | 0.15                                | 0.15                     | 0.32        |
| 3                                                                                            |        | 0.18         | 0.04                                | 0.05                     | 0.28        |
| 4                                                                                            |        | 0.19         | 0.16                                | 0.16                     | 0.30        |

Table 5.52: Prediction comparison for Henry Hub Natural Gas Spot Price - local data, 1-866, predicting 12 values

| Model                                                                                         | ACTUAL | ARIMA        | ARIMAX                             | Transfer Function        | ECM         |
|-----------------------------------------------------------------------------------------------|--------|--------------|------------------------------------|--------------------------|-------------|
| Prediction intervals for g_HH_w Est. period 1-Aug-97-7-Mar-14 (obs nr 1-866) ; pred.12 values |        |              |                                    |                          |             |
| Pred<br>nr                                                                                    |        | ARIMA(4,1,2) | variabl<br>3, 7, 8, 11, 12, 17, 19 | lags<br>2                | ECM r=1 k=2 |
| 1                                                                                             | 4.58   | 6.30         | 5.81                               | 5.80                     | 6.34        |
| 2                                                                                             | 4.43   | 6.33         | 5.86                               | 5.85                     | 6.34        |
| 3                                                                                             | 4.46   | 6.50         | 6.04                               | 6.02                     | 6.32        |
| 4                                                                                             | 4.45   | 6.51         | 5.89                               | 5.86                     | 6.31        |
| 5                                                                                             | 4.63   | 6.42         | 5.97                               | 5.93                     | 6.30        |
| 6                                                                                             | 4.65   | 6.36         | 6.00                               | 5.96                     | 6.28        |
| 7                                                                                             | 4.8    | 6.35         | 5.71                               | 5.66                     | 6.27        |
| 8                                                                                             | 4.76   | 6.38         | 5.61                               | 5.55                     | 6.26        |
| 9                                                                                             | 4.74   | 6.42         | 5.48                               | 5.42                     | 6.25        |
| 10                                                                                            | 4.46   | 6.44         | 5.47                               | 5.41                     | 6.24        |
| 11                                                                                            | 4.52   | 6.46         | 5.44                               | 5.37                     | 6.22        |
| 12                                                                                            | 4.52   | 6.45         | 5.35                               | 5.27                     | 6.21        |
| Err<br>nr                                                                                     |        | ARIMA<br>err | ARIMAX<br>err                      | Transfer function<br>err | ECM<br>err  |
| 1                                                                                             |        | 1.72         | 1.23                               | 1.22                     | 1.76        |
| 2                                                                                             |        | 1.90         | 1.43                               | 1.42                     | 1.91        |
| 3                                                                                             |        | 2.04         | 1.58                               | 1.56                     | 1.86        |
| 4                                                                                             |        | 2.06         | 1.44                               | 1.41                     | 1.86        |
| 5                                                                                             |        | 1.79         | 1.34                               | 1.30                     | 1.67        |
| 6                                                                                             |        | 1.71         | 1.35                               | 1.31                     | 1.63        |
| 7                                                                                             |        | 1.55         | 0.91                               | 0.86                     | 1.47        |
| 8                                                                                             |        | 1.62         | 0.85                               | 0.79                     | 1.50        |
| 9                                                                                             |        | 1.68         | 0.74                               | 0.68                     | 1.51        |
| 10                                                                                            |        | 1.98         | 1.01                               | 0.95                     | 1.78        |
| 11                                                                                            |        | 1.94         | 0.92                               | 0.85                     | 1.70        |
| 12                                                                                            |        | 1.93         | 0.83                               | 0.75                     | 1.69        |

Table 5.53: Prediction comparison for Henry Hub Natural Gas Spot Price - global data, 1-866, predicting 12 values

| Model                                                                                      | ACTUAL | ARIMA        | ARIMAX                              | Transfer Function        | ECM         |
|--------------------------------------------------------------------------------------------|--------|--------------|-------------------------------------|--------------------------|-------------|
| Prediction intervals for g_HH_w Est. period 1-Aug-97-7-Mar-14 (obs nr 1-866) ; pred.12 val |        |              |                                     |                          |             |
| Pred<br>nr                                                                                 |        | ARIMA(4,1,2) | variabl<br>3, 9, 10, 13, 14, 18, 19 | lags<br>2                | ECM r=1 k=2 |
| 1                                                                                          | 4.58   | 6.30         | 5.87                                | 5.87                     | 6.34        |
| 2                                                                                          | 4.43   | 6.33         | 6.02                                | 6.02                     | 6.34        |
| 3                                                                                          | 4.46   | 6.50         | 6.39                                | 6.39                     | 6.32        |
| 4                                                                                          | 4.45   | 6.51         | 6.25                                | 6.24                     | 6.31        |
| 5                                                                                          | 4.63   | 6.42         | 6.19                                | 6.18                     | 6.30        |
| 6                                                                                          | 4.65   | 6.36         | 6.26                                | 6.26                     | 6.28        |
| 7                                                                                          | 4.8    | 6.35         | 6.02                                | 6.01                     | 6.27        |
| 8                                                                                          | 4.76   | 6.38         | 5.93                                | 5.92                     | 6.26        |
| 9                                                                                          | 4.74   | 6.42         | 5.75                                | 5.74                     | 6.25        |
| 10                                                                                         | 4.46   | 6.44         | 5.73                                | 5.73                     | 6.24        |
| 11                                                                                         | 4.52   | 6.46         | 5.62                                | 5.61                     | 6.22        |
| 12                                                                                         | 4.52   | 6.45         | 5.53                                | 5.52                     | 6.21        |
| Err<br>nr                                                                                  |        | ARIMA<br>err | ARIMAX<br>err                       | Transfer function<br>err | ECM<br>err  |
| 1                                                                                          |        | 1.72         | 1.29                                | 1.29                     | 1.76        |
| 2                                                                                          |        | 1.90         | 1.59                                | 1.59                     | 1.91        |
| 3                                                                                          |        | 2.04         | 1.93                                | 1.93                     | 1.86        |
| 4                                                                                          |        | 2.06         | 1.80                                | 1.79                     | 1.86        |
| 5                                                                                          |        | 1.79         | 1.56                                | 1.55                     | 1.67        |
| 6                                                                                          |        | 1.71         | 1.61                                | 1.61                     | 1.63        |
| 7                                                                                          |        | 1.55         | 1.22                                | 1.21                     | 1.47        |
| 8                                                                                          |        | 1.62         | 1.17                                | 1.16                     | 1.50        |
| 9                                                                                          |        | 1.68         | 1.01                                | 1.00                     | 1.51        |
| 10                                                                                         |        | 1.98         | 1.27                                | 1.27                     | 1.78        |
| 11                                                                                         |        | 1.94         | 1.10                                | 1.09                     | 1.70        |
| 12                                                                                         |        | 1.93         | 1.01                                | 1.00                     | 1.69        |





## Chapter 6

# Conclusion

Natural gas market in the U.S. is the proper subject of my study and represents commodity markets, with known economic taxonomy and methodology. In the recent years mathematical models became more and more popular and nowadays are successfully applied in the economic prediction. Proposed stochastic models describe commodity markets and can be used by policy agencies and induce market players decisions. Understanding of above-mentioned models is crucial for comprehension of the commodity market dynamics. Thus, for the purpose of the model building, various econometrical techniques were applied in this research.

As a reflection on the conducted analysis, some conclusion can be drawn. First of all, the prediction results do not depend solely on the applied method but also on a multitude of other factors. Those factors are, for example, time horizon (for model identification and forecast), not only the length of the given time horizon but also the actual time (and its circumstances, such as hurricane, hot summer, cold winter, economical instability etc.).

Another important factor is selection of exogenous variables. As can be seen from the correlation matrix (Table 5.19) and ARIMAX models, high correlation between variables (like Henry Hub Natural Gas Spot Price and Shut-in production) does not always indicate that the variable will be prominent for prediction of response variable. Also a matter of significant importance is not only variable selection, but also data selection. As can be seen, local data predicted much more precise than global data (although variables, in a meaning of data representation were the same).

From conducted analysis, best models for certain data can be selected. The best models occurred ARIMAX/Transfer Function models. This fact could be caused by usage of the full data set for prediction (all the variables). In comparison, in the Error Correction Model only West Texas Intermediate Crude Oil Spot Price has been used, which might explain its weak predicting power.

As future research, a closer look into the Error Correction Model and also the Vector Error Correction Conditional model would be advised. Also Artificial Neural Network seem to be an appropriate model that could give satisfactory results in predicting natural gas price.

Another reflection concerns ARIMA models. Since ARIMA models depend only on previous values of explanatory variables, there is no need of additional factors and this implicates that the time horizon can be longer (since not all data are available from the same time point). This means that using longer identification and estimation period, ARIMA models can give better results than ARIMAX/Transfer Function models or Error Correction Model (see Appendix B).



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# Appendix A

## Data

### A.1 Variables description

#### A.1.1 Henry Hub natural gas Spot Price

Describing the amount of natural gas consumed by an entire country or a single residential appliance can be confusing, since natural gas can be measured in several different ways.

The energy content of natural gas and other forms of energy (i.e., the potential heat that can be generated from the fuel) is measured in Btus (British thermal units). The number of "therms" that residential natural gas customers consume each month is listed on their gas bills.

Quantities of natural gas are usually measured in cubic feet. For example, a typical natural gas futures contract is a financial instrument based on the value of about 10 million cubic feet (Mmcf) of natural gas.

Here are some frequently used units for measuring natural gas:

1 cubic foot (cf) = 1,027 Btu  
100 cubic feet (1 ccf) = 1 therm (approximate)  
1,000 cubic feet (1 Mcf) = 1,027,000 Btu (1 MMBtu)  
1,000 cubic feet (1 Mcf) = 1 dekatherm (10 therms)  
1 million (1,000,000) cubic feet (1 Mmcf) = 1,027,000,000 Btu  
1 billion (1,000,000,000) cubic feet (1 bcf) = 1.027 trillion Btu  
1 trillion (1,000,000,000,000) cubic feet (1Tcf) = 1.027 quadrillion Btu

To put this in context:

1,000 cubic feet of natural gas is approximately enough to meet the natural gas needs of an average home (space-heating, water-heating, cooking, etc.) for four days.

In 2009, the average American home consumed 70,500 cubic feet of natural gas (or 72.6 million Btu). On a daily basis, the average U.S. home used 193 cubic feet of natural gas.

Most natural gas household bills display the amount of natural gas consumed by the 'therm' (100,000 Btu) or in hundreds of cubic feet (100 cubic feet - 1 Ccf).

Looking at larger quantities, 1 billion cubic feet (Bcf) of natural gas is enough to meet the needs of approximately 10,000 - 11,000 American homes for one year.

5 trillion cubic feet of natural gas is enough to meet the needs of 5 million households for 15 years.

## A.1.2 Natural gas Storage Differential

Natural gas is stored in various locations throughout the United States, ranging from depleted oil and/or gas fields to LNG storage facilities. The EIA collects and sums the data on storage levels on a weekly basis and reports it on its website. The STOR\_ch variable is the difference between a weekly gas storage level and the 5-year running average for that date, reported in billion cubic feet (Bcf, equivalent to 1.03 million mmBtu). The average storage level from 1997-2010 was 2,332 Bcf (2.4 billion mmBtu) on any given week, with a standard deviation of 744.8 Bcf (767.53 million mmBtu). The average storage differential from 1997-2010 was 157 Bcf (162.1 million mmBtu), and the average amount of time that storage remained out of sync with the normal storage level was about 39 weeks. The median duration of a storage differential was 12 weeks. Figure A6 shows a graph of this variable.

## A.1.3 Shut-in production

(Following Ramberg and Persons in [4]). The amount of natural gas production capacity that is curtailed in the Gulf of Mexico in million cubic feet (mmcf, equivalent to 1,030.5 mmBtu). This is always a result of hurricane activity. The shut-in natural gas production in the Gulf (SHUTIN) serves as an alternative to a ‘hurricane dummy’ variable in order to model the one-off effects of a hurricane’s impact on the Gulf of Mexico gas industry. However, the shut-in production figure could be considered superior in one aspect: prolonged curtailments in production, perhaps due to damage caused by the hurricane to drilling rigs or gathering infrastructure, can be tracked according to their severity better than through the use of a binary dummy variable. The Minerals Management Service (MMS) posts weekly shut-in production statistics on its Gulf of Mexico webpage under Press Releases/Reports whenever hurricane activity prompts oil and gas producers to halt production at their offshore platforms. Gulf of Mexico natural gas production is roughly 10% of total U.S. gas production over the course of the year (according to the Energy Information Administration (EIA) of the Department of Energy). Over the 1997-2010 period, Gulf of Mexico natural gas production ranged from about 7 to 10 Bcf (7.2 to 10.3 million mmBtu) per day (MMS website). Hurricanes on the scale of the Katrina/Rita event managed to shut in over 80% of Gulf production (MMS website press releases). Hurricane disruptions to Gulf gas production have a characteristic pattern: a large spike in the week preceding hurricane impact, followed by a gradual decline in shut-in production. This reflects the initial evacuation before hurricane impact, the immediate return to production of undamaged wells after the hurricane has passed, and the gradual return to production of rigs that were damaged to different degrees by the hurricane. The shortest such disruption in our dataset was 34 weeks in response to Hurricane Ivan in 2004. The other two hurricane impacts in the dataset were actually combinations of two hurricane impacts in each case - Hurricanes Katrina and Rita in 2005 and Hurricanes Gustav and Ike in 2008. We thus cannot reliably calculate an average production impact for a single hurricane from our dataset - we have only one observation.

## A.2 Data statistics

| date_w          | g_HH_w           | o_WTI_w         | HDD_WSC_w       | HDDdev_WSC_w    |
|-----------------|------------------|-----------------|-----------------|-----------------|
| 1-Apr-05: 1     | Min. : 1.340     | Min. : 11.00    | Min. : 0.00     | Min. : -85.000  |
| 1-Apr-11: 1     | 1st Qu.: 3.072   | 1st Qu.: 28.09  | 1st Qu.: 0.00   | 1st Qu.: -8.000 |
| 1-Aug-03: 1     | Median : 4.280   | Median : 56.07  | Median : 19.00  | Median : 0.000  |
| 1-Aug-08: 1     | Mean : 4.788     | Mean : 57.07    | Mean : 43.24    | Mean : -2.685   |
| 1-Aug-97: 1     | 3rd Qu.: 6.080   | 3rd Qu.: 85.85  | 3rd Qu.: 74.00  | 3rd Qu.: 3.000  |
| 1-Dec-00: 1     | Max. : 14.490    | Max. : 142.52   | Max. : 228.00   | Max. : 88.000   |
| (Other) : 872   |                  |                 |                 |                 |
| HDD_USA_w       | HDDdev_USA_w     | CDD_WSC_w       | CDDdev_WSC_w    | CDD_USA_w       |
| Min. : 0.00     | Min. : -88.000   | Min. : 0.00     | Min. : -35.000  | Min. : 0.00     |
| 1st Qu.: 9.00   | 1st Qu.: -14.000 | 1st Qu.: 5.00   | 1st Qu.: -3.000 | 1st Qu.: 2.00   |
| Median : 71.50  | Median : -2.000  | Median : 31.00  | Median : 1.000  | Median : 10.00  |
| Mean : 89.52    | Mean : -5.226    | Mean : 51.22    | Mean : 4.719    | Mean : 25.34    |
| 3rd Qu.: 162.75 | 3rd Qu.: 3.000   | 3rd Qu.: 101.00 | 3rd Qu.: 13.000 | 3rd Qu.: 47.00  |
| Max. : 272.00   | Max. : 59.000    | Max. : 174.00   | Max. : 61.000   | Max. : 107.00   |

| CDDdev_USA_w    | STOR_PROD_w    | STOR_USA_w    | STOR_PROD_ch_w   | STOR_USA_ch_w     |
|-----------------|----------------|---------------|------------------|-------------------|
| Min. : -19.000  | Min. : 198.0   | Min. : 642    | Min. : -0.53850  | Min. : -0.54710   |
| 1st Qu.: -1.000 | 1st Qu.: 599.5 | 1st Qu.: 1772 | 1st Qu.: 0.01903 | 1st Qu.: -0.00860 |
| Median : 0.000  | Median : 762.0 | Median : 2472 | Median : 0.14340 | Median : 0.06875  |
| Mean : 2.454    | Mean : 768.6   | Mean : 2409   | Mean : 0.13830   | Mean : 0.07193    |
| 3rd Qu.: 4.000  | 3rd Qu.: 938.0 | 3rd Qu.: 3033 | 3rd Qu.: 0.25972 | 3rd Qu.: 0.15995  |
| Max. : 34.000   | Max. : 1297.0  | Max. : 3929   | Max. : 0.84330   | Max. : 0.67190    |

| STORAGE.EAST_w  | STORAGE.WEST_w | HH_shutin_w   | WTI_shutin_w   |
|-----------------|----------------|---------------|----------------|
| Min. : 277.0    | Min. : 114.0   | Min. : 0.0    | Min. : 0       |
| 1st Qu.: 896.5  | 1st Qu.: 258.2 | 1st Qu.: 0.0  | 1st Qu.: 0     |
| Median : 1342.0 | Median : 341.5 | Median : 0.0  | Median : 0     |
| Mean : 1303.1   | Mean : 337.3   | Mean : 220.3  | Mean : 46927   |
| 3rd Qu.: 1723.8 | 3rd Qu.: 409.8 | 3rd Qu.: 0.0  | 3rd Qu.: 0     |
| Max. : 2101.0   | Max. : 555.0   | Max. : 8111.6 | Max. : 1650000 |

## Appendix B

# ARIMA models for longer time horizon

Table B.1: ARIMA models for Henry Hub Natural Gas Spot Price - weekly data, longer dataset, predicting 4 values

| Series                                                       | Models              | AIC      | BIC      | RMSE  | MAPE   |
|--------------------------------------------------------------|---------------------|----------|----------|-------|--------|
| HH_w.csv; Jan10, 1997 – Apr11, 2014 (900 obs); pred.4 values |                     |          |          |       |        |
| 1                                                            | <b>ARIMA(5,1,3)</b> | 1168.117 | 1216.140 | 0.158 | 3.279  |
| 2                                                            | ARIMA(4,1,4)        | 1168.430 | 1216.454 | 0.158 | 3.306  |
| 3                                                            | ARIMA(5,1,0)        | 1171.561 | 1205.178 | 0.145 | 2.950  |
| 4                                                            | ARIMA(5,1,2)        | 1172.717 | 1215.938 | 0.114 | 2.328  |
| 5                                                            | ARIMA(4,1,2)        | 1175.079 | 1213.498 | 0.047 | 0.724  |
| HH_w.csv; Aug19, 2005 – Apr11, 2014 (451 obs); pred.4 values |                     |          |          |       |        |
| 1                                                            | ARIMA(5,1,2)        | 538.159  | 575.162  | 0.783 | 15.822 |
| 2                                                            | ARIMA(4,1,4)        | 540.056  | 581.171  | 0.281 | 5.253  |
| 3                                                            | ARIMA(4,1,5)        | 540.791  | 586.017  | 0.325 | 6.246  |
| 4                                                            | ARIMA(5,1,0)        | 542.105  | 570.885  | 0.417 | 8.806  |
| 5                                                            | <b>ARIMA(6,1,5)</b> | 542.386  | 595.835  | 0.431 | 8.627  |
| 6                                                            | ARIMA(1,1,5)        | 546.059  | 578.950  | 0.384 | 8.056  |
| HH_w.csv; Jun26, 2009 – Apr11, 2014 (251 obs); pred.4 values |                     |          |          |       |        |
| 1                                                            | ARIMA(3,1,2)        | 71.041   | 95.719   | 0.167 | 3.334  |
| 2                                                            | ARIMA(3,1,5)        | 71.448   | 106.702  | 0.103 | 1.929  |
| 3                                                            | ARIMA(5,1,2)        | 71.853   | 103.582  | 0.284 | 4.993  |
| 4                                                            | ARIMA(1,1,2)        | 72.159   | 89.786   | 0.100 | 1.811  |
| 5                                                            | <b>ARIMA(0,1,0)</b> | 72.870   | 79.921   | 0.113 | 2.096  |
| 6                                                            | ARIMA(1,1,3)        | 73.554   | 94.707   | 0.223 | 4.318  |
| 7                                                            | ARIMA(5,1,0)        | 75.222   | 99.900   | 0.153 | 3.137  |
| HH_w.csv; May27, 2011 – Apr11, 2014 (151 obs); pred.4 values |                     |          |          |       |        |
| 1                                                            | ARIMA(1,1,5)        | 15.822   | 39.960   | 0.376 | 7.701  |
| 2                                                            | ARIMA(2,1,5)        | 17.770   | 44.926   | 0.375 | 7.691  |
| 3                                                            | ARIMA(5,1,4)        | 17.966   | 51.156   | 0.366 | 7.505  |
| 4                                                            | <b>ARIMA(1,1,0)</b> | 24.687   | 33.739   | 0.381 | 7.841  |
| 5                                                            | ARIMA(0,1,5)        | 26.116   | 47.237   | 0.379 | 7.804  |
| 6                                                            | ARIMA(2,1,3)        | 27.277   | 48.398   | 0.372 | 7.662  |
| 7                                                            | ARIMA(3,1,2)        | 27.480   | 48.601   | 0.377 | 7.745  |



Table B.2: ARIMA models for Henry Hub Natural Gas Spot Price - weekly data, longer dataset, predicting 12 values

| Series                                                                | Models              | AIC      | BIC      | RMSE  | MAPE   |
|-----------------------------------------------------------------------|---------------------|----------|----------|-------|--------|
| HH_w.csv; <i>JJan10, 1997 – Feb14, 2014</i> (892 obs); pred.12 values |                     |          |          |       |        |
| 1                                                                     | ARIMA(5,1,3)        | 1140.153 | 1188.087 | 1.442 | 28.915 |
| 2                                                                     | ARIMA(4,1,4)        | 1140.359 | 1188.294 | 1.474 | 29.551 |
| 3                                                                     | ARIMA(4,1,5)        | 1141.480 | 1194.208 | 1.462 | 29.317 |
| 4                                                                     | ARIMA(5,1,4)        | 1141.562 | 1194.290 | 1.464 | 29.352 |
| 5                                                                     | ARIMA(3,1,5)        | 1142.319 | 1190.254 | 1.444 | 28.986 |
| 6                                                                     | ARIMA(5,1,1)        | 1145.828 | 1184.176 | 1.442 | 28.917 |
| 7                                                                     | ARIMA(1,1,5)        | 1145.913 | 1184.260 | 1.403 | 28.142 |
| 8                                                                     | ARIMA(2,1,4)        | 1145.955 | 1184.303 | 1.486 | 29.731 |
| 9                                                                     | <b>ARIMA(4,1,1)</b> | 1147.189 | 1180.743 | 1.428 | 28.612 |
| HH_w.csv; <i>Aug19, 2005 – Feb14, 2014</i> (443 obs); pred.12 values  |                     |          |          |       |        |
| 1                                                                     | ARIMA(5,1,3)        | 502.590  | 543.526  | 1.177 | 23.799 |
| 2                                                                     | ARIMA(5,1,2)        | 502.593  | 539.435  | 1.272 | 25.560 |
| 3                                                                     | ARIMA(4,1,4)        | 503.229  | 544.165  | 1.247 | 25.143 |
| 4                                                                     | ARIMA(5,1,4)        | 504.137  | 549.166  | 1.195 | 24.146 |
| 5                                                                     | ARIMA(4,1,5)        | 504.233  | 549.263  | 1.218 | 24.591 |
| 6                                                                     | <b>ARIMA(6,1,5)</b> | 507.030  | 560.246  | 1.232 | 24.842 |
| 7                                                                     | ARIMA(1,1,5)        | 513.670  | 546.419  | 1.107 | 22.219 |
| HH_w.csv; <i>Jun26, 2009 – Feb14, 2014</i> (243 obs); pred.12 values  |                     |          |          |       |        |
| 1                                                                     | ARIMA(1,1,2)        | -14.921  | 2.544    | 1.891 | 38.356 |
| 2                                                                     | ARIMA(1,1,3)        | -12.439  | 8.519    | 1.715 | 34.651 |
| 3                                                                     | ARIMA(3,1,0)        | -14.962  | 2.503    | 1.653 | 33.293 |
| 4                                                                     | ARIMA(3,1,1)        | -14.420  | 6.538    | 1.808 | 36.590 |
| 5                                                                     | ARIMA(2,1,3)        | -14.015  | 10.436   | 1.906 | 38.574 |
| 6                                                                     | <b>ARIMA(2,1,2)</b> | -13.875  | 7.083    | 1.879 | 38.003 |
| HH_w.csv; <i>May27, 2011 – Feb14, 2014</i> (143 obs); pred.12 values  |                     |          |          |       |        |
| 1                                                                     | <b>ARIMA(2,2,3)</b> | -77.361  | -56.621  | 0.669 | 12.681 |
| 2                                                                     | ARIMA(2,1,2)        | -82.673  | -64.896  | 0.519 | 10.392 |
| 3                                                                     | ARIMA(1,1,8)        | -81.769  | -49.178  | 2.885 | 57.096 |
| 4                                                                     | ARIMA(6,1,8)        | -81.697  | -34.292  | 0.553 | 10.718 |
| 5                                                                     | ARIMA(8,1,0)        | -80.163  | -50.535  | 2.879 | 57.180 |